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Fast and Stable Virtual Human Video Generation: A Survey

Anonymous

Abstract Virtual-human video generation must produce timely output while maintaining identity, appearance, and motion during sustained use. Achieving both goals requires decisions about representation, temporal state, and the complete generation pipeline. This survey reviews these decisions through two perspectives, *Fast* and *Stable*, focusing on talking portraits and selected upper- and full-body systems. We organize methods into GAN, diffusion, visual-autoregressive, and rendering routes according to the component responsible for generated appearance, and explain how hybrid systems distribute control, state, and decoding across modules. We compare acceleration through compact representations, fewer model evaluations, computation reuse, and execution optimization. We then examine temporal failures and the roles of reference anchors, bounded memory, position handling, history-aware training, and boundary-state transfer. Representative studies illustrate how decoder choices, parallel execution, and appearance reuse affect efficiency and temporal behavior. Finally, we review datasets and evaluation measures and discuss open challenges in complete-pipeline measurement, long-session consistency, and controllable generation. This organization connects method design with the practical requirements of fast and stable virtual humans.

1 Introduction

Virtual-human video generation aims to synthesize a human subject whose appearance and behavior follow specified driving signals, such as speech, text, pose, expression, or video. The output may show a face, an upper body, or a full body, while the subject may be specified by one reference image, an enrollment recording, or a learned identity representation. Across these settings, the central requirements are to preserve the intended appearance and reproduce the motion conveyed by the input [134, 183, 212, 237].

Practical deployment introduces two further requirements. The first is computational efficiency: generation must fit the available compute and latency budget, particularly when serving many users or following a live signal [62, 81, 209]. The second is long-horizon stability: identity, appearance, synchronization, and motion should remain consistent as generation continues [113, 165]. We refer to these perspectives as *Fast* and *Stable*. Short-clip visual quality alone establishes neither property. We therefore examine how each family of methods reduces computation, maintains temporal consistency, and trades these objectives against representation quality and deployment cost.

Figure 1 illustrates the distinction: delayed output limits responsiveness, whereas unintended changes in identity or scene limit continuity. A useful evaluation must measure both, while allowing changes explicitly requested by the driving signal.

Figure 2 places representative methods on the four ownership routes over time. Recent work reduces different costs. Distillation can reduce the number of diffusion evaluations [127, 174, 224, 225]. Caching can reuse computation across denoising steps or adjacent frames [118, 129, 132]. Autoregressive models can generate cross-frame visual state sequentially and reuse cached history [27, 102, 208]. Subject-specific renderers can reuse an enrolled scene or reference representation across frames [62, 63, 205, 209]. These mechanisms change different cost terms and should not be treated as equivalent: a measured frame rate does not by itself establish first-output latency or long-session stability, and a local speedup can move a bottleneck to the decoder, the

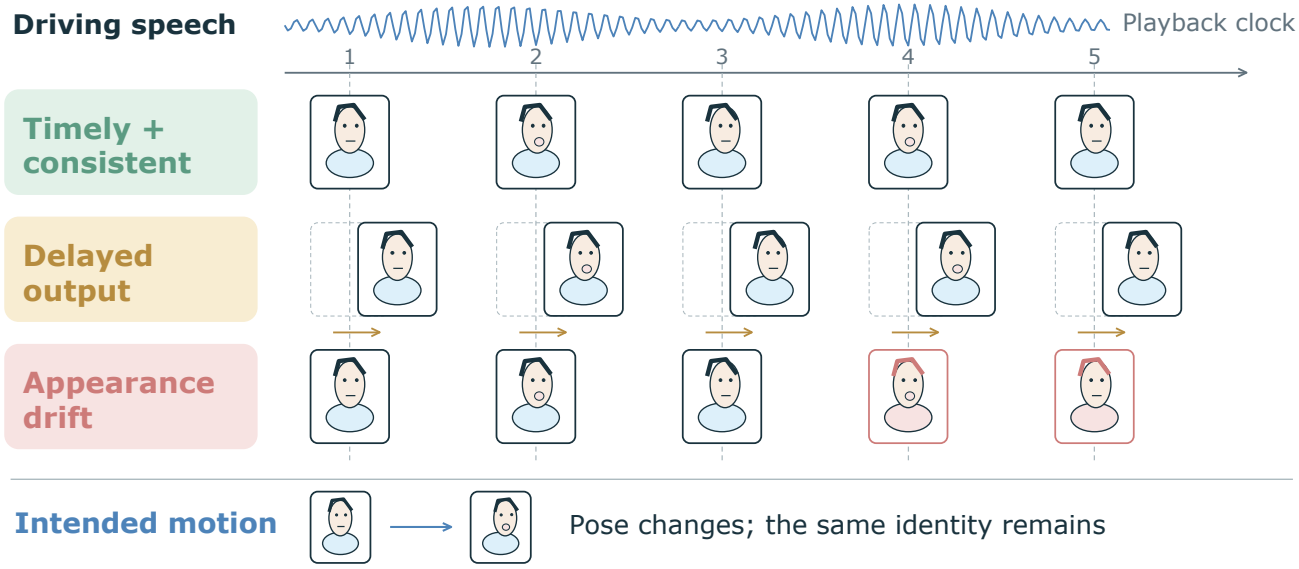


Fig. 1 Fast and Stable are distinct, coupled requirements. Aligned playback timelines distinguish timely, consistent generation from delayed output and unintended appearance drift. Requested pose changes should be preserved rather than penalized as drift. Portraits and timing marks are schematic.

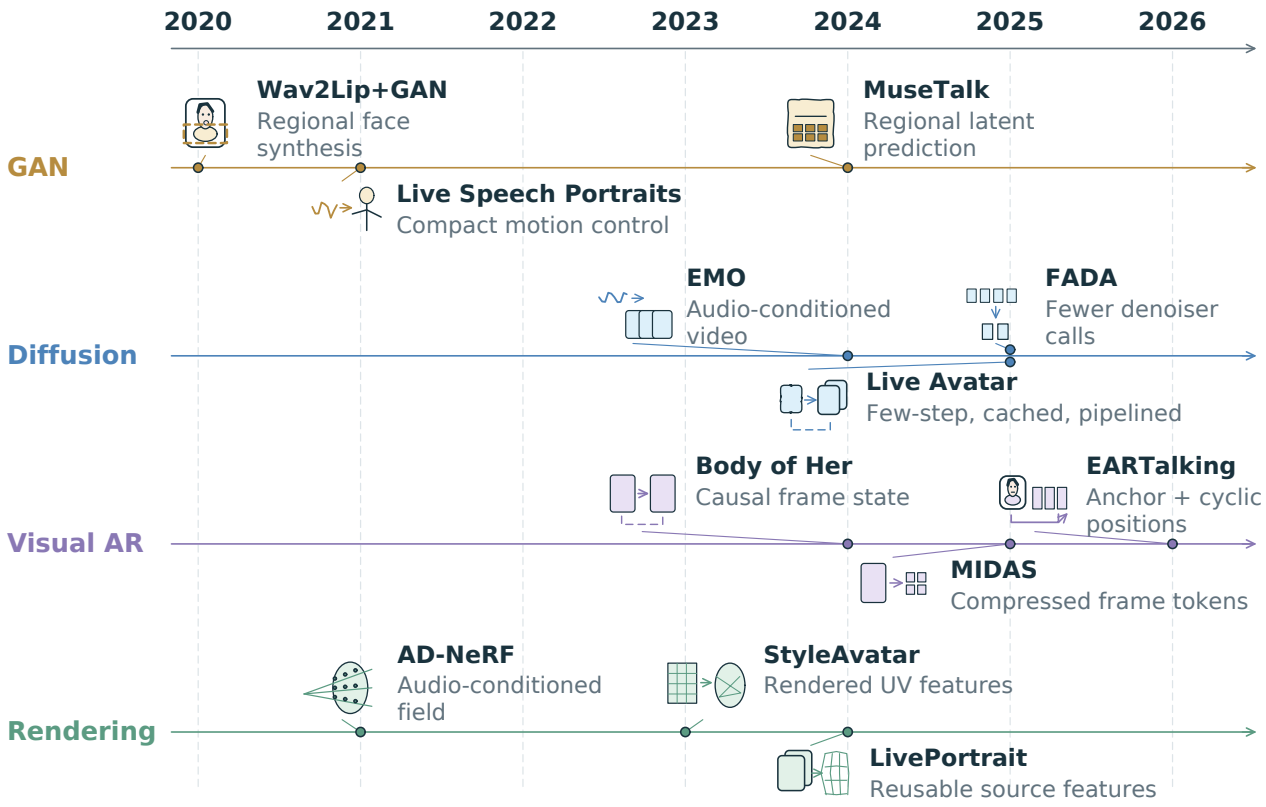


Fig. 2 Selected mechanisms along the four routes: regional and latent face synthesis; fewer diffusion evaluations and pipelined serving; compressed causal visual state; and reusable fields or geometric appearance [5, 27, 62, 63, 81, 124, 150, 183, 193, 203, 239, 254]. Dates indicate the first public edition, not priority or comparative performance.

condition encoder, or the serving system rather than remove it.

These developments motivate an organization based on the component responsible for generated appearance, which we call the *visual owner*. Systems with similar outputs can have different visual owners and computational costs, whereas systems covering different body regions can share the same acceleration mechanism. Identifying this component provides a common starting point for discussing efficiency and stability within each method family.

1.1 Scope: Fast and Stable

The main subjects of this survey are systems that produce human-centered RGB video and introduce mechanisms for reducing runtime cost or supporting extended generation. Our coverage centers on talking portraits, with selected upper- and full-body systems included to examine representation and deployment differences. We cover generalizable and subject-specific methods without claiming exhaustive coverage of whole-body generation. Render-based systems are included when they contribute to efficient appearance construction or frame production, for example through neural radiance fields (NeRFs) or 3D Gaussian splatting (3DGS) [91, 137]. Methods whose final output consists only of a mesh, motion coefficients, or vertex trajectories are discussed as supporting components when they clarify the design of an RGB-generation pipeline.

We distinguish the systems under comparison from foundational and contextual references. Foundational methods explain the origin of a representation or mechanism; contextual references clarify an interface, such as motion prediction followed by rendering. Neither role implies inclusion in a route's deployment comparison. Inclusion in a primary route requires a sufficiently described architecture to identify the visual owner. Works with undisclosed ownership, or with a visual synthesis mechanism outside the four families considered here, are omitted from the route comparisons. Missing timing or stability measurements do not prevent an otherwise supported architectural assignment; such measurements are marked not reported (NR).

The following two cross-cutting lenses organize the analysis.

Fast concerns the computation and time required to produce visible output. We examine representation size, model-call count, cost per call, computation reuse, and decoding or rendering overhead. Throughput and first-output latency are recorded separately under the source's hardware and timing conditions.

Stable concerns consistency over an evaluated continuous duration. We examine identity and appearance drift together with the memory, training, and recovery mechanisms intended to control it. A long demonstration establishes an observed duration; quantitative stability additionally requires measurements over that duration.

Interactivity is a related but distinct property: a system can respond to new conditions during generation, such as a dialogue turn, interruption, or change of pose [81, 113, 215, 230, 234]. This capability does not establish stability over a long session; an interactive system may operate briefly, reset frequently, or accumulate drift. We include interactive systems when their generation mechanisms fall within the stated scope and examine their efficiency and stability separately. Interaction behavior has its own surveys and evaluation protocols [8, 15, 49, 228] and is not ranked here.

Fast and Stable are distinct evaluation axes whose design choices can be coupled. A fast method may drift within a minute; a stable long-horizon method may run far below playback speed. Both properties therefore require evaluation under the intended operating conditions. Section 2 defines the terms and measurement boundaries used throughout the survey.

1.2 Literature Selection

We consider literature available through 13 October 2026, prioritizing additions that clarify a mechanism, a classification boundary, or an evaluation limitation rather than treating the bibliography as an exhaustive census.

This survey takes a selective, mechanism-oriented view of human-centered RGB generation. We include methods that clarify how appearance representation, computation reuse, or temporal state affects efficiency and stability. The primary comparison covers talking portraits and selected upper- and full-body systems with sufficiently described generation architectures. Foundational work on generative modeling, acceleration, and rendering supplies the technical background for these methods.

Motion-only and coefficient-only models are discussed when they explain a control interface used by an RGB-generation pipeline. We distinguish these supporting components from the visual generator when assigning a route. Different releases and configurations of a method are identified where they change the architecture or reported results. Runtime figures retain the settings of the cited paper, and NR denotes a measurement not reported there. The companion catalogue provides a searchable guide to the literature.

Table 1 Complementary survey perspectives, emphasizing the tasks, organizing principles, and practical questions addressed by each review.

Survey / scope	Organizing principle	Efficiency and temporal emphasis	Relation to the present analysis
Deep Person Generation [161]	Face, pose, and clothing generation; shared technical components	Identity/attribute control, generation quality, and task-specific challenges	Connects human-generation tasks; our comparison unit is the deployed RGB pipeline.
Talking Head Synthesis [134]	Portrait generation, driving mechanisms, and editing	Task-dependent synthesis, control, and evaluation	Complements our separation of appearance, control, and persistent state.
Human Motion Video Generation [212]	Input, motion planning, video generation, refinement, and output; modality and task	Motion/video methods, datasets, evaluation, and temporal challenges	Provides broad human-motion coverage; we focus on cost, state, enrollment, and runtime and temporal behavior.
Efficient Diffusion Models [164]	Algorithm, system, and framework levels	Sampling and implementation efficiency across diffusion applications	Supplies acceleration concepts; we track which stage is accelerated and include GAN, visual AR, and rendering pipelines.
Interactive Real-Time Visual Avatar Generation [49]	Method families and latency, anatomy, personalization, and input modality	Production constraints and head–body integration; drift as a challenge	Closely related deployment perspective; our analysis connects module roles with computation and temporal behavior.
This survey	Visual owner plus independent control/state/decoder roles	Fast/Stable lenses; timing boundaries, enrollment, seven stability dimensions	Cross-route mechanism analysis and comparison of complete-pipeline cost and temporal consistency.

1.3 Relation to Existing Surveys

Prior surveys provide complementary views of the area. Work on efficient generative modeling reviews distillation, caching, attention, and inference optimization [164]. Long-video surveys organize temporal extension around generation paradigms and resource or quality challenges [103]. Human-centered surveys organize talking heads, portrait animation, and human video by task or representation [134, 156, 161, 212]. Graphics-oriented reviews cover avatar construction and learned appearance priors [263]. Recent surveys consider interactive generative video, organized around the interaction loop [228], and real-time interactive avatars, organized by method family and by production axes such as rendering latency, anatomical coverage, and personalization [49].

Table 1 compares these organizing emphases. Our analysis connects three questions: which component produces appearance, which operations reduce its computational cost, and how does it maintain consistency over time? Describing the modules of hybrid systems makes these connections explicit and complements taxonomies based on task, acceleration, or interaction.

Figure 3 summarizes these complementary organizing perspectives. Our emphasis is on connecting the component that generates appearance to its computational cost and temporal behavior.

1.4 Ownership Taxonomy

Figure 4 assigns each included system to one of four routes by identifying its visual owner. The rules are applied in order: render-based, autoregressive, diffusion-based, then GAN-based. This convention resolves hybrids, such as an autoregressive motion controller driving a subject-specific renderer. It is not intended to classify every possible video generator. Section 2.3 defines the rules and illustrates how they apply to hybrid architectures.

- **GAN-based methods** (Section 3) use an adversarially trained feed-forward generator for dense or regional imagery [58, 89, 150, 178, 239]. Regional prediction and compact control reduce per-frame work, while temporal consistency depends on the generator's conditioning and any temporal components.
- **Diffusion-based methods** (Section 4) use diffusion or flow matching to generate dense pixels, visual latents, or visual chunks [69, 73, 117, 183]. Acceleration reduces iterative calls, per-call cost, or repeated work; stability mechanisms include reference anchors, bounded memory, and training with generated history.
- **Autoregressive methods** (Section 5) predict evolving visual tokens or latents from previous visual state [5, 27, 203, 208]. Token volume and serial prediction determine much of the cost, while retained history can support continuity and propagate errors.

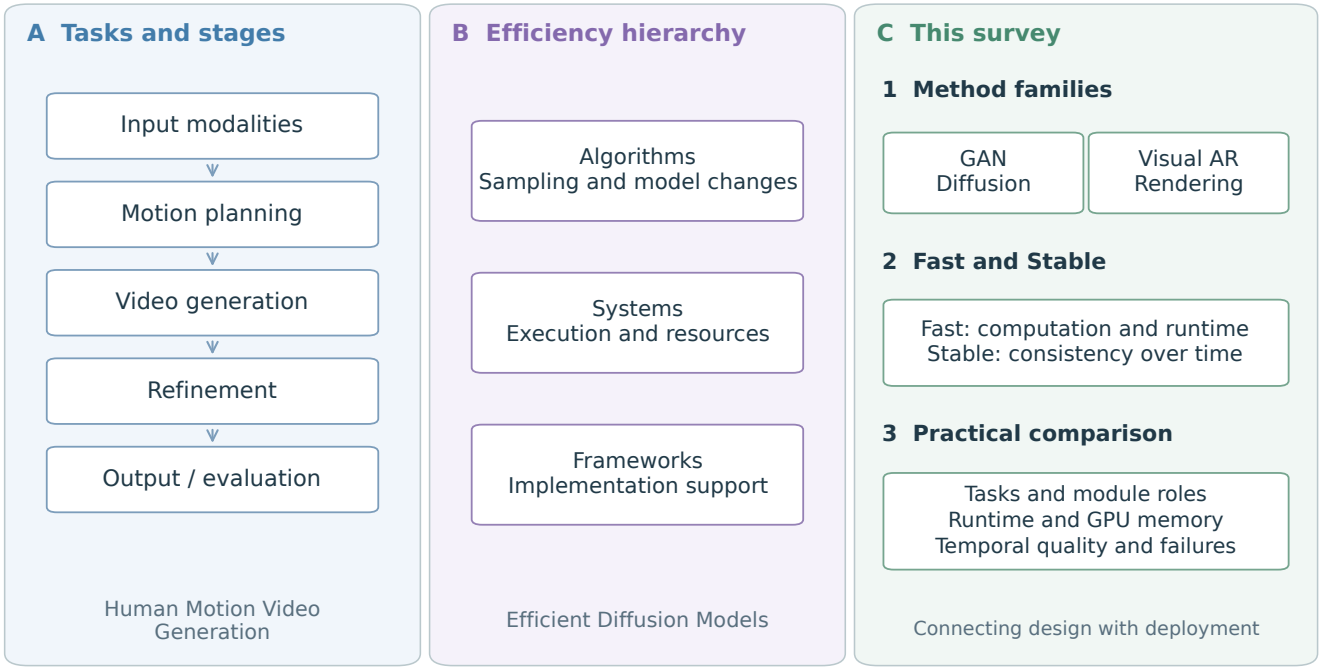


Fig. 3 Complementary views of the literature: human-motion tasks and generation stages [212]; diffusion efficiency across algorithms, systems, and frameworks [164]; and the method families, Fast/Stable perspectives, and practical comparisons developed in this survey. The three views emphasize different aspects of generation. Author-created schematic.

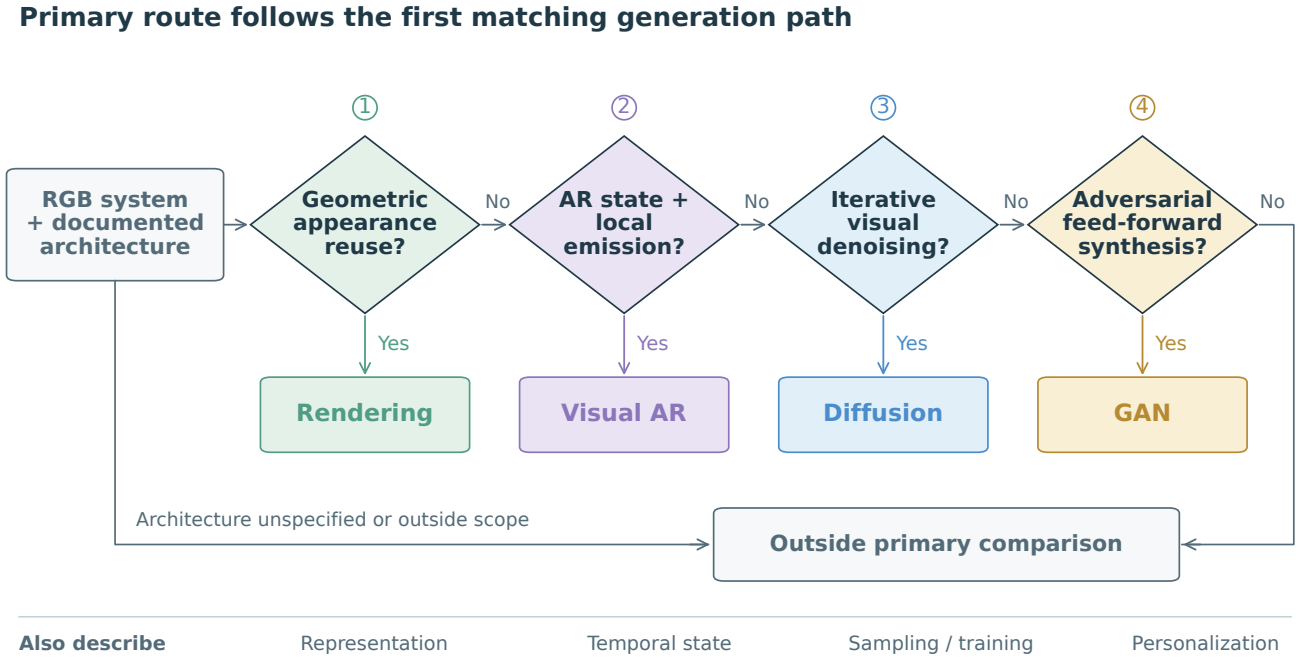


Fig. 4 Assignment to the four visual-generation routes. For methods within the survey scope, the first satisfied rule determines the primary route. Table 2 separately describes auxiliary modules and temporal-state responsibilities; runtime bottlenecks depend on the complete pipeline. The order resolves hybrid cases without ranking quality or efficiency. Methods lacking a supported assignment are excluded from route comparisons. AR denotes autoregressive; VAE denotes variational autoencoder.

– **Render-based methods** (Section 6) reuse a subject-specific mesh, neural field, Gaussian set, or identity-conditioned warp [62, 91, 109, 137, 209]. Reused appearance can reduce repeated synthesis and anchor identity, with enrollment, tracking, deformation, and compositing contributing additional costs and failure modes.

Driving modality, anatomical coverage, personalization, representation, and deployment mode are comparison axes rather than additional routes. Secondary diffusion, autoregressive, GAN, or rendering modules are recorded as boundary components and are not double-counted in route totals.

1.5 Contributions and Main Findings

The survey connects a taxonomy of visual-generation methods with their computational costs and temporal behavior. Its main contributions are a consistent treatment of hybrid architectures, an analysis of acceleration and stability mechanisms across four routes, and a review of datasets and evaluation methods. Section 7 develops four practical observations from representative studies:

1. **Decoder choice can change the speed–quality trade-off.** PersonaLive's within-source ablation quantifies this effect; it does not isolate the runtime bottleneck.
2. **Throughput gains need not shorten startup.** Live Avatar's serving configurations illustrate different effects of call reduction, extra GPUs, decoder parallelism, and bundled kernel optimizations.
3. **Memory management and demonstrated stability are separate results.** A fixed cache gives a resource bound; history and anchor ablations test output failures over specified horizons.
4. **Appearance reuse leaves other failure paths active.** The architectural identity anchor does not remove motion, tracking, deformation, or composition failures; DyStream and GaussianHeadTalk supply partial empirical examples.

These observations concern the configurations and tasks studied in the cited papers. We also plan to measure all open-source methods covered by the survey on a common machine, recording frames per run, resolution, preprocessing, generation and postprocessing time, and peak GPU memory (Section 8.7). The resulting comparison will be added after the measurements are available.

1.6 Organization

Section 2 defines the task, route ownership, and the cost and measurement language. Sections 3–6 review the four routes under common Fast and Stable lenses, each ending with route-specific design guidance. Section 7 analyzes cross-route trade-offs. Section 8 examines datasets, metrics, and runtime and temporal evaluation. Section 9 develops cross-route design guidance and future directions.

2 Preliminaries and Common Evaluation Language

This section introduces a common formulation for the systems reviewed below. We first distinguish the generated representation from its decoder or renderer, then define route assignment and the measurements used to compare efficiency and stability.

2.1 Problem Formulation and System Boundary

Let $V = (I_1, \dots, I_T)$ denote the generated human video. A system may use enrollment data e (e.g., subject video, calibrated views, or a pre-built avatar), runtime conditions $c_{1:T}$ (e.g., audio, text, pose, expression, or a driving video), and persistent deployment state h_t. We write

$$(I_{t:t+k-1}, h_{t+k}) = \mathcal{G}(e, c_{\mathcal{V}_t}, h_t; \theta), \tag{1}$$

where k is the emitted frame or chunk length and $\mathcal{V}_t$ is the portion of the conditioning stream consumed at emission time. This formulation covers one-shot, subject-specific, and generalizable systems without assuming that a reference image is supplied at every runtime step.

Three output requirements recur across tasks. *Identity fidelity* asks whether the intended subject remains recognizable; *driving fidelity* asks whether speech, expression, pose, gaze, and gesture follow their controls; and *temporal fidelity* asks whether motion and appearance remain coherent across emitted units. Their measurement depends on anatomy, modality, personalization, and the availability of a paired reference, so no single metric is valid for every method.

For analysis, we expose an intermediate representation r and a decoding or rendering operation D:

$$\begin{aligned}(r_{t:t+k-1}, h_{t+k}) &= G(e, c_{\mathcal{V}_t}, h_t),\\ I_{t:t+k-1} &= D(r_{t:t+k-1}).\end{aligned} \tag{2}$$

Equation (2) is a bookkeeping view, not a mandatory two-network architecture. In some systems, G and D are fused; in others, e is compiled into a subject-specific

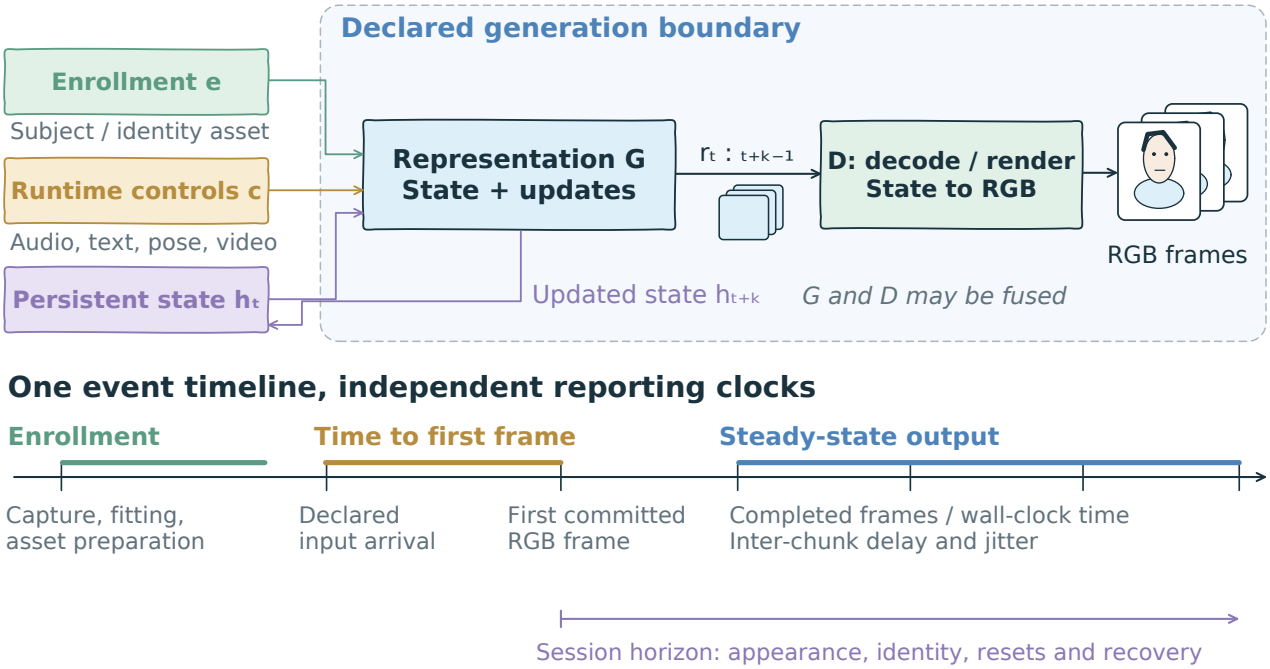


Fig. 5 Analytical generation boundary and independent reporting clocks. The representation owner G updates visual state; decoder or renderer D produces observable RGB frames. Enrollment, time to first frame, steady-state output, and the evaluated session horizon retain their own measurement boundaries. G and D need not be separate networks.

scene or the parameters of D before runtime. Any subject information consumed at decoding is therefore included in r or in those precompiled parameters under this notation. The distinction remains useful because it identifies the representation and state-update owner, the decoder or renderer boundary, and the cost paid during enrollment versus repeated inference.

Figure 5 makes this accounting explicit: G owns the generated representation and persistent-state update, while D maps that representation to observable frames. The clocks are separate reporting boundaries, so a steady-state throughput result cannot substitute for enrollment, first-output, or session-duration measurements.

2.2 Efficiency and Stability

The Fast and Stable perspectives measure different aspects of a system. Efficiency is assessed from the computation and measured timing of a declared pipeline, including conditioning, synthesis, decoding, and system overhead. Stability is assessed from changes in identity, appearance, synchronization, and motion over a declared generation horizon, together with the state and recovery policy used during that run.

For long-horizon results, we distinguish an author's duration claim, a qualitative demonstration, and quantitative measurements over time. These support different conclusions. A bounded cache establishes a bound on stored context, while drift must be measured from the generated output. Similarly, sequential emission and responsiveness to new input describe operating capabilities; neither establishes long-horizon stability. Enrollment duration is recorded separately from generation duration.

2.3 Ordered Route Ownership

For a system within the scope of Section 1.1, we inspect the documented representation, its temporal update, and its path to RGB. We then apply the following rules in order and assign the first matching route (Figure 4). This precedence resolves hybrid architectures; it does not rank quality or efficiency.

1. **Render-based.** A subject-specific explicit or neural scene, including a mesh, neural field, Gaussian set, or identity-conditioned warp, owns appearance and is queried or updated before rendering. A learned model that predicts compact motion, geometry, deformation, or Gaussian attributes remains a controller.
2. **Autoregressive.** Otherwise, a native autoregressive model owns cross-frame dense or discrete visual state, such as visual tokens or visual latents. A downstream

diffusion process or renderer may be a local emission head, but it does not own the evolving visual sequence.

3. **Diffusion-based.** Otherwise, diffusion or flow matching owns dense pixels, dense visual latents, or sequential dense visual chunks. A generic variational autoencoder (VAE) decoder does not displace this ownership.
4. **GAN-based.** Otherwise, an adversarially trained feed-forward generator owns final dense or regional visual synthesis. Adversarial loss attached only as a regularizer is insufficient.

Route assignment follows the architecture described in the primary paper or its accompanying technical description. A method enters the route comparison when its generation path can be identified and one of the four rules applies. The taxonomy is therefore an organization of the reviewed method families, not an exhaustive partition of all video synthesis algorithms.

Supporting modules are identified by their role in the complete pipeline. A diffusion motion controller driving a subject-specific renderer remains part of a render-based system; an autoregressive dialogue model emitting pose controls does not become a visual AR method. Missing performance fields are marked NR independently of this architectural decision. Historical and contextual examples are identified explicitly and do not determine the performance of another route.

An operational test for hybrid systems. The primary route is an editorial index, not a claim that adversarial learning, diffusion, autoregression, and rendering are disjoint properties. We separately record appearance representation, temporal factorization, training/sampling mechanism, and personalization. The visual owner can differ from both the latency bottleneck and the component that causes a temporal failure.

For the AR–diffusion boundary, inspect where the noisy current sample enters the inference graph. A visual AR backbone computes a next-unit state $h_t = F(c_{\le t}, z_{<t})$ independently of the local sampling noise; a head then samples z_t conditioned on that fixed state. The sampled unit is committed to the next backbone step. A chunkwise diffusion backbone instead repeatedly evaluates $v_\theta(z_t^{(s)}, s, z_{<t}, c)$ while denoising the current chunk, with the current noisy sample and noise level entering the temporal visual backbone. Both constructions factor generation causally. We index the former as visual AR and the latter as diffusion, retaining local iterative sampling in the cost breakdown in either case.

For the rendering boundary, inspect whether an explicit identity-specific spatial representation is queried or transported under geometric control before neural image decoding. Learned decoding and adversarial losses do not negate that path. Applying this rule consistently places StyleAvatar, with its rendered facial UV features and compositional decoder, in the render route alongside LivePortrait's source-volume warping. StyleAvatar remains a useful contextual example of adversarial synthesis in Section 3.4; it is counted only once in route comparisons. Static features without a documented geometric query or transport path would not alone satisfy this test.

Figure 6 contrasts the three inference paths through six representative systems. The decisive distinction is where noisy samples or geometric controls enter the path to visible appearance.

2.4 Module Roles in Hybrid Systems

Hybrid systems distribute appearance, temporal state, motion control, decoding, and composition across modules. Table 2 shows this division for ten representative systems. The primary route describes how appearance is generated, while the other modules explain how the system is controlled and served. Their costs depend on the execution schedule: a component that delays the first output may differ from the stage that limits steady-state throughput.

The route chapters discuss the corresponding computational and temporal trade-offs. For runtime comparison, preprocessing, generation, and postprocessing will be measured separately under the common setup described in Section 8.7.

2.5 Route-Specific Cost Structures

The same visible frame can arise from different cost structures, so each route chapter opens with a route-specific cost decomposition. The following summaries fix the shared vocabulary.

Throughout these decompositions, C denotes an additive work or resource quantity in one declared unit, such as FLOPs, GPU-seconds, or measured serial-stage milliseconds. Different units cannot be added. If stages overlap, their sum is not elapsed latency. For ideal pipeline stages with times $t_1, \ldots, t_m$, first-output processing time is approximately $\sum_i t_i$ plus input buffering, whereas steady-state chunk cadence is bounded below by $\max_i t_i$; communication and contention can increase both. Enrollment is charged separately or amortized over a declared number of frames.

GAN-based synthesis. After training, a GAN generator can synthesize dense appearance without an iterative sampling loop [58, 89, 90]. One feed-forward evaluation

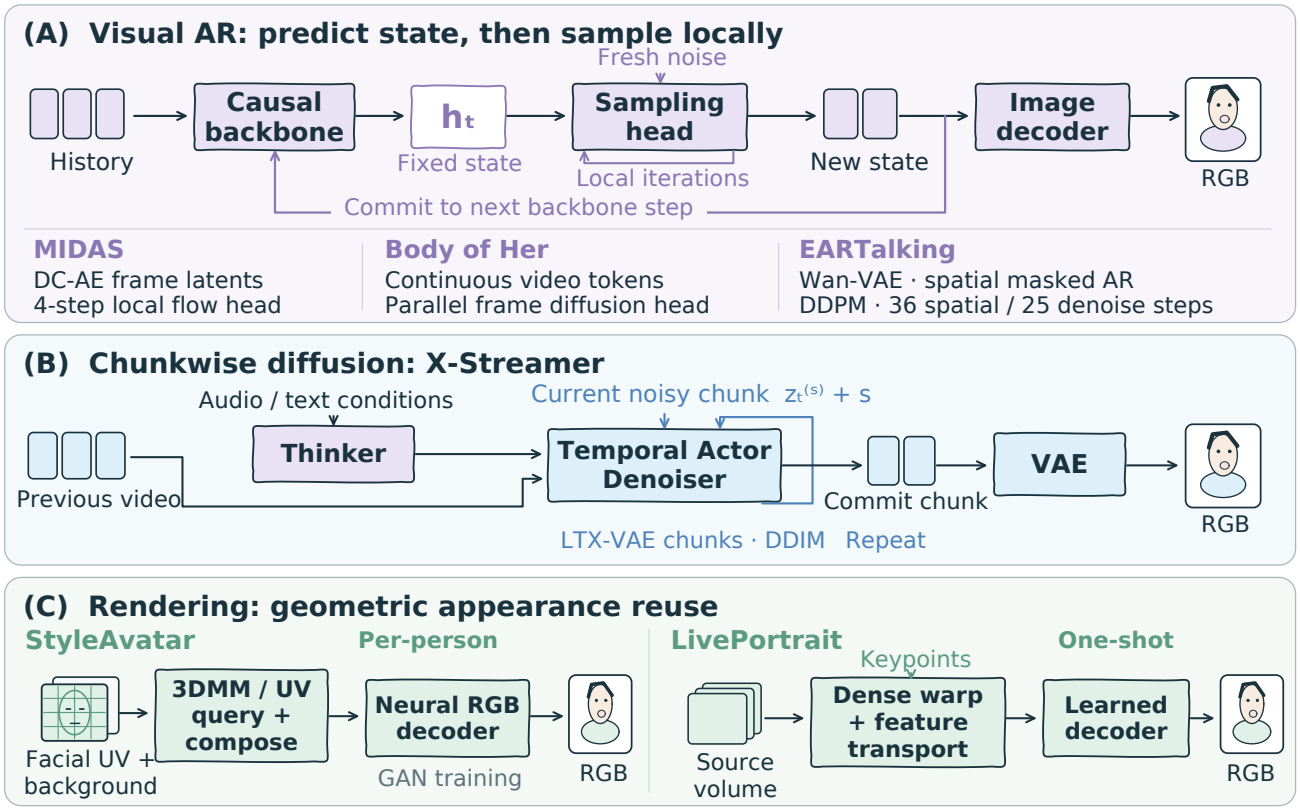


Fig. 6 Inference paths that resolve hybrid assignments. (A) MIDAS, Body of Her, and EARTalking separate causal visual-state prediction from local sampling [5, 27, 203]. (B) X-Streamer's temporal visual backbone repeatedly consumes the current noisy sample [207]. (C) StyleAvatar and LivePortrait reuse appearance through geometric queries or feature transport before learned image decoding [62, 193]. Representation, personalization, and auxiliary modules remain separate annotations; primary route does not identify the runtime bottleneck.

describes call count, not absolute latency: preprocessing, network size, output resolution, compositing, and hardware still determine runtime. Route-specific speed claims therefore require a measured boundary rather than the presence of a discriminator or a single-pass architecture.

Diffusion and flow matching. Diffusion [69] and flow-matching [53, 117] models traverse a noise-to-data path through repeated function evaluations. Their cost depends on the generated unit (frame, clip, or chunk), strict number of function evaluations (NFE), cost per evaluation, conditioning passes, latent codec [96, 149, 158], and system scheduling. Step reduction [127, 160, 174], caching, and parallelism reduce different terms and should not be collapsed into a single speed label.

Autoregressive visual state. Autoregressive models factor a sequence and predict future units from prior state [14, 155, 208], over discrete tokens [139] or continuous latents [24]. Runtime depends on token or block granularity, serial depth, attention, cache updates, local decoding, and output rate. Key–value (KV) caching can avoid recomputing past projections while still growing memory; bounded windows, recurrent summaries, compression, eviction, or resets are separate Stable mechanisms. Serial depth itself can be attacked with parallel or speculative emission [102].

Render-based synthesis. Render-based systems amortize a subject-specific appearance representation and repeatedly update or query it using compact controls. Their costs include enrollment or scene construction, upstream tracking, controller evaluation, deformation or field queries, rasterization or volume integration, neural decoding, compositing, and input/output (I/O). A renderer call is not universally cheap or one-step: neural radiance field (NeRF) ray sampling and Gaussian sorting expose different cost terms [91, 137, 138]. Under our accounting rule, a fixed scene can anchor identity without proving long-session stability of the complete pipeline.

2.6 Conditions and Intermediate Representations

Runtime conditions specify what the person should do. Common signals include speech audio, text, pose, ex-

Table 2 Module roles in representative systems. Inputs list driving conditions in addition to the source identity or reference. Portrait output includes a face crop where specified.

Method; task / input	Visual owner; route / supporting component	Temporal state and motion/control owner	Decoder / renderer; compositor
Wav2Lip+ GAN [150]; face crop / audio + source video	RGB face generator; GAN / training-only sync expert	No recursive visual state; masked source pose and audio condition each frame	Feed-forward face decoder; crop insertion is a separate output stage
MuseTalk [239]; lower face / audio + source	Regional latent predictor; GAN / generic VAE	Framewise visual prediction; audio conditioning and reused blend mask	VAE image decoder; mask-based blending
PersonaLive [113]; portrait / driving video	Dense visual diffusion; diffusion / VAE	Selected visual history and reference; motion conditions drive chunk synthesis	VAE variants
Live Avatar [81]; portrait / audio	Dense chunk denoiser; diffusion / causal VAE	Reference sink and rolling KV; audio-conditioned denoising	Causal VAE on a separate GPU in full system
MIDAS [27]; human video / multimodal controls	Continuous frame tokens; visual AR / local diffusion	AR backbone context; separate codec history; condition tokens	Four-step local head, then DC-AE
EARTalking [203]; portrait / audio + frame controls	Frame visual state; visual AR / local denoising	Causal context, reference sink and cyclic positions; frame-level control	Local sampling and 3D-VAE
StyleAvatar [193]; portrait / tracked motion	Enrolled facial UV features; rendering / adversarial decode	Fixed subject appearance; 3DMM tracking drives per-frame geometry	OpenGL UV rendering and StyleUNets; learned regional masks
LivePortrait [62]; portrait / driving video	Source feature volume; rendering / learned decode	Source identity and keypoint-driven deformation; motion extractor	Feature warping and neural decoder; paste-back stage
DyStream [22]; dyadic portrait / audiovisual input	Reused appearance volume; rendering / motion generation	Motion controller with anchor; RGB appearance persists outside controller	Portrait feature warping and rendering
OmniResponse [126]; dyadic portrait / audiovisual conversation	Pretrained portrait renderer; rendering / AR response control	Multimodal AR model predicts facial coefficients and text; TempoVoice predicts speech	Coefficient decoder followed by portrait renderer

pression, gaze, and a driving video; pose and expression may themselves be estimated by upstream trackers whose cost belongs to the deployed pipeline [19, 221]. Enrollment data specifies who or what is rendered and may be a single reference, a short monocular recording, calibrated multiview capture, or a pre-built asset.

Intermediate representations determine the interface between control and appearance. Landmarks and keypoints are interpretable, but they omit appearance. Facial action units [52], three-dimensional morphable models (3DMMs), Faces Learned with an Articulated Model and Expressions (FLAME), the Skinned Multi-Person Linear Model (SMPL) and its expressive extension SMPL-X, and blendshape parameters expose compact geometric controls [109, 123, 145]. Flow fields and identity-conditioned warps transport source features. Learned motion latents trade semantics for capacity, whereas visual tokens or latents carry dense appearance state. Neural fields and Gaussian primitives encode subject-specific scenes [91, 137]. Dimensionality alone does not determine speed. Compact controls may feed an expensive renderer, while a dense latent may use a fast decoder.

For comparison, we record anatomical coverage (face, upper body, full body, hands), personalization regime, representation owner, decoder/renderer, and evaluation boundary. These axes remain orthogonal to the four primary routes.

2.7 Common Acceleration and Measurement Terms

Generic acceleration acts on distinct Fast terms [164]. The following categories are the shared skeleton of every route's Fast analysis; each category still requires a measured system boundary:

- **Fewer evaluations:** distillation, consistency objectives, solver reduction, block prediction, or speculative emission reduce strict NFE or serial calls [102, 127, 174, 224].
- **Cheaper evaluations:** representation compression, sparse computation, lightweight backbones, quantization, and optimized kernels reduce the cost

of each call. For attention, I/O-aware FlashAttention reorganizes exact-attention computation in 16-bit floating point (FP16) or bfloat16 (BF16) to reduce high-bandwidth-memory traffic [44, 45], whereas FlashAttention-3 also provides a Hopper 8-bit floating-point (FP8) path whose speed lever is lower-precision Tensor Core execution and whose block quantization and incoherent processing control quantization error [162]. These standalone kernel results do not establish an end-to-end virtual-human speedup.

- **Reuse:** feature, condition, reference, or KV caches reuse a precisely named object across denoising steps, frames, chunks, or autoregressive history [118, 129, 132].
- **Decoder/renderer reduction:** efficient codecs, bounded ray samples, primitive pruning, rasterizer design, and regional synthesis reduce visible-output cost.
- **System optimization:** compilation, mixed precision, memory layout, I/O, pipelining, and parallel execution change the measured pipeline boundary.

The following terms are not interchangeable. *Throughput* is completed output per wall-clock time after the disclosed operating condition is reached. *Frames per second (FPS)* is frame throughput at a stated resolution and boundary; it is not content frame rate or latency. *Real-time factor (RTF)* is wall-clock time divided by generated media duration under a stated convention, so values below one are faster than playback. *First-output latency* is the umbrella term for the interval from a declared input-arrival event to the first committed output of any type. *Time to first frame (TTFF)* is reserved for the case in which that first committed output is an observable frame. *Inter-chunk latency* measures successive committed outputs and includes the declared buffering policy. *Number of function evaluations (NFE)* counts evaluations of the iterative vector field or denoiser for the stated generated unit.

Real-time means meeting a stated playback or application deadline under the declared load; average FPS alone can conceal startup delay or stalls. *Streaming* means committing outputs incrementally as a session proceeds. *Causal* additionally restricts access to conditions and generated history: report the input lookahead, buffering interval, and causal unit (token, frame, or chunk). A chunk-causal visual backbone may still consume future audio within that chunk. *Long-horizon* denotes a declared duration beyond the method's training window or ordinary evaluation clip; report both durations rather than assign a universal threshold. A long streaming run can therefore be noncausal in its input, miss real-time deadlines, or exhibit unstable output.

Every reported number should identify hardware, device count, resolution, precision, batch or concurrency, warm-up, measurement window, and included components. When these fields or the evaluated duration are absent from the primary source, we report them as not reported (NR) rather than infer deployment capability from architecture or title.

Matched timing quantities can be converted. If N completed frames take w seconds and the media rate is f frames/s, then $\mathrm{FPS} = N/w$ and $\mathrm{RTF} = wf/N = f/\mathrm{FPS}$, provided both use the same measured interval and pipeline boundary. Neither conversion yields TTFF. For example, a 12-frame chunk generated in 0.30 s has throughput 40 FPS and RTF 0.625 at 25 FPS playback, yet collecting 0.48 s of audio before processing already imposes a different startup delay. This is an illustrative calculation, not a measured system result.

2.8 Failure Modes and Stability Mechanisms

Stability means preserving the intended attributes while allowing the requested changes. We distinguish five observable failures: *appearance drift*, including identity, color, and mouth-detail degradation; *temporal discontinuity*, including frame jitter and chunk-boundary jumps; *control failure*, including lip desynchronization and suppression of requested motion; *geometric failure*, including tracking wobble, deformation, disocclusion, and compositing errors; and *service failure*, including stalls, memory growth, and resets. Only some of these require recursive visual feedback. A fixed appearance asset can remain recognizable while its motion tracker jitters, and an unchanged but incorrect face can have zero measured drift.

For measurement, we separate seven dimensions (Table 3). These refine the failure vocabulary rather than form an additive stability score. Identity and clothing/background are distinct appearance attributes; motion continuity and audiovisual synchronization can fail independently; state accumulation describes an error pathway that needs a controlled test. The responsible component may be a tracker, controller, compositor, or serving process outside the visual owner.

Five recurring mechanisms help organize the streaming-generation literature. Their names describe operations, not guarantees. They are particularly common in streaming diffusion, with related designs appearing in other representations.

L1: Persistent identity anchor. Retain a source image, appearance asset, or designated reference latent

Table 3 Dimensions of temporal stability and the components that can affect them. Interpret measurements together with generation duration, input conditions, and reset behavior. Low variation alone is insufficient if the output has poor quality or fails to follow intended changes.

Dimension	Observable and measurement	Candidate responsible components	Interpretation limit
Identity preservation	Reference-aligned identity similarity over time; failed-detection fraction	Appearance representation, visual generator, identity conditioning	A consistently wrong identity can have little drift; pose changes affect recognition.
Appearance preservation	Region-specific clothing/background/color or detail consistency; artifact ratings	Generator, decoder, appearance asset, compositor	Mask intended lighting, clothing, or scene changes according to the test condition.
Motion continuity	Pose/landmark trajectories, frame/chunk jumps, target error where paired	Motion controller, state update, tracker	Low movement may indicate frozen or suppressed motion.
Audiovisual synchronization	Fixed-window lip-sync measures over time; shift/error distribution	Audio encoder, motion control, audiovisual scheduling	Synchronization does not establish identity or expression fidelity.
Geometry/rendering stability	Wobble, disocclusion, deformation and seam errors in a declared coordinate frame	Tracking, deformation, scene representation, renderer, compositor	Static geometry or a fixed identity asset does not exclude visible artifacts.
State/error accumulation	Error trajectories under continuous rollout; matched state/anchor/history ablations	Visual or controller memory, eviction, boundary transfer, training history	A trend alone cannot identify its cause; bounded memory is a resource result.
Service reliability	Completion/failure rate, stalls, queue growth, resets, recovery time, memory trajectory	Scheduler, cache allocation, decoding and I/O	High average FPS can coexist with missed deadlines or repeated resets.

across updates. This supplies a stable appearance reference. A fixed anchor can also oppose intended changes; that trade-off requires a change-following test rather than an identity score alone.

L2: Bounded multiscale temporal memory. Preserve recent state and selected or compressed older temporal state under a documented capacity and update policy. We reserve L2 for this combination. A rolling KV window, a codec feature cache, and a learned detail-retrieval bank serve different purposes. PersonaLive selects historical frames but does not specify the memory capacity or eviction rule [113]. SoulX-LiveAct specifies fixed reference, long-term, and short-term slots; InteractiveAvatar additionally specifies promotion and replacement in its bounded long-term bank [173, 252].

L3: Position re-indexing. Reassign temporal positions as a cache advances, for example by storing keys before rotary position embedding (RoPE) or using cyclic indices. This controls positional extrapolation; it does not restore information removed by eviction. EARTalking's cyclic positions and Live Avatar's Rolling RoPE instantiate different policies [81, 203].

L4: History-robust training. We distinguish *L4-G*, training on model-generated histories, from *L4-C*, corrupting ground-truth histories with noise or dropped patches. Self-Forcing exemplifies L4-G; MIDAS, EARTalking, and Hallo2 provide L4-C examples [27, 41, 80, 203]. The former reduces a training–inference mismatch over the training rollout distribution. Neither subtype establishes distribution matching for arbitrarily long or differently conditioned deployment.

L5: Explicit boundary-state transfer. Pass a designated motion latent, overlap state, or boundary representation from one chunk to the next. Motion Condition Injection in JoyStreamer-Flash (the third arXiv version of JoyAvatar) and Avatar V's motion propagation illustrate the operation [105, 114]. Ordinary attention to previous frames is insufficient for an L5 annotation. Transfer can improve continuity or propagate an error; its direction and magnitude are empirical questions.

The effect of a stability mechanism depends on the duration and conditions of its evaluation. A component ablation can show how a reference or history policy changes quality over the tested clips, while measurements at several time points reveal how quality evolves during longer generation. EARTalking evaluates its anchor and history corruption through ablations and demonstrates output over three training-window lengths; Live Avatar measures sampled windows during extended generation with repeated audio [81, 203]. These tests address complementary aspects of temporal behavior.

Mouth-detail degradation, pose drift, tracking wobble, and service interruptions also warrant separate evaluation. A long demonstration may contain such failures between displayed samples, while an aggregate score can conceal a late decline. Section 8.8 discusses how to measure these changes over continuous sessions.

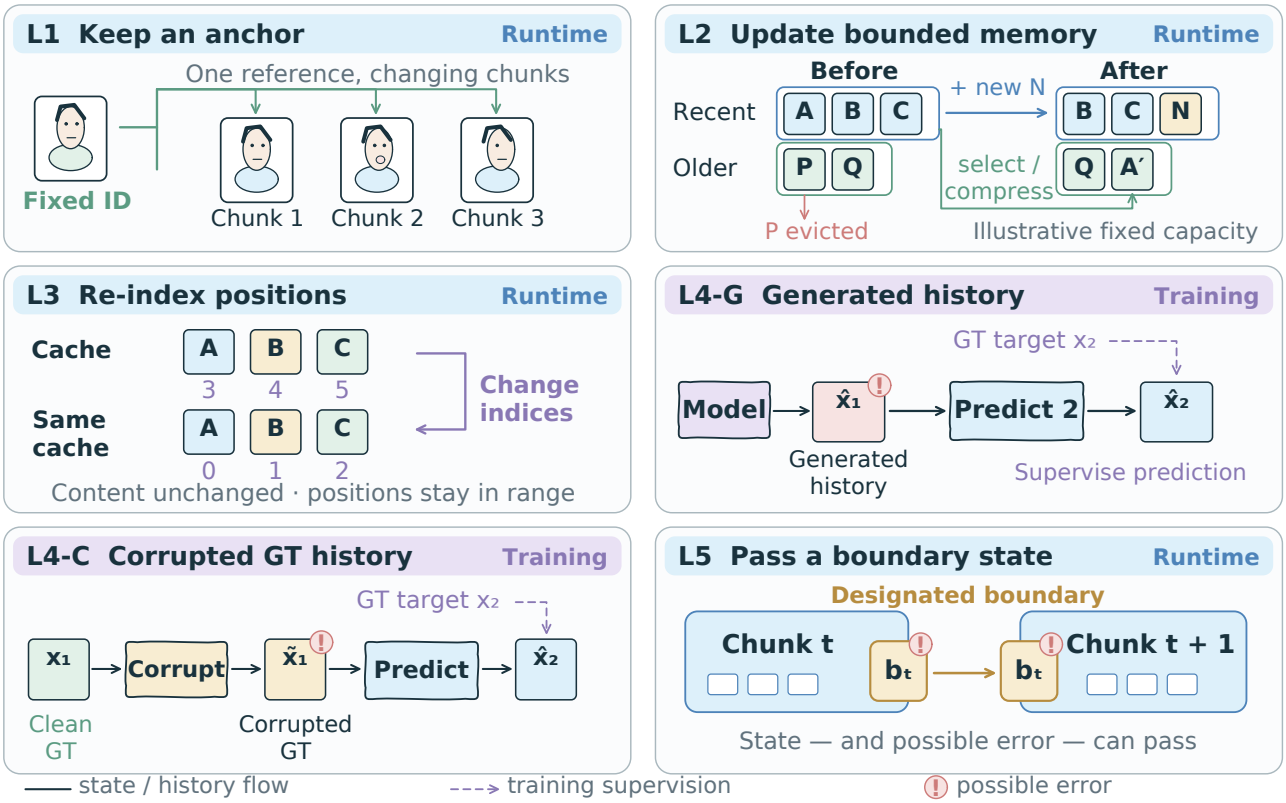


Fig. 7 State operations behind the stability vocabulary. Runtime memory updates and explicit boundary transfer are separate from the two training constructions. A memory budget bounds storage, whereas output measurements establish whether the retained state is useful.

3 GAN-Based Methods

Adversarially trained feed-forward generators produce a frame or image region in a single generator evaluation [58, 89, 90]. This structure avoids an iterative sampling loop, but runtime still depends on the network, conditioning, and output-processing stages. We review three ways of reducing this work: regional synthesis, compact control, and reusable appearance. We then examine temporal consistency, which requires particular attention when the visual generator operates independently on each frame.

3.1 Scope, Ownership, and Cost Boundary

A method is assigned to this route when an adversarially trained feed-forward generator owns final dense or regional visual synthesis after the render-, autoregressive-, and diffusion-ownership tests in Section 2.3. A discriminator is normally used only during training, so removing it from the deployed graph is not a route test. Conversely, an adversarial loss may regularize a warper, neural field, or diffusion decoder without making that system GAN-primary. The relevant questions are what representation is generated, which module owns appearance, and how RGB is reconstructed.

The training objective and inference path must both be checked when assigning a method. Wav2Lip reports a base generator without the visual-quality adversarial term and a Wav2Lip+GAN variant with a separate quality discriminator [150]; only the latter is included in the GAN comparison. MuseTalk reuses a pretrained variational autoencoder (VAE) and a U-Net architecture associated with latent diffusion, but performs one-call regional prediction with reconstruction, synchronization, and face/lip adversarial losses [239]. It has no diffusion objective or iterative sampler. Thus, architectural ancestry alone does not determine the route.

For a feed-forward visual path, a useful first-order deployment decomposition is

$$C_{\text{frame}} = C_{\text{condition}} + C_{\text{generator}} + C_{\text{decode/compose}} + C_{\text{system}}. \tag{3}$$

The first term covers audio, identity, pose, expression, and reference encoding. The second covers the network that predicts pixels or dense visual features. The third covers decoding, parsing, blending, refinement, and super-resolution. The final term covers data movement,

precision conversion, scheduling, and I/O. The methods below each attack a different term; none establishes time to first frame (TTFF) or steady-state throughput unless the source defines the timer and system boundary.

3.2 Regional Synthesis: Shrinking $C_{\text{generator}}$

The most direct saving is to restrict synthesis to the region that actually changes. Wav2Lip conditions a feed-forward face decoder on identity, a lower-half-masked pose prior, and speech, and generates a dense RGB face crop frame by frame; a frozen five-frame synchronization expert supplies a training loss rather than recurrent inference state [150]. Under the paper's LRS2-derived protocol, the base and GAN variants expose a small synchronization–fidelity trade-off measured by lip-sync error distance (LSE-D), lip-sync error confidence (LSE-C), and Fréchet Inception Distance (FID): 6.386/7.789/4.887 versus 6.469/7.781/4.446. The source reports no runtime measurements, but the design pattern of synthesizing the mouth and keeping the canvas became the template for a large lip-editing family, including VideoReTalking's editing pipeline [28], DINet's deformation-inpainting for high-resolution dubbing [242], landmark- and appearance-prior variants [255], memory-augmented mouth retrieval [144,180], and style-space lip synchronization [61, 94], with ReSyncer unifying style-based synchronization across tasks [60].

MuseTalk moves regional synthesis into a pretrained VAE latent space, which shrinks the generated tensor a second time: its multimodal U-Net predicts a lower-face latent in one call, and a precomputed mask is reused for compositing [239]. The authors report 30 FPS for 256×256 output on one V100 with preloaded data (component inclusion NR). MuseTalk's supplementary analysis identifies occasional jitter from single-frame generation, a temporal weakness of stateless regional synthesis discussed in Section 3.6.

Training-only critics can improve a regional generator without adding a deployed inference module. TalkLip uses a frozen AV-HuBERT-derived lip-reading expert and reports improved word intelligibility over Wav2Lip on its LRW/LRS2 protocols [190]. Synchronization experts, lip-reading experts, and quality discriminators therefore offer additional supervision without a corresponding critic evaluation at runtime. Their training cost and the generator's own inference cost remain separate considerations.

3.3 Structured Control and Codebooks: Shrinking $C_{\text{condition}}$ and Bounding the Output Space

A second family separates identity from compact motion controls. MakeItTalk splits content and speaker-aware landmark motion [258], while PC-AVS separates pose, content, and identity in latent space [256]. EDTalk factorizes mouth, pose, and expression into orthogonal basis banks; its shared adversarial image generator produces RGB, and a normalizing flow samples compact pose-bank weights [178]. Emotional adaptation provides another use of disentangled control [57].

StyleGAN-based reenactment instead controls a pretrained generator by editing latent directions, masking style channels, or refining inversions [11–13, 21]. Variants address continuous emotion [7], semantic expression [143], cross-domain heads [88], and text-guided editing [17]. These methods make the control interface compact, but do not necessarily make control extraction inexpensive: inversion and feature estimation can add work, and the full image generator remains on the inference path.

Codebooks represent generated appearance through a learned set of reusable features. FlowVQTalker combines autoregressive normalizing-flow controllers for compact 3DMM expression and pose coefficients with an adversarially trained vector-quantized image generator, which decodes a spatial code sequence into a 512×512 face [179]. A grid-based codebook provides a related efficiency-oriented representation for person-specific reenactment [192].

Retrieval reuses detail compatible with the source identity. FixTalk compresses fourth-layer detail features from $512 \times 32 \times 32$ to a 1×512 token and retrieves aligned detail before generation; its Enhanced Motion Indicator suppresses identity leakage from driving features [177]. The reported throughput is 27.601 FPS on an A100 (80 GB) at 512×512, with complete timer composition NR. This supports the reported operating point, but does not independently quantify the speed contribution of retrieval or compression.

3.4 Personalized Compositional Generators: Amortizing Appearance

Appearance reuse can be made stronger through subject-specific training, trading enrollment cost for per-frame savings. Adversarial synthesis can also be combined with geometric appearance reuse; the primary route then follows the first matching rule. Live Speech Portraits is the route's early real-time landmark: a person-specific pipeline from streaming audio through predicted motion to adversarial image synthesis, reported at 30 FPS

with over 27 FPS attributed to generation on a consumer GPU [124]. StyleAvatar is a contextual render–GAN hybrid, counted in Table 11 under the consistent render-first rule. It scales the idea to a compositional generator: two StyleGAN modules for static facial-UV and background features, a StyleUNet for non-facial foreground, Neural Textures with a learned mask for regional composition, and a second StyleUNet for final RGB [193]. With two 16-bit TensorRT StyleUNets, the authors report a source-labeled "live system" rate of 35 FPS (28 ms/frame) on one RTX 3090, with 20 ms for the model-only GPU boundary. The cost structure is instructive: static regional features are computed once (enrollment, reported at roughly two to three hours of person-specific training plus about four more for teeth convergence), so the per-frame path touches only the dynamic StyleUNets, a direct reduction of $C_{\text{generator}}$ purchased with C_{enroll}. RAM-Avatar extends person-specific adversarial synthesis to full-body control with real-time intent [48].

3.5 Fast Design Guidance

Table 4 summarizes representative methods. Four levers recur, each acting on a different term of Equation (3):

1. **Synthesize only what changes** ($C_{\text{generator}} \downarrow$). Face-crop or lower-face prediction reduces the generated area; latent-space prediction, as in MuseTalk, further compresses the representation. The saving depends on crop size and codec compression. Compositing and boundary blending remain necessary, with seams and mouth interiors requiring separate quality evaluation.
2. **Drive with compact controls** ($C_{\text{condition}} \downarrow$). Basis banks, 3DMM coefficients, latent directions, and codebook indices replace dense driving features. Cost: expressiveness is bounded by the control space; motion outside it cannot be generated.
3. **Reuse appearance features** ($C_{\text{generator}} \downarrow$). Person-specific static feature banks, as in StyleAvatar, trade enrollment for repeated savings. Reference-conditioned retrieval, as in FixTalk, can instead support one-shot operation. Their preparation costs and generalization requirements should therefore be reported separately.
4. **Optimize the fixed path** ($C_{\text{system}} \downarrow$). Compilation, reduced precision, and kernel fusion can reduce deployment overhead. StyleAvatar provides an FP16 TensorRT example. The gains depend on the hardware and implementation; its model-only 20 ms remains distinct from system latency.

These mechanisms address different components and can be combined, although their total benefit depends on the remaining pipeline. Face detection, alignment, audio encoding, and I/O may become substantial once generation is inexpensive. The FPS values in Tables 4 and 11 use different resolutions, devices, personalization regimes, and timing boundaries, so they describe individual systems rather than a controlled ranking.

3.6 Stable Design Guidance

Framewise generation avoids feedback of a previously synthesized frame when that frame is not reused as an input. This removes one source of recursive error, but does not enforce temporal consistency. MuseTalk reports single-frame jitter; variation in pose estimates or reference features can also produce inconsistent output. Recurrent controllers and retrieval memories introduce additional state whose errors must be evaluated independently of the image generator.

The drift-control patterns in Section 2.8 help distinguish these cases. A fixed reference or person-specific feature bank serves as an identity anchor (L1). FixTalk's retrieval memory and FlowVQTalker's coefficient history supply persistent information, although they do not by themselves establish bounded multiscale temporal memory (L2). Position re-indexing (L3) is relevant only where a sequential component uses positions. Self-history training (L4) and explicit motion inheritance (L5) are less developed in the systems reviewed here. None of the papers in Table 4 reports a joint minute-scale, time-binned evaluation of identity, synchronization, motion, and visual degradation. The long-session benefit of avoiding visual feedback therefore remains a hypothesis to test.

3.7 Synthesis and Open Questions

Regional methods require evaluation that separates synthesized content from preserved source pixels, including artifacts at their boundary. Compact controls should be assessed for expressive coverage and extraction cost, while personalized generators require enrollment reporting alongside runtime. For stability, a duration-binned comparison could test whether avoiding visual feedback reduces drift and how much framewise jitter remains. The reviewed sources do not provide that joint comparison. Reporting the generated region, resolution, personalization, preparation cost, and timing boundary would make these trade-offs easier to assess.

GAN-based methods reduce repeated synthesis through feed-forward prediction, regional output, compact control, and reusable appearance. Their temporal

Table 4 Representative GAN-based visual generators. Timing values retain the source paper's hardware, resolution, and component boundary and are not directly comparable. NR denotes not reported in the primary paper.

Method	Visual owner / scope	Primary Fast mechanism	Source timing boundary	Stable boundary
Wav2Lip+ GAN [150]	Feed-forward RGB face-crop generator; identity-general	Face-crop rather than full-canvas synthesis; training-only sync and quality critics	FPS, hardware, resolution, TTFF, and timer NR	Stateless framewise generation; evaluated duration NR
MuseTalk [239]	U-Net-generated lower-face VAE latent; no subject adaptation	One-call regional latent prediction; reusable blend mask	30 FPS, 256^2, one V100, preloaded data; included stages NR	Single-frame jitter reported; quantitative long-horizon evaluation NR
TalkLip [190]	Feed-forward 96^2 RGB face-crop generator; identity-general	Training-only lip-reading expert improves intelligibility	FPS, latency, hardware, parameters, and timer NR	Stateless framewise generation; long-horizon evaluation NR
Live Speech Portraits [124]	Person-specific adversarial portrait generator	Compact predicted motion; person-specific appearance	Source-reported 30 FPS system rate on a consumer GPU; timer boundary NR	Recurrent motion prediction; long-horizon evaluation NR
EDTalk [178]	Adversarial RGB generator; mouth, pose, and expression banks; one-shot	Compact reusable bases and progressive freezing	Inference timing NR; about 7 h disentanglement training on two 3090 GPUs	Compact controller state only; long-horizon evaluation NR
FlowVQ Talker [179]	Adversarial VQ image generator; full 512^2 portrait	Compact coefficient control and learned spatial codebook	Inference FPS, latency, hardware, memory, and timer NR	Coefficient history in the controller; long-horizon evaluation NR
FixTalk [177]	Adversarial dense portrait generator; one-shot	Identity-suppressing EMI and compressed memory-retrieved EDI detail	27.601 FPS on A100 80 GB; 512^2 data, full timer boundary NR	Learned detail retrieval (not temporal L2); TTFF and long-horizon evaluation NR

EDI: Enhanced Detail Indicator; EMI: Enhanced Motion Indicator. Timing values are taken from the cited papers.

Table 5 GAN-route design options and their trade-offs. Potential extensions require evaluation in the intended setting.

Target	Design options	Trade-offs to check
Low per-frame latency, general identity	regional synthesis; one-call generator	Seams, mouth-interior quality; preprocessing untimed
High throughput at high resolution	compact controls and retrieval; Potential: transfer compiled FP16 serving	Control space bounds expressiveness; hardware-specific
Repeated use of one subject	person-specific appearance reuse	Hours-scale enrollment; no zero-shot use
Minute-scale stability	reference and temporal control; Potential: long-session deployment	Framewise jitter and controller drift; joint long-horizon evaluation NR in the reviewed systems

behavior depends on both the image generator and its conditioning components. The reviewed sources demonstrate several efficiency mechanisms, but their heterogeneous timing boundaries and limited longitudinal evaluation leave the joint efficiency–stability trade-off unresolved.

4 Diffusion-Based Methods

Diffusion and flow models support generation conditioned on identity, speech, text, pose, and previous video [69, 73, 116, 117, 183]. Their iterative inference motivates methods that reduce sampling calls, compress the generated representation, or reuse computation. Extending generation across chunks introduces a second challenge: the model must maintain consistency while consuming its own outputs. We first review the main acceleration mechanisms, then discuss streaming generation and five recurring approaches to drift control. Architecture alone does not determine their speed relative to the other routes.

4.1 Scope and Cost Model

The reverse chain of a conventional denoising diffusion probabilistic model (DDPM) serially invokes the de-

noising network many times [69]. Flow matching avoids simulating an ordinary differential equation (ODE) while constructing its training targets and can learn straighter probability paths, but generation still requires integrating the learned vector field [53, 117]. With a diffusion Transformer (DiT) in video-latent space, each invocation is itself governed by the number of tokens, model width and depth, and the temporal and spatial extent of attention [146, 220]. A useful first-order decomposition of deployed generation cost is

$$C_{\text{gen}} = N_{\text{FE}} C_{\text{denoise}} + C_{\text{condition}} + C_{\text{decode/render}} + C_{\text{system}}. \quad (4)$$

Here, the first term counts the number of function evaluations (NFE) and depends on sampling steps, classifier-free guidance (CFG) branches, solver stages, and any extra network calls used to refresh a cache. The second covers speech, text, reference-image, and motion encoders; the third includes VAE decoding, super-resolution, and any separate neural renderer; and the last includes kernel scheduling, device communication, data transfer, playback buffering, and I/O. Equation (4) explains why few-step sampling does not by itself establish end-to-end speed. Once N_{FE} is small, the bottleneck can move to a single large DiT call, the codec, condition encoding, or communication. Component-level gains from caching, quantization, and parallelism therefore cannot be multiplied mechanically into a system-level speedup.

For deployed systems, two service clocks matter beyond raw throughput. Time to first frame (TTFF) controls when playback can begin; it includes input acquisition, condition encoding, first-chunk denoising, first decode, and buffering, so audio buffering imposes a TTFF floor that no denoiser acceleration can remove. Steady-state inter-chunk latency and jitter control whether playback continues without stalls. A nominal playback rate and a real-time factor are neither of these. We project *Fast* onto NFE, per-call computation, decoding, and system overhead, with TTFF and steady-state throughput as its service observables; we project *Stable* onto bounded state and identity or appearance drift under self-generated history. Reported speeds retain the source timing boundary.

Dense visual diffusion generates pixels or video latents through a noise-conditioned visual backbone; a codec then reconstructs RGB. FADA, RAP, StreamAvatar, SoulX-FlashTalk, and Avatar V illustrate this path [51, 114, 165, 176, 254]. Compact motion generators followed by identity-conditioned warping or scene rendering belong at the render boundary, including IF-MDM, INFP, AudioRTA, Avatar Forcing, EmbodiedHead, FloodDiffusion, and OmniTalker [16, 93, 100, 202, 217, 240, 262]. Figure 6 resolves the harder distinction between an iterative temporal backbone and a fixed causal visual state with a local iterative head.

The terminology around "flow" needs one clarification. Normalizing flows use invertible transformations with tractable Jacobians; their connection to diffusion lies at the level of probability-density evolution [142]. In current virtual-human systems, flow matching or rectified flow means regressing a velocity field from noise toward data and sampling it with an ODE solver or a distilled approximation; "simulation-free" characterizes the training objective, not one-call inference [117]. Sampling steps convert to strict NFE only when solver stages, CFG branches, and auxiliary cache calls are counted.

Across these architectures, a stable task prior organizes many of the mechanisms below: *appearance changes slowly while the driving signal changes quickly.* Identity, clothing, background, and reference style are low-frequency state that can be compressed, cached, or kept as persistent anchors. Phonemes, mouth shape, transient expressions, and gesture must track the current input. READ aligns a temporally compressed visual latent with a SpeechAE [188]; RAP introduces hybrid audio attention to compensate for strongly compressed video tokens [51]; SyncCache reuses heavy DiT residuals within one reverse trajectory while recomputing light audio blocks at every step [128]. The same asymmetry motivates long-video systems to preserve a reference sink and recent motion while compressing or evicting older history.

The main acceleration mechanisms can now be located in Equation (4). Few-step distillation, consistency training, and distribution matching reduce N_{FE}. Compressed latents, smaller backbones, sparse attention, and low-precision execution reduce C_{denoise}. Pre-encoding and cross-step, cross-block, or cross-modal caches remove repeated conditioning and attention work. Lightweight or causal codecs reduce $C_{\text{decode/render}}$. Fixed windows, fused kernels, and asynchronous pipelines reorganize C_{system}. Deployed systems usually combine several levers, and meaningful evaluation must identify the active levers and track where the bottleneck moves, rather than assigning the joint system result to a single module [81, 128, 254].

Figure 8 separates component optimization from runtime state updates. The mechanism vocabulary in Section 2.8 makes it possible to ask which failure each state operation addresses.

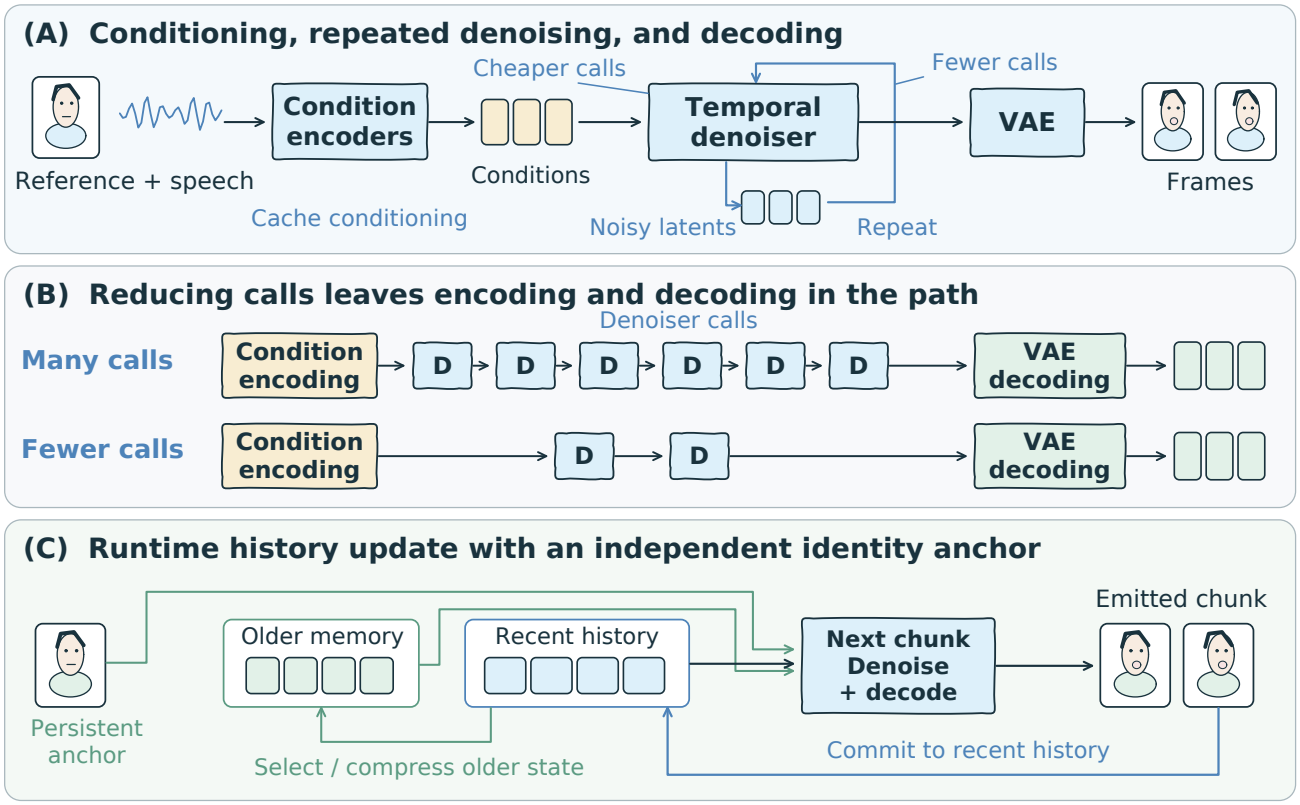


Fig. 8 Diffusion generation as a runtime path and a state-update process. (A) Acceleration acts on distinct components. (B) After denoiser-call reduction, other stages still contribute to runtime; Section 7 gives a measured decoder trade-off. (C) The committed chunk updates recent history, while older state is selected or compressed under the declared policy. Block sizes are schematic and do not represent measured stage durations. Training constructions are shown separately in Figure 7.

4.2 Fast I: Compact Representations and Efficient Backbones

Representation-level acceleration follows two fundamentally different paths. One diffuses in a low-dimensional motion, expression, or parametric space and delegates appearance to a separate decoder or renderer. Its per-step state is small and easy to retain, but its expressiveness is bounded by the intermediate representation and often by an identity-specific renderer. The other continues to generate dense visual content, but applies a highly compressed spatiotemporal VAE so that the DiT processes far fewer video tokens. It preserves a more open appearance model, while paying for audio–visual alignment, recovery of local high frequencies, and VAE decoding. FPS measured for these two paths does not denote the same computation.

Compact state can move the expensive stochastic process out of pixel space. IF-MDM diffuses in a 20-dimensional implicit facial-motion space before an RGB generator synthesizes the portrait. INFP uses a conditional DiT with only four blocks to generate speaking and listening motion latents (no inference timing reported). AudioRTA distills a 40-step DDIM teacher into one-step expression-latent prediction and renders through the mesh/3D-Gaussian decoder of URAvatar. Avatar Forcing applies block-causal diffusion forcing to a 512-dimensional motion latent per frame and predicts reactive dyadic behavior from both parties' signals. EmbodiedHead conditions on the per-frame listening/speaking state, predicts a 115-dimensional FLAME motion vector with a rectified-flow DiT, and passes it to GAGAvatar [93, 100, 217, 240, 262]. FloodDiffusion studies streaming, time-varying text conditions in skeleton-motion space [16]. These methods show why compact state is efficient: the diffusion process runs over compact motion vectors rather than dense visual latents. Their motion-module or renderer-inclusive speeds should nevertheless not be compared directly with open-domain dense-video pipelines, and their appearance ceiling is set by the downstream renderer (Section 6).

A compact controller does not necessarily imply a render-based visual owner. Motar predicts 512-dimensional motion with a causal Transformer and a local diffusion head, but its downstream decoder performs four-step block-causal diffusion over dense visual

content [3]. The complete RGB pipeline therefore remains in the diffusion route. This contrasts with compact controllers that transport an identity-specific appearance representation before decoding; the word "renderer" alone does not resolve the distinction.

The dense path reduces cost by compressing space and time. Stable Video Diffusion shows that latent-video quality depends not only on the denoiser, but also on image pretraining, video-data curation, and temporal-layer training [10]. HunyuanVideo, Wan, and CogVideoX organize large video generation around a spatiotemporal VAE, DiT, and text encoder [98, 186, 220]; they expose the shared computational substrate: VAE compression determines token volume, full spatiotemporal attention determines much of the per-call cost, and chunked decoding determines whether frames can be emitted before all latents are available. LTX-Video co-designs a high-compression VAE with a token-matched DiT, illustrating that reducing token count can change the computation more substantially than shrinking the network [65]. For virtual humans, however, aggressive compression is especially risky around teeth, lip boundaries, eyelids, and rapid expressions.

If temporal and spatial compression factors are r_t and r_s, the video-token count scales approximately as $THW/(r_t r_s^2)$ [65, 146]. Global self-attention is quadratic in this count, and audio cross-attention depends on the number of audio tokens. Greater compression reduces attention, MLP computation, KV storage, and communication. It also increases the phonetic changes represented by each temporal token and allocates fewer spatial tokens to small facial structures. Compression must therefore be co-designed with audio downsampling, detail recovery, and decoder throughput. READ and RAP illustrate two solutions: READ uses a Temporal VAE to extend each visual token over a spatiotemporal block and trains a SpeechAE at the corresponding time scale (eight solver steps; the reported 4.421 s is backbone time for one 4.84 s, 121-frame sample) [188]; RAP combines an LTX-VAE latent, hybrid audio attention, and latent inheritance, reaching 42.41 and 39.87 FPS on one A800 at 512×512 under the authors' protocol [51]. Both show that once the visual timeline is compressed, frame-level speech features cannot simply be injected without accounting for how several phonetic events map to one visual token.

Reference-guided decoding provides another way to recover detail after compression. Xu et al. inject reference features into a deep-compression VAE decoder while a rectified-flow Transformer generates the video latents [211]. The decoder's appearance guidance complements dense visual synthesis rather than replacing it with geometric rendering. Their component timing assumes precomputed reference features, illustrating why reference preparation and per-chunk decoding must be separated when interpreting a reported generation rate.

Dense generation can also reduce the cost of a long reference. Avatar V lets generated tokens attend to a long reference video through asymmetric sparse reference attention; reference K/V projections and their rotary position embedding (RoPE) results are computed in the first denoising step and reused for the remaining 23, followed by a one-step adversarial super-resolution stage. The paper reports an approximately threefold latency reduction relative to its unoptimized implementation, with no absolute latency or FPS [114]. The case shows that long-reference tokens can become a second attention bottleneck and that cross-step conditioning caches matter even after denoising is shortened.

After DiT distillation, VAE cost remains on the inference path (Figure 8B). PersonaLive offers TinyVAE for higher decoder throughput, but records degradation in identity, motion, and temporal metrics; MotionStream's highest reported 29.5 FPS configuration on one H100 also uses TinyVAE [113, 166]. Wan's causal VAE can encode and decode arbitrarily long video in chunks while caching adjacent codec features [186]. Compact representations should be evaluated with reconstruction quality, decoder latency, mouth detail, identity, synchronization, and the actual end-to-end timing boundary, not only compression ratio or denoiser floating-point operations (FLOPs).

Conditioning cost and expression control. Audio conditioning is not a single objective. EmotiveTalk separates audio information relevant to lip motion from information associated with emotion and expression, then uses emotional video diffusion to expand the temporal and expressive signal. The design makes explicit a distinction that matters for both Fast and Stable evaluation: phonetic accuracy, motion naturalness, and identity fidelity are separate targets, and an audio encoder or diffusion sampler may improve one while leaving the others unchanged. Conditioning-side disentanglement therefore belongs in the diffusion cost breakdown, including its extra encoders, attention paths, and temporal expansion, rather than being counted as a free quality module [189]. The same principle motivates emotion-specialized or mixture-of-expert controllers, but routing complexity and expert capacity must be reported separately from the visual generator.

4.3 Fast II: Few-Step and One-Step Distillation

Few-step methods shorten the reverse trajectory, but at least three operations are often conflated. Trajectory

rectification or consistency training makes distant noise states easier to connect; distribution matching distillation (DMD) trains a short-path generator by contrasting real- and generated-distribution scores; and CFG distillation folds conditional and unconditional branches within a step into one student call. These mechanisms alter the path, student distribution, and calls per step, respectively, and should not all be called one-step generation. In a video DiT, a sampling step may invoke several CFG branches, while a streaming method may add a clean-latent call solely to update KV. Strict NFE must count actual inference-time network evaluations.

The generic lineage requires precise objectives. Consistency Models map points on the same probability-flow ODE trajectory to a common output; consistency distillation depends on a pretrained diffusion trajectory, whereas standalone consistency training learns directly from data [174]. Latent consistency models and their LoRA form made few-step sampling broadly practical for latent image diffusion [127]. DMD estimates a Kullback–Leibler gradient direction from the difference between real-data and generated-distribution scores and adds a perceptual regression term anchoring the student to its teacher's conditional sample [224]; DMD2 addresses stale generated-score estimates and unstable optimization through two-timescale updates, adversarial supervision on real data, and backward simulation of inference-relevant states [223]. These generic image results motivate virtual-human students, but do not settle identity, synchronization, or self-history errors.

A reproducible account of model calls should separate the main denoiser, auxiliary score networks used only during training, and the decoder. For each inference step, the main-network count is the product or sum induced by CFG branches, solver substages, and extra cache-refresh calls. Teacher and critic evaluations during training do not belong in deployed NFE, whereas an inference-time clean-latent forward does. If a paper reports steps but not solver or guidance calls, the survey retains "steps" rather than inventing NFE; if it reports an NFE ratio without wall-clock measurements on matched hardware, the ratio is not rewritten as a system speedup.

FADA provides a relatively explicit call count. Its PeRFlow student reduces a 25-step DDIM teacher to six denoising steps. Because the teacher executes three multi-CFG branches per step, the authors count 75 denoising-network evaluations (NFE-D). The Balanced student retains the three branches and uses 18 NFE-D, whereas the Fast student replaces them with a learnable CFG token and uses 6 NFE-D. The corresponding 4.17× and 12.5× reductions refer to denoiser calls; the paper reports neither inference hardware nor wall-clock time [254]. Trajectory shortening and branch folding are thus distinct levers: a six-step model can still repeat a large network three times at every step.

Live Avatar illustrates why cache calls also belong in this accounting. Its undistilled baseline uses 80 NFE. DMD removes CFG and yields four denoising steps, but a conventional Self-Forcing implementation adds one clean-cache refresh and therefore totals five NFE. History Corrupt instead keeps historical KV at the same noise level as the current block, eliminating that clean refresh and reducing the deployed count to exactly four NFE [81]. The mechanism simultaneously avoids one call and produces local, step-matched noisy KV for a pipeline that assigns one timestep to each device.

LLIA and TurboTalk represent direct few-step training and progressive one-step distillation. LLIA trains two- and four-step students with consistency and adversarial objectives, then combines them with TensorRT, 8-bit integer (INT8) execution, and VAE–U-Net overlap; at 384×384 on an RTX 4090D, 78 FPS is measured for a 12-frame steady-state chunk, with a source-defined approximately 140-ms initial/state-switch response using a four-frame startup segment [227]. TurboTalk first obtains a four-step DMD student and progressively distills three-, two-, and one-step students; its claimed 120× ratio compares approximately 120 NFE per InfiniteTalk chunk with the final author-counted 1 NFE per chunk, with no wall-clock result [120]. NFE ratios describe reductions in model calls; elapsed time also depends on the cost of each call and the rest of the pipeline.

The unit attached to "one-step" also matters. Avatar-Forcing by Cui et al. applies one joint denoising update per sliding-window advance; under its default four-block window, each block is updated four times before emission [43]. This is different from a single evaluation on an isolated output block and from the motion-space Avatar Forcing of Ki et al. [93]. LeapTalk instead distills a flow-matching teacher into a reference-anchored Brownian-bridge student using an SNR-aligned time transformation [236]. These approaches change, respectively, the streaming update schedule and the generative path. Their step labels cannot be compared without the generated unit, guidance branches, and window schedule.

Block-autoregressive distillation must additionally close the gap between ground-truth histories used in training and self-generated histories seen at deployment. This is a Stable concern (pattern L4-G in Section 2.8) that surfaces inside the distillation objective. MotionStream's Self Forcing, StreamAvatar's and LiveTalk's on-policy rollouts, SoulX-FlashTalk's multi-block retrospective self-correction, and AsymTalker's asymmetric distillation all expose the student to some of its own

errors [29, 125, 165, 166, 176]. SoulX-FlashTalk trains a four-NFE student over rollouts of up to five blocks and reports 32 FPS with a 0.87 s startup time at 720 × 416 on eight H800 GPUs [165]. AsymTalker gives the teacher a ground-truth continuity reference and the student a self-generated one, then reports FID and ArcFace curves through 600 s [125]. A streaming student must match not only an isolated teacher clip, but also the state distribution induced by its own rollout.

Virtual-human-specific rewards or timestep schedules can protect signals that generic DMD discards. Lip Forcing observes that teacher CFG affects fidelity and lip synchronization differently along the trajectory; it activates CFG only in an intermediate teacher-score window and weights DMD updates with SyncNet confidence. The deployed student performs two denoising NFE without CFG plus one clean-latent cache forward; its 1.3B and 14B variants report 31.58 and 15.11 FPS on one H100 at 512 × 512, excluding speech preprocessing, detection, and paste-back [31]. Hallo-Live uses VideoAlign, AudioBox, and a SyncNet proxy in preference-based DMD and reports joint audio–video throughput of 20.38 FPS with 0.94 s reported latency on two H200 GPUs [104]. Such rewards may preserve synchronization or identity while inheriting evaluator bias. A convincing comparison should report sampling steps, CFG branches, actual NFE, matched-hardware wall-clock time, teacher–student quality gaps, and long-horizon degradation under self-generated history.

Maintaining appearance is insufficient if distillation suppresses intended motion. DynaForcing observes diminished dynamics in its self-forcing setup and combines data-anchored rollouts, reward-weighted updates, and reference perturbation [82]. Routed Forcing further separates semantic regions and noise stages: real-data supervision targets non-mouth person regions at high noise levels, while distribution matching is retained elsewhere and during detail refinement [175]. These results identify a failure mode in the tested training regimes, not an inevitable outcome of all distribution matching. Motion amplitude and diversity should therefore accompany identity and synchronization scores when evaluating a distilled avatar.

4.4 Fast III: Computation Reuse and System Parallelism

When several denoising calls remain, adjacent steps contain exploitable redundancy. LightningCP runs the full U-Net at selected key timesteps, reuses high-level decoder features at the others, and assigns finer attention to the foreground; on its Hallo setup, fixed-clip U-Net latency falls from 23.692 to 7.528 s (3.15×), though the baseline uses one A800 whereas the proposed system uses an unspecified number of A800 GPUs [122]. Follow-Your-Emoji-Faster uses a timestep-aware information cache under the same 30-step DDIM schedule, reducing U-Net denoising for 64 frames at 256 × 256 from 6.28 to 2.36 s (2.66×; hardware NR) [131]. Both results are fixed-clip denoiser acceleration rather than FPS. The same lever generalizes: flow caching skips recomputable segments of the velocity field in autoregressive video generation [132], and speculative decoding, which drafts cheap candidate continuations that the full model then checks, has been transplanted from language models [102] to autoregressive video diffusion [76].

The slow-appearance/fast-speech asymmetry permits spatially and modally selective reuse. SyncCache's spatial-aware probe applies a soft portrait mask to the difference between outputs of the first Transformer block at adjacent denoising steps and accumulates that error; only a threshold crossing triggers a full forward and refresh. Its modality-dynamic cache operates within one reverse trajectory: after a full-computation step, selected heavy DiT inter-block residuals are reused at later steps, while lightweight audio blocks are recomputed at every step. On eight A800 GPUs, the authors report a HunyuanVideo-Avatar path decreasing from 524 to 127 s (4.12×) and a Wan path from 113 to 30.15 s (3.75×) [128].

Two cache axes are orthogonal, and matching each cache to the update frequency of the state it stores is the route's general reuse rule. *Denoising-time reuse* shares features between sampling steps of the same output chunk; static references, background, and slowly varying high-level features are plausible candidates, but feature distributions change from the high-noise to detail-refinement regime. *Sequence-time reuse* carries KV or clean latents from one output chunk to the next; it enables continuation, but can propagate a mouth, pose, or texture error. Spatially, the background tolerates less frequent refresh than the mouth; across modalities, identity references tolerate longer reuse than current speech. Avatar V caches reference K/V that is invariant across denoising steps; SoulX-LiveAct's Neighbor Forcing aligns historical and current chunks at the same diffusion step so that existing KV can be reused across chunks, while ConvKV compresses older history [114, 252]. Hybrid KV compression schemes extend the same idea for autoregressive video diffusion [86]. Cache evaluations should expose key-step selection, refresh thresholds, hit rates, warm-up cost, peak memory, and quality as cache depth increases [81, 128, 252].

Cache lifetime determines what a reuse mechanism can preserve. Avatar-Forever's ForeverCache computes historical features at the first denoising step, reuses

them within the current chunk, and resets them for the next chunk; it is a denoising-time cache rather than an L2 memory bank [108]. Its separately trained recovery adapter instead addresses accumulated rollout errors. Vidu S1's TwinCache maintains noisy and clean historical states and selects them at different denoising stages [234]. These designs distinguish feature reuse from the choice of temporal information available to the model.

System design determines whether algorithmic savings become a stable output pipeline. LLIA overlaps VAE and U-Net execution under TensorRT and INT8. StreamAvatar overlaps DiT and VAE on two H800 GPUs; its approximately 1.20 s time to first output decomposes into 0.48 s input buffering, 0.33 s for the first DiT output, and 0.39 s for the first VAE output, and its 25 FPS is the target playback rate rather than measured full-pipeline throughput [176]. FlowAct-R1 combines a three-NFE base streaming cycle without CFG, selective FP8, frame-level hybrid parallelism, kernel fusion, and asynchronous DiT/VAE execution, reporting 25 FPS and about 1.5 s TTFF at 480p on an "NVIDIA A100 platform" (GPU count NR) [194]. SoulX-FlashTalk assigns pipeline stages to speech processing, four DiT steps, VAE, and history encoding [165].

Live Avatar's Timestep-forcing Pipeline Parallelism (TPP) partitions denoising time rather than network layers. In the v6 paper, four H100 GPUs each execute one DiT timestep with step-matched rolling KV; a fifth H100 decodes with a causal VAE. Section 7 analyzes the measured serving configurations. Table 3 reports 45.2 FPS and 1.21 s TTFF excluding network transfer, while Appendix C.4, Table 10 reports 0.94 s for the stated default three-latent block. We use the Table 3 setting when discussing this ablation [81]. Pipeline throughput depends on the slowest active stage, whereas startup includes the first traversal and buffering; neither follows from device count alone.

Table 6 makes the comparison problem explicit. Motion-parameter prediction is lighter than dense-video synthesis, TinyVAE and device count can alter the throughput of the same student, and NFE ratios exclude decoders and I/O. The useful object of comparison is therefore a quality–speed Pareto curve under a fixed task, frame size, hardware budget, and timing boundary, with algorithmic call reduction, component reuse, and system parallelism reported separately.

4.5 Streaming Generation: Where Fast Meets Stable

Extending a clip generator to continuous output is the point where the route's Fast and Stable problems become one design problem. Chunkwise emission bounds per-chunk latency (Fast), but it forces the model to condition on its own committed history (Stable), and the chunk size trades the two directly: smaller chunks reduce buffering delay and TTFF while increasing scheduling, cache-update, and decoder-startup overhead; larger chunks improve parallel efficiency while raising TTFF. Equal FPS can therefore coexist with very different service behavior, which is why TTFF and inter-chunk latency are separate columns in our tables.

Diffusion Forcing provides the mechanism-level starting point. It independently samples a noise level for every token, unifying prediction, filtering, and imputation under a causal sequence model; at inference, a configurable two-dimensional noise schedule controls the denoising progress at each position [23]. Self-Forcing changes the training distribution: the student consumes its own generated history during an on-policy rollout and matches the distribution of the full autoregressive trajectory while maintaining a rolling KV cache [80]. This addresses the discrepancy between ground-truth history in training and self-history at deployment (L4-G in Section 2.8) and is not itself an NFE-reduction method. CausVid distills a bidirectional video-diffusion teacher into a student that generates chunks in order, using ODE initialization, asymmetric DMD, and rolling KV (9.4 FPS with four sampling steps on one H100, generic video) [225]. Causal Forcing first constructs an autoregressive teacher aligned with the deployed architecture and then applies asymmetric DMD with self-rollout (17.0 FPS on one H100, generic video) [260]. Rolling Forcing pushes the same lineage to real-time long-video generation with rolling-window joint denoising [119]. Foundation-model streaming variants now package these mechanisms at scale: Wan-Streamer targets end-to-end real-time interactive generation, with its second version raising resolution at unchanged latency [77,79]. Vidu S1 combines DMD with a perceptual consistency objective and stage-dependent noisy/clean history reuse [234].

Virtual-human systems add task state to this lineage. PersonaLive advances micro-chunks through four noise levels [113]. StreamAvatar and LiveTalk distill a bidirectional teacher into a chunkwise student and add reference or identity sinks plus self-generated history [29, 176]. EchoTorrent generates chunks in order with a reference sink and tail-only forcing [135]. SoulX-FlashTalk uses multi-block self-correction; SoulX-LiveAct maintains fixed multiscale KV; Lip Forcing introduces synchronization-aware distillation with an explicit clean-cache call; and FlowAct-R1 extends chunkwise generation to full-body behavior with MLLM action planning [31,165,194,252]. Sliding-window systems organize the same state for offline long inputs: StableAvatar's

Table 6 Acceleration mechanisms and reported runtimes of diffusion systems. Values use the settings of the cited papers and are not directly comparable. NFE-D counts denoiser calls; NR denotes not reported.

Method	Sampling / model calls	Reported timing	Resource and boundary
FADA [254]	6 steps; 75 → 18/6 NFE-D	4.17/12.5× call reduction	Wall-clock time and hardware NR
LLIA [227]	2/4 steps; strict NFE NR	78 FPS; ∼140 ms response	RTX 4090D, 384^2; different startup/steady chunks
RAP [51]	Strict NFE NR	42.41 / 39.87 FPS	One A800, 512^2; full boundary NR
LightningCP [122]	Steps/NFE NR	U-Net: 23.692→7.528 s/clip	One-GPU baseline; multi-A800 method
Follow-Your-Emoji-Faster [131]	30 DDIM steps	U-Net: 6.28→2.36 s	64 frames, 256^2; device NR
PersonaLive [113]	4 steps	15.82 FPS; 0.253 s/chunk	One H100, 512^2; default decoder
MotionStream [166]	3 NFE/chunk	29.5 FPS; 0.39 s latency	One H100, 480p, TinyVAE; general motion
StreamAvatar [176]	3 steps; strict NFE NR	∼1.20 s first output	Two H800, 928 × 704; 25 FPS is media rate
Hallo-Live [104]	Steps/NFE NR	20.38 FPS; 0.94 s latency	Two H200; full boundary NR
Lip Forcing [31]	2 denoising + 1 cache call	31.58 / 15.11 FPS	One H100, 512^2; speech preprocessing excluded
FlowAct-R1 [194]	3 base NFE; refiner NR	25 FPS; ∼1.5 s TTFF	A100 platform, 480p; device count/full boundary NR
Live Avatar [81]	4 NFE	45.2 FPS; 1.21 s TTFF	Five H100, 720 × 400, v6; network excluded
SoulX-FlashTalk [165]	4 NFE	32 FPS; 0.87 s startup	Eight H800, 720 × 416
SoulX-FlashHead [229]	Steps/NFE NR	Lite: 96; Pro: 10.81 FPS	Lite device differs across paper sections: H20 / RTX 4090
SoulX-LiveAct [252]	3 steps; strict NFE NR	20 FPS; 0.94 s latency	Two H100/H200; latency definition NR
Avatar V [114]	24 base steps + 1 SR step	∼ 3× relative latency gain	Eight unspecified GPUs; absolute timing NR

Dynamic Weighted Sliding Window (2.32 min per 81 frames at 480 × 832) and InfiniteTalk's 81-frame chunks with nine context frames extend duration without targeting service latency [184, 216].

Streaming control increasingly includes changes to the appearance reference itself. Vidu S2 supports reference switching, video editing, and spatial-video workflows [233]; FlowAct-R2 couples dynamically updated multimodal references to an agent that prepares behaviors and schedules reactions and interruptions [83]. Evaluation must distinguish attributes intended to persist from those the user requests to change. Successful interaction examples or reported session duration alone do not quantify the transition delay, residual influence of old references, or subsequent drift.

One Fast implication deserves explicit statement: the conditioning pipeline sets a latency floor that no visual-path acceleration can cross. Speech encoders with wide receptive fields, block-level audio buffering (Teller collects a 200 ms speech block before computing [251]), and window-based speech pre-encoding all contribute delay ahead of the first DiT call; StreamAvatar's 1.20 s first output includes 0.48 s of pure input buffering [176]. Reports that time only the denoiser therefore understate TTFF by construction, and reproducible streaming protocols should jointly report input-chunk duration, TTFF, steady-state inter-chunk latency and jitter, playback buffer, device count, and module-specific timing endpoints.

4.6 Stable: Streaming State Management

The patterns in Section 2.8 act on different failure paths. We now examine their implementations and observed effects in streaming diffusion. Generic long-video work clarifies how these patterns interact. Self-Forcing++ samples five-second segments from long self-generated rollouts for distributional supervision (L4-G at scale), exposing training to states far from the sequence origin; its standardized quantitative horizon is 100 s [42]. LongLive combines short-window attention, a persistent frame sink (L1), KV recaching, and streaming long tuning; its recache is event-triggered at a prompt-switch boundary, rebuilding KV from recent generated video and the new prompt to remove stale semantics [215]. The reported 20.7 FPS uses a 1.3B model at 832×480 on one H100, with quantitative tests at 30 s (single prompt) and 60 s (scheduled prompt switches) and a 240 s demonstration. Mapped to virtual-human state: a short window preserves immediate mouth and motion continuity, an identity sink stabilizes appearance, compressed long-term KV recovers information lost by local attention, event-triggered recaching removes stale semantics, and RoPE re-indexing controls positional extrapolation. Together, these mechanisms show that computation can continue with finite memory; they do not by themselves demonstrate recovery of identity, synchronization, and motion over a long session [176, 215, 252].

From a state-estimation perspective, the patterns have complementary roles: the anchor (L1) supplies identity and scene priors, the recent window preserves

velocity and mouth continuity, and long-term memory (L2) retrieves poses or events forgotten locally. An ever-growing cache increases computation and error contamination, and a constant cache can forget remote events. Recent systems make the memory-update policy explicit, but how to detect and recover from corrupted state across varied sessions remains open. Stable evaluation should measure identity, appearance, synchronization, and motion in predeclared duration bins, and report error trends, legitimate changes blocked by the anchor, generation faults, and recovery length.

Bounded memory can retain older information without treating every generated frame as equally useful. InteractiveAvatar combines a recent FIFO buffer with a fixed-capacity long-term bank, promoting and replacing entries according to semantic diversity [173]. TaoMate separates bounded active context from fixed-capacity audiovisual memory and regulates recurrent appearance updates through agreement with persistent anchors [56]. LiveAnimate combines a permanent sink, pose-matched retrieval from a bounded bank, and a rolling window; it also performs a separate clean-KV update [241]. These policies address different retrieval and contamination risks. A fixed storage budget establishes bounded resource use, while ablations and duration-specific output measurements assess whether the retained information remains useful.

Live Avatar illustrates the joint design of sampling, deployment, and bounded state. It combines an Adaptive Attention Sink (L1), Rolling RoPE (L3), and History Corrupt, which uses step-matched noisy KV and removes a clean-cache call. Its window retains four recent blocks of three latents each and one persistent sink; after the first chunk, the first generated latent replaces the original reference sink. The long-horizon test loops roughly seven-minute speech recordings and measures windows at 0–10, 100–110, 1000–1010, and 10000–10010 s. The reported Q-Align aesthetic score, image quality, Sync-C, and DINOv2 cosine similarity do not show marked degradation in those windows [81]. This supports stability at sampled points under repeated audio. The static-scene prior, unsampled intervals, and absence of an external baseline under the same protocol limit broader conclusions.

Appearance anchoring also varies with the information that must persist. Gloria uses scene, viewpoint, and expression anchors to support character consistency across changing poses [219]. EverAnimate distinguishes background texture degradation from semantic identity drift, combining persistent latent propagation with restorative flow matching [111]. Their evaluations cover different tasks and durations (Table 7); neither mechanism alone implies a particular streaming speed. Conditioning can also be revised at inference: TT-SAC re-encodes early generated frames and performs a second generation pass with updated conditioning, without parameter optimization [245]. Its short-clip, model-dependent results should be distinguished from recovery during a continuous long session, and the additional pass belongs in the runtime cost.

Interactive systems provide further examples of state management, although interaction capability alone does not establish stability. LPM 1.0 supports speaking, listening, idling, and reflecting and combines a 2-NFE generator with a 1-NFE refiner over pre-RoPE KV (L1+L3). Its benchmark accepts inputs up to one hour and its demonstration timeline reaches 60,000 s, but duration-binned quantitative metrics are NR [230]. FlowAct-R1 uses fixed structured memory and generated-history replacement during training (L2+L4-G) for full-body behavior with MLLM action planning [194]. As a contextual example from the render route, ARIG maintains 512 interaction summaries and reports improved source metrics relative to a capacity of four; this cache capacity is not translated into a measured generation duration [64].

4.7 Design Guidance

Table 8 summarizes three practical considerations. First, the benefit of further step reduction depends on how much runtime remains in the denoiser; once decoding or conditioning dominates, optimizing those stages may be more effective. There is no universal NFE threshold for this transition. Second, pipeline overlap primarily improves steady-state throughput, whereas smaller initial chunks and shorter input buffers can reduce TTFF. Third, rolling caches, chunk overlap, and memory compression affect both per-chunk cost and the history available for maintaining consistency. Their efficiency and stability effects should therefore be evaluated together.

4.8 Synthesis and Open Questions

Online diffusion-based virtual humans combine sampling, representation, and execution choices, and a gain at one stage can be limited by the next bottleneck. Future work should examine the quality–speed–horizon trade-off, reporting model calls (sampler, steps, CFG branches, strict NFE), component timings, TTFF, steady-state chunk time, jitter, and peak memory separately. Cache methods should also describe warm-up, refresh policy, and memory growth with duration.

For stability, the reviewed studies lack a matched-compute comparison of all five patterns and their combinations. Such a comparison should examine both drift

Table 7 State management and reported generation durations in diffusion systems. Quantitative tests and demonstrations are distinguished in the final column; their duration and input conditions differ across papers.

Method	State management	Reported evaluation and duration
Hallo2 [41]	Patch-drop/noisy history (L4-C)	Quantitative: 4 min, 512^2, enhancer off; hour-long claim separate
StableAvatar [184]	Reference anchor; overlapping windows	Aggregate metrics on Long100: 100 clips, 2–5 min; offline generation
PersonaLive [113]	Selected history; capacity/eviction NR	Quantitative on LV100: at least 1 min; 2700-frame demonstration separate
MotionStream [166]	Sink (L1), rolling KV, L3, L4-G	Quantitative: 241 frames (∼15 s); general-motion task
JoyStreamer-Flash [105]	Boundary transfer (L5), L3	Demonstration: 120 s; no time-binned quantitative test
StreamAvatar [176]	Sink (L1), L3, L4-G	Quantitative: 20 s; ∼150-s demonstration
LiveTalk [29]	Three sinks, two recent slots, L4-G	100-s demonstration; several-minute claim
EchoTorrent [135]	Reference sink; tail-only forcing	1000-s demonstration
Live Avatar [81]	Sink (L1), L3, L4-G; noisy KV	Sampled quantitative windows up to 10000–10010 s; repeated audio, static-scene prior
LPM 1.0 [230]	Sink/reference; pre-RoPE KV	60,000-s demonstration timeline; time-binned metrics NR
AsymTalker [125]	Asymmetric GT/generated history	Time-specific FID/ArcFace at 5/50/150/300/600 s
Lip Forcing [31]	Sink, dynamic positions, L4-G	Quantitative: up to 6 min; speech pre-encoded
SoulX-FlashTalk [165]	Retrospective self-correction (L4-G)	Quantitative: over 5 min; 1000-s demonstration
SoulX-FlashHead [229]	Reference anchor; L4-G	60-s demonstration
SoulX-LiveAct [252]	Fixed reference/long/recent slots (L2)	Hour-scale claim; duration-binned metrics NR
Avatar V [114]	Appearance anchor; motion transfer (L5)	Unbounded-length claim; quantitative duration NR
FlowAct-R1 [194]	Structured memory; generated-history replacement	Longest quantitative rollout NR; full-body scope
Gloria [219]	Scene/view/expression content anchors	Aggregate consistency on 20 clips of 5–10 min; not a time-binned drift curve
EverAnimate [111]	Latent propagation; restorative flow	Quantitative target durations of 10/30/60/90 s; pose-driven animation
Avatar-Forever [108]	Recovery rollout training; within-chunk cache	Quantitative: 5 and 30 s; cache reset for each chunk
LiveAnimate [241]	Fixed sink; bounded pose retrieval and rolling KV	Three-minute evaluation; reported block timing excludes VAE decoding

reduction and restrictions on motion diversity or legitimate scene changes. Reliability-aware updates now have concrete instances such as TaoMate; general recovery criteria under unexpected inputs and reference changes remain to be established [56]. Evaluation should track identity, synchronization, visual quality, background drift, motion, and failures in predeclared time bins; training studies should compare ground-truth, generated, and mixed histories under matched conditions. Section 9 also considers authorization and provenance for long-running systems with cached identity state.

Diffusion-based virtual humans combine representation compression, fewer sampling evaluations, computation reuse, and deployment optimization. Streaming methods additionally manage generated history through anchors, memory, position handling, training, and motion transfer. These mechanisms offer complementary benefits, but their joint effect depends on the task and measurement protocol. When a native autoregressive backbone instead predicts the evolving visual state and diffusion supplies only a local emission head, the system belongs to the autoregressive route discussed next.

5 Autoregressive Methods

The term *autoregressive* (AR) appears at several levels of a virtual-human system. A dialogue model may predict speech, a controller may predict motion, and a visual model may predict frame tokens. This section reserves the AR route for the last case. A native AR model must own cross-frame dense or discrete visual state, as in MIDAS and Body of Her [5, 27]. An AR speech, language, pose, landmark, or motion branch does not establish visual ownership when another component owns appearance, and a sequential scheduler does not make a dense diffusion process autoregressive. This narrow definition separates system organization from the mechanism that produces visible temporal content.

The literature nevertheless advances along two largely independent axes. The first is *state ownership*: causal state progresses from language and behavior, through motion and sparse control, to decodable visual representations. The second is *interaction scope*: one-way animation expands toward online dyadic response, full-duplex interaction, and persistent world grounding. A world-model label belongs to the second axis unless

Table 8 Diffusion-route design options and their trade-offs. Potential extensions require evaluation in the intended setting.

Target	Design options	Trade-offs to check
Faster fixed clips	step caches, CFG folding, compressed latents	Cache staleness near detail steps; compression harms mouths and teeth
Playback-rate streaming	few-step DMD, rolling KV, DiT/VAE overlap, FP8 and fused kernels	Multi-GPU budgets; Tiny-VAE quality loss; L4-G training cost
Low TTFF	smaller initial chunks and staged startup	Input buffering floor; startup-quality gap
Minute-scale stability	anchors, generated-history training, position management	Static-scene bias; motion-diversity loss; memory policy requires validation
Hour-scale operation	bounded memory, anchors, positions; Potential: adaptive refresh/recovery	Check output between sampled windows and under varied input

the causal model also predicts the evolving visual representation. This distinction permits highly interactive systems to inform AR system design without relabeling their flow-, diffusion-, or renderer-owned visual generation paths.

Autoregressive visual state connects efficiency directly to history management. Reusing context can avoid repeated computation, while a long context increases memory and may retain generation errors. KV caching, quantization, parallel decoding, and speculative execution provide candidate acceleration mechanisms [76,102]. Their benefits for a virtual-human pipeline depend on the visual representation, local decoder, and quality constraints, which must be evaluated alongside temporal drift.

5.1 Scope, State Ownership, and Cost Boundary

The ordered test in Section 2.3 first excludes methods whose appearance is owned by a subject-specific mesh, neural field, Gaussian set, or identity-conditioned warp. Among the remaining systems, the AR route is selected only when the evolving visual tokens or latents are predicted natively from prior visual state; a downstream diffusion process may refine the current representation without changing that assignment.

For one emitted frame or block, a useful route-specific accounting is

$$C_t = C_{\text{condition}} + C_{\text{AR}}(h_t) + C_{\text{local-emission}} + C_{\text{decode/render}} + C_{\text{system}}, \tag{5}$$

where h_t names the persistent visual state available at step t. The AR term includes token prediction, attention, and state update; a local continuous-output head retains its own denoising or flow evaluations; and the final terms include codec decoding, rendering, transfer, compositing, and input/output (I/O). Consequently, one AR-backbone update per frame does not imply only one network evaluation or one complete system call per visible frame, and a reported frame rate is interpretable only with the hardware, resolution, precision, and included stages defined in Section 2.7.

5.2 Mechanism Map: Compression, State, and Emission

AR visual generation exposes three coupled design choices [5, 27, 208]. First, the representation fixes serial granularity. A frame may contain many discrete tokens [139], a compact set of continuous tokens [24], or one latent block. Spatial compression and within-frame parallel prediction reduce different work, but tokenizer and decoder costs remain. Second, persistent state controls repeated computation and resource growth. KV reuse avoids recomputing past projections; windows, summaries, eviction, and resets can bound the retained state. A codec feature cache is not interchangeable with an AR-backbone context. Third, local emission converts predicted state into an observable frame. A diffusion or flow head may improve continuous output, but its calls and final decoder remain on the critical path [5, 27].

These choices create a characteristic efficiency tension. More aggressive visual compression shortens the AR sequence but may move cost or reconstruction error into the codec. A longer cache supplies more context while increasing memory and preserving erroneous history; truncation bounds memory but can discard identity, pose, or conversational events. Parallel prediction inside one frame reduces serial depth, yet cross-frame self-conditioning still exposes the model to its own errors. Fast and Stable therefore require separate measurements rather than an architectural label.

5.3 Autoregression at Different Pipeline Levels

The predicted variable determines the role of an AR module. UniTalker models responses and compact facial behavior; Teller, ARIG, DyStream, Talker-T2AV, and

UniLS predict motion or audiovisual controls that a renderer maps to visible appearance [22, 36, 64, 75, 222, 251]. These systems motivate causal control, with rendering retained as a separate cost. OmniResponse provides a detailed example of how autoregression coordinates conversational behavior across modalities.

Native visual AR instead predicts dense visual latents or appearance-bearing tokens. UniMo interleaves 2D video and quantized 3D motion tokens, providing a general human-video bridge; MIDAS, Body of Her, and EARTalking target conditioned human video with framewise visual state [5, 27, 141, 203]. The inference-graph test and worked cases in Figure 6 replace a second taxonomy here.

5.3.1 OmniResponse: Autoregressive Conversational Responses

OmniResponse generates verbal and facial responses causally during a dyadic conversation [126]. A Phi-3.5-based multimodal model predicts the listener's next text token and facial embedding from the preceding speaker and listener states. Chrono-Text Markup represents silence and word duration with `[PAUSE]` and `[LASTING]` tokens, aligning the text stream with facial time steps. TempoVoice then predicts audio tokens from the aligned text representations. The visual decoder recovers 52 facial blendshape coefficients and 12 head-pose parameters; a pretrained portrait renderer converts these controls into images. Thus, the recurrent state describes facial behavior, while the renderer supplies appearance. Under our ownership convention, OmniResponse is an AR response controller within a renderer-dependent system.

The ablations show why temporal alignment matters for this form of autoregression. On ResponseNet, removing Chrono-Text Markup increases lip–speech error distance (LSE-D) from 9.56 to 11.51; replacing TempoVoice with a direct audio prediction head increases it to 11.91 [126]. These within-source comparisons support the alignment design. They evaluate audiovisual synchronization, however, rather than identity preservation or drift over a long session.

The supplementary runtime study reports 15.62 FPS on one NVIDIA A100 80 GB and states a 64 ms latency without deployment optimizations [126]. The source does not specify the timing endpoints, timed output resolution, precision, or whether portrait rendering and input/output are included. The 64 ms value therefore cannot be interpreted as TTFF. OmniResponse illustrates the value of compact causal state and synchronized modality generation, while leaving a complete visible-output timing boundary and time-binned stability evaluation open.

5.4 Representative Visual-State Autoregression

UniMo, MIDAS, Body of Her, and EARTalking satisfy the visual-state criterion under different task and decoding assumptions. UniMo supplies a general human video–motion example, whereas the other systems target digital humans or talking heads. Table 9 summarizes their state representations, reported runtime, and temporal evaluation.

MIDAS illustrates compressed AR visual state followed by local iterative emission. Its Qwen2.5-3B backbone predicts a continuous frame representation with at most 60 visual tokens, using a Deep Compression Autoencoder (DC-AE) with 64× spatial compression [24, 27]. An approximately 0.5-billion-parameter PixArt-α head performs four denoising steps before DC-AE decoding. The six-frame logical block exposes its condition tokens to subsequent frame tokens, so visual autoregression does not imply zero input look-ahead. Five frames of codec history and eight blocks of AR context belong to different state owners.

MIDAS's 480 ms block length and 25 FPS output denote media span and rate, while measured throughput and inference hardware are NR. Its four-minute qualitative singing example is distinct from an author claim of up to one hour after identity-specific post-training; neither supplies time-binned stability measurements [27]. Its reported operating conditions are summarized in Table 9.

Body of Her provides a framewise counterpart. Its Transformer decoder conditions on human/agent audiovisual histories, text or trajectory controls, and identity. A compact diffusion head samples the current frame's continuous tokens in parallel, followed by video decoding [5]. This separates cross-frame prediction from within-frame sampling; both contribute to visible-output cost.

The reported 42 ms next-frame time at 640×360 lacks an inference device and complete component boundary, and the 24 FPS figure is a media rate. Cache/reset policy and quantitative long-session duration are NR. The demonstrated agents also use 9 and 5.5 hours of subject-specific data [5]. These conditions support a personalized architectural example while leaving a controlled speed and generalization comparison open.

EARTalking provides a stronger framewise case. Its GPT-style causal model carries prior visual context and generates the next 3D-VAE frame representation with frame-level control; a sink-frame anchor, sliding KV window, and cyclic position policy make its persistence contract more explicit than those of the earlier retained systems [203]. The frame-local generation and final de-

Table 9 Representative visual-autoregressive systems. Timing includes local sampling only where stated in the cited paper; NR denotes not reported. Figure 6 illustrates the generation paths.

Method	State / local emission	Reported runtime and resources	Temporal evaluation
MIDAS [27]	Eight AR blocks; five codec frames; four-step local head	Measured FPS, TTFF, hardware NR; 25 FPS/480 ms are media rate/span	One-hour author claim; four-minute qualitative demonstration; time-binned metrics NR
Body of Her [5]	Frame-causal state; compact parallel diffusion head; cache bound NR	42 ms/next frame at 640 × 360; device and included stages NR; 24 FPS media rate	Long-session quantitative duration NR
UniMo [141]	Visual and motion tokens; deployment cache contract NR	General human video–motion bridge; conversational service timing NA	Conversational long-session evaluation NA; general-task results separate
EARTalking [203]	Ten-latent-frame window; 36 spatial AR and 25 denoising steps	512^2 setup; numeric FPS, TTFF and inference device NR; source notes real-time limitation	Anchor/history-corruption ablations; demonstration at 3× training window; minute-scale binned metrics NR

coder still belong in the cost breakdown, but the evolving decodable visual state is owned by the causal backbone. EARTalking consequently passes the route test without implying direct pixel-space next-token prediction.

The route boundary also matters when interpreting runtime. DyStream reports 29.24 FPS and 34.2 ms audio-packet-to-motion delay on one H200, but these are motion-level measurements for a render-owned system [22]. Conversely, native visual owners may disclose a causal state contract without publishing a complete end-to-end timing breakdown. Table 9 therefore separates the state representation from the reported runtime and temporal evaluation.

Full-duplex systems as contextual interfaces. X-Streamer and the Wan-Streamer series separate interaction state from dense visual generation. X-Streamer's Thinker conditions a video Actor whose backbone iteratively denoises dense chunks [207]. Wan-Streamer v0.2 separates a token-causal Thinker from a context-parallel flow-matching Performer; v0.3 makes relatively persistent world context and time-localized events explicit [77, 78]. These diffusion-route examples show how serving and interaction can be modular without transferring visual ownership to the dialogue model. These successive Wan releases retain diffusion or flow-based visual generation.

5.5 Fast Design Guidance

The Fast levers below are tied to the cost terms in Equation (5). Examples from native visual AR illustrate representation and state choices; contextual motion and diffusion systems illustrate related execution strategies.

- **Compress the serial visual state** ($C_{\mathrm{AR}}\downarrow$). MIDAS limits a frame to at most 60 DC-AE tokens, while UniMo interleaves compressed video and quantized motion tokens. Compression can reduce serial work but may move cost or error into the codec, particularly for fine facial detail and rapid articulation.
- **Parallelize local emission** (serial depth ↓). Body of Her samples all continuous tokens of the current frame in parallel through a compact diffusion head, and MIDAS uses a four-step local head after the AR frame state. EARTalking similarly separates cross-frame causal state from frame-local realization. The local head's strict number of function evaluations (NFE), tensor shape, and decoder cost remain part of end-to-end latency.
- **Reuse and bound persistent state** ($C_{\mathrm{AR}}(h_t)$ growth control). Standard KV reuse avoids recomputing past projections. MIDAS truncates AR context to eight blocks while separately retaining five codec frames; EARTalking combines a sink-frame anchor with a sliding KV window and cyclic positions. These mechanisms bound repeated work, but their eviction, correction, and reset costs require direct measurement.
- **Separate control-path and emission-path serving** ($C_{\mathrm{system}}\downarrow$). Wan-Streamer v0.2 keeps its token-causal Thinker on one GPU and places high-resolution flow-matching generation in a Ulysses context-parallel Performer [77]. This diffusion system places the causal state update and the expensive local emitter on distinct serving paths organized by a pipelined unit schedule, without changing the route assignment.
- **Test speculative visual decoding as a transfer hypothesis**. Speculative decoding is established for language models and has been studied for general AR video [76, 102]. A virtual-human deployment must additionally verify identity, lip synchronization, and visual-state acceptance when drafted tokens or frames are committed; none of the retained native visual systems reports such a comparison.

The route boundary also exposes bottleneck migration. DyStream reports motion-level packet delay while

its warp, decoder, and display boundary remain undisclosed; X-Streamer reports a multimodal system rate without a matched decomposition of its dense diffusion Actor; and Wan-Streamer v0.2 reports both model-side and network-inclusive interaction latency for a flow-owned visual path [22, 77, 207]. These values answer different questions. A reproducible AR report should provide AR calls per visible frame, token or block count, local-head NFE, cache-update cost, decoder time, I/O boundary, and peak state memory under one declared pipeline.

5.6 Stable Design Guidance

Visual autoregression repeatedly consumes committed visual state, which can preserve continuity and propagate error. The common patterns therefore require method-specific checks. EARTalking documents an L1 reference sink, L3 cyclic positions, and L4-C noise-corrupted histories. MIDAS also uses L4-C; its five-frame codec history and eight-block AR context are different state objects and do not alone establish multiscale L2 memory. Body of Her's dataset-history training is distinct from generated-history training. None of these facts should be relabeled as an L4-G rollout study.

Explicit boundary transfer (L5) likewise requires a designated motion or boundary variable; ordinary access to preceding visual tokens is insufficient. Generated-history training, adaptive reference refresh, and hierarchical state are research directions whose effects on long-session output remain to be measured. EARTalking's component ablations provide a concrete example, analyzed in Section 7; the retained systems do not yet provide a complete joint time-binned session protocol.

5.7 Synthesis and Open Questions

The reviewed visual AR systems span video–motion tokens, compressed frame representations, and framewise controlled generation. Their tasks, personalization requirements, tokenizers, and local emission heads differ. No system in Table 9 provides both complete end-to-end timing and a joint time-binned stability evaluation. Consequently, architectural feasibility is supported more broadly than a common deployment comparison.

The most useful next step is interface-level co-design: visual tokenizers should report how compression changes both AR work and decoder cost; state managers should expose cache growth, eviction, correction, and reset; local heads should disclose strict NFE; and serving evaluations should charge condition encoding, decoding, and I/O to the same event timeline. Sub-linear-state sequence models [59, 218] are a natural research direction for the route's Stable problem, offering constant-size state with different forgetting behavior than KV eviction. Response-, motion-, and world-level AR can be excellent deployment choices, but their appearance owner and emission costs remain assigned to the diffusion, flow, or rendering routes that produce visible content.

Table 10 Autoregressive design options and their trade-offs. Potential extensions require evaluation in the intended setting.

Target	Design options	Trade-offs to check
Low serial latency	compression and parallel within-frame emission	Codec error on facial detail
High throughput	KV reuse; Potential: speculative visual decoding	Draft-model identity fidelity unknown
Minute-scale stability	anchors and bounded history; Potential: generated-history training and adaptive refresh	Accumulated error; anchor over-constraint; refresh discontinuity; matched long-session comparison still needed
Hour-scale operation	Potential: hierarchical state and declared eviction/reset/recovery	Long-range forgetting; reset discontinuity; unmeasured memory and error trajectories

Autoregression can organize behavior, motion, or the visible sequence itself. Only the last role defines the visual AR route. Its main design choices concern representation compression, history reuse, and local frame emission; each affects both runtime and error propagation. Future comparisons should evaluate these choices through explicit state-update policies and matched efficiency and stability measurements.

6 Render-Based Methods

6.1 Scope, Ownership, and Cost Boundary

Render-based methods separate a reusable appearance representation from the time-varying controls that animate it. We assign a system to this route when a subject-specific mesh, neural field, Gaussian set, or identity-conditioned image-space representation owns appearance and its renderer produces the visible frame [62, 63, 91]. The controller may itself be deterministic, autoregressive, diffusion-based, or flow-based; generating compact motion with one of these models does not change which component owns image synthesis. This ownership test also distinguishes the present

route from direct pixel or video-latent generation in Sections 3–5.

Reusing appearance connects the route's efficiency and stability mechanisms. An enrolled asset reduces repeated appearance construction and supplies a persistent identity reference, corresponding to L1 in Section 2.8. Frame production still requires deformation, rendering, and often neural decoding or inpainting. Enrollment, tracking, and compositing must therefore be included when assessing the complete pipeline. For a session of T observable frames, we write

$$\begin{aligned} C_{\text{session}} = C_{\text{enroll}} + \sum_{t=1}^{T} \big[& C_{\text{track/cond}}(t) + C_{\text{motion}}(t) \\ & + C_{\text{repr}}(t) + C_{\text{render}}(t) \\ & + C_{\text{compose}}(t) + C_{\text{system}}(t) \big]. \end{aligned} \tag{6}$$

Here, C_{enroll} covers capture, calibration, tracking, reconstruction, rigging, per-subject optimization, and asset storage; $C_{\text{track/cond}}$ covers runtime acquisition and estimation of pose, camera, gaze, audio, or other controls; and C_{motion} is the cost of predicting the compact time-varying state. The remaining terms cover appearance deformation or attribute updates, rendering, masks/inpainting/super-resolution, and system overhead such as transfer, scheduling, encoding, and I/O. Thus, renderer-only throughput cannot be reported as end-to-end latency, and a high per-frame rate cannot hide a costly enrollment stage.

Figure 9 projects Equation (6) onto four recurring interface choices. The representations expose different computational boundaries: enrollment and reused appearance are session-level, whereas the control-to-output path repeats per frame, and individual methods may omit or relocate the optional source-pixel and compositor paths.

Codec avatars provide an early example of person-specific learned appearance for real-time telepresence under studio capture conditions, including deep appearance models [121], expressive extensions [33], and audio- and gaze-driven animation [157]. Subsequent work explores different enrollment budgets, from multi-view capture to monocular video and single-image reconstruction. These settings offer different trade-offs between acquisition effort, subject fidelity, and rendering cost.

6.2 Explicit Controllable Appearance Representations

Explicit backends expose a geometric or scene representation that is updated by pose, expression, viewpoint, or audio-derived controls. Parametric models such as 3DMM, FLAME, SMPL, and the expressive SMPL-X provide interpretable control interfaces [9, 109, 123, 145]; they do not, on their own, establish photorealism or real-time operation. RC-SMPL sits at the lightweight end of this parametric interface: it animates an SMPL-X body mesh whose texture and normal maps are accumulated in real time from a single RGBD camera, trading multi-camera capture for progressive per-subject enrollment; the source claims real-time operation but reports neither frame rate nor inference hardware [170]. Likewise, NeRF and 3D Gaussian Splatting (3DGS) define rendering families [91, 137], while capture coverage, representation size, deformation, compositing, and hardware determine the operating point of a particular avatar.

6.2.1 Neural-field backends

Early talking-head neural fields made the appearance asset target-specific. AD-NeRF learns separate audio-conditioned head and torso fields from a 3–5-minute target video sampled at 25 FPS, under a static-camera/background setting [63]. In its reported five-minute setting, training takes about 36 hours and rendering one 450×450 frame takes about 12 seconds on an RTX 3090. This is a frame-rendering measurement, not end-to-end throughput. HeadNeRF instead renders a 32×32 neural feature map and uses a 2D neural renderer to produce a 256×256 image [71]; it reports about 25 ms per frame (over 40 FPS), but the inference GPU is unspecified and the timing begins from supplied latent codes and camera parameters, excluding fitting, tracking, and I/O. FLAME-in-NeRF attaches parametric expression control to the field [6].

Hash-grid and occupancy structures then removed most field-query work without removing subject enrollment. This is the route's adoption of the Instant-NGP lever, in which a multiresolution hash encoding replaces deep MLP queries [138]. RAD-NeRF combines a 3D spatial hash, an audio feature grid, occupancy pruning, and a separate 2D torso stage; it reports 40 FPS and about 7 hours of target training on one V100, with output resolution and timer coverage not reported in the source runtime statement [182]. ER-NeRF uses tri-plane hash encodings, regional attention, and a separate 2D torso model [107]; its 34 FPS result is reported for a 450×450 Obama comparison on one RTX 3080 Ti, with included runtime modules unspecified. These reports illustrate efficient personalized field designs, but their different devices, timing boundaries, and enrollment procedures do not isolate a common speedup over AD-NeRF.

SyncTalk retains the personalized neural field but makes audio–visual synchronization the central design target. A tri-plane hash head field is driven by a Face-Sync controller aligning lip motion to audio, a 3D facial-

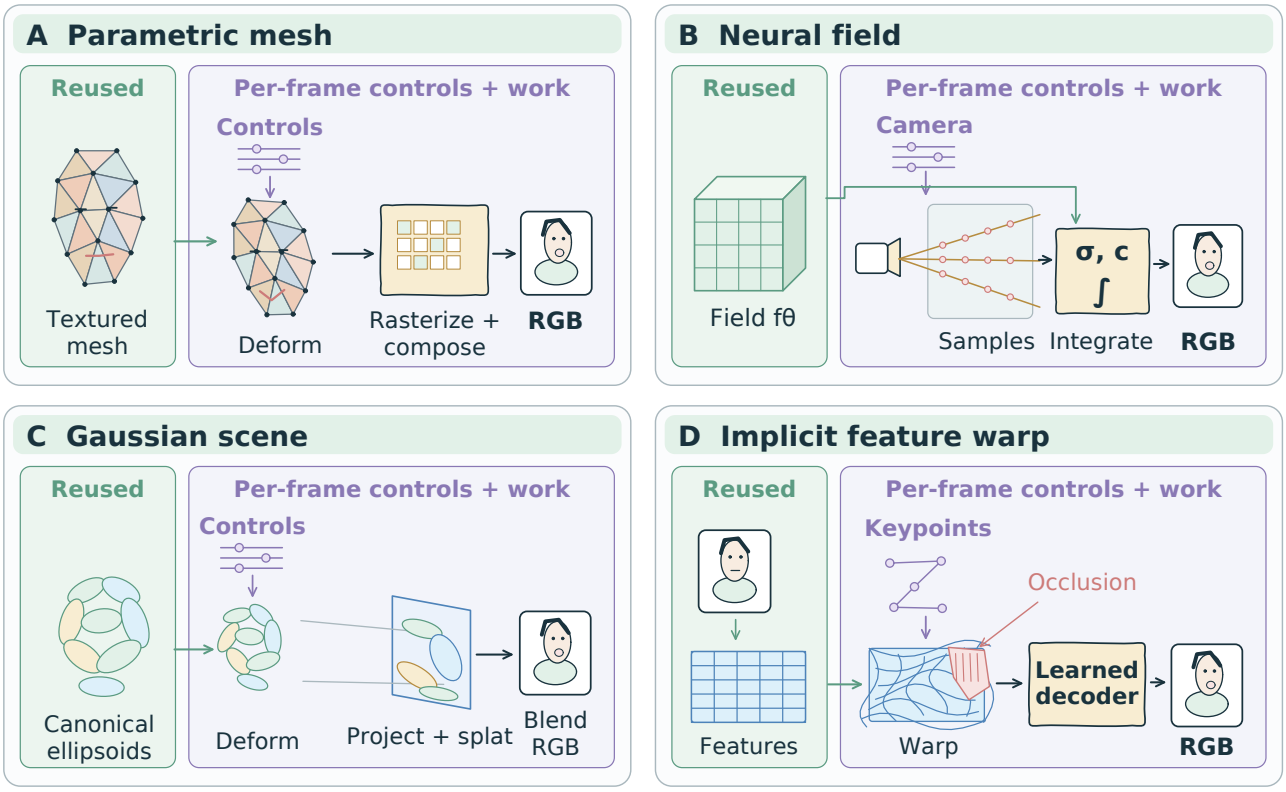


Fig. 9 Reusable appearance in four representations: (A) a textured parametric mesh, (B) a neural field sampled along rays, (C) Gaussians projected and rasterized, and (D) a source feature volume transported by a learned warp [62, 63, 107, 109, 168, 205, 226, 246]. Each panel distinguishes the stored appearance from the control and rendering operations repeated for each frame. The diagrams explain representations rather than rank speed, quality, or stability.

blendshape expression model, and a Head-Sync stabilizer for pose, and a Portrait-Sync generator restores hair detail and blends the head with the torso [148]. It reports 52 FPS without and 50 FPS with the Portrait-Sync stage at 450 × 450 on one RTX 3090 with preloaded data, after about two hours of per-subject training. The synchronization machinery is a quality lever on the conditioning path rather than a change to C_{render}; its Gaussian successor SyncTalk++ appears in Section 6.2 below.

Beyond talking heads, AvatarReX extends the personalized neural field to expressive full-body avatars with separately controllable body, hands, and face, learned from multi-view video [253]. Its efficiency lever is a deferred rendering pipeline built on surface rather than dense volume rendering; the source reports 25 FPS at 1024 × 1024 on one RTX 3090 for animation and rendering, characterized as roughly two orders of magnitude faster than prior NeRF avatars. Enrollment is studio-scale (22 synchronized cameras, about 2000 frames per sequence), the opposite end of the C_{enroll} spectrum from the one-shot methods below.

6.2.2 Gaussian scene representations

Gaussian avatars replace repeated field sampling with splattable primitives, but they differ substantially in acquisition and animation. GaussianTalker binds a target-specific Gaussian face to FLAME and predicts speaker-specific blendshapes; Gaussian rasterization and an inpainting generator synthesize the face, while pixels outside the facial mask are retained from the source video [226]. Its two training stages take roughly 6 and 4 hours on one RTX 4090. The paper reports 130 FPS on that GPU with the sequence preloaded and 36 FPS for an MNN/OpenGL implementation on an Apple M1, an early sign that splatting fits mobile compute budgets. GaussianAvatars optimizes a subject from multi-view observations and assigns Gaussians to FLAME triangles so that they inherit mesh motion [152]. SplattingAvatar uses a registered mesh for low-frequency deformation and walkable barycentric Gaussians for higher-frequency appearance; its Unity implementation reports over 300 FPS on an RTX 3090 and a steady 30 FPS on an iPhone 13 (resolution and timer window unreported) [163]. FlashAvatar narrows the personalized

head to 13,453 surface-embedded Gaussians deformed by tracked FLAME motion and a residual MLP, reporting 304 FPS at 512×512 on one RTX 3090 for animation and rendering only [205]. A broader family varies the binding and deformation design without changing the cost structure: HeadGaS's blendshape-latent formulation [50], PSAvatar's point-based shape model [250], deformable body avatars [153], emotional control [214], progressive construction [171], and real-time full-pipeline avatar systems [99], text-driven toonification combining a conditional tri-plane with a Gaussian deformation field (TextToon: 48 FPS on one RTX 4090 and 15–18 FPS on an Apple M1) [172], and real-time hand avatars fit from as few as 5–20 views (GaussianHand: up to 125 FPS, inference hardware NR) [247]. In every case, sparse, rasterizable appearance shrinks C_{render}, while enrollment, tracking, and topology handling remain.

Audio-driven per-subject splatting has become its own competitive subfamily. GaussianTalker (Cho et al.) encodes canonical Gaussian attributes into a shared spatial-aware feature and predicts per-frame Gaussian offsets through a spatial–audio attention module driven directly by speech, reporting 98 FPS (and up to 120 FPS for a lightweight variant) at 512×512 after about one to 1.5 hours of per-subject training, with the inference GPU NR [30]. This is a distinct method from the FLAME-bound GaussianTalker of Yu et al. [226] despite the shared name: the former drives raw Gaussians from audio, whereas the latter animates FLAME-bound blendshapes. SyncTalk++ ports the synchronization-centric SyncTalk design onto a 3D-Gaussian dynamic portrait renderer and adds an out-of-distribution audio expression generator and a torso restorer, reporting 101 FPS at 512×512 on one RTX 4090 after about 1.5 hours of per-subject training [147]. The pair illustrates a transition from neural fields to Gaussian rendering within a related synchronization-oriented line of work. The changed hardware, resolution, and auxiliary modules prevent attributing the difference in reported frame rate solely to the renderer.

Recent audio-driven Gaussian work makes the controller–appearance boundary more expressive. GaussianSpeech couples an audio-conditioned Transformer to personalized 3D Gaussians, including expression-dependent attributes and multi-view supervision; DGTalker instead navigates a disentangled Gaussian latent space that separates lip motion from broader talking expression. Both retain a Gaussian appearance owner, so their learned audio or latent modules are controllers rather than diffusion-route owners. GGTalker addresses the enrollment–generalization trade with a generalizable Gaussian prior followed by identity-specific adaptation, making explicit the cost of replacing per-subject optimization with a reusable prior. AudioAvatar extends the same direct audio-to-deformation idea to a personalized whole-body avatar: audio-conditioned particle/deformation fields drive facial, hand, and body dynamics before differentiable Gaussian rendering. Its feature-level distillation from a video diffusion model is a training or controller mechanism; the Gaussian representation remains the primary appearance owner. These systems therefore move beyond the conventional audio–pose–renderer interface, but they also move stability risk into OOD audio, large motion, multi-view consistency, and deformation quality [4, 74, 101, 115].

The deformation representation itself can reduce recurring computation. EmbedTalk replaces a triplane encoder with per-Gaussian learned embeddings that are deformed using audio and facial controls [159]. This removes a feature-lookup stage while retaining per-identity optimization of the Gaussian appearance model. Its enrollment requirement therefore differs from one-shot reconstruction, even though both approaches can use the same rasterization primitive.

GaussianHeadTalk studies a complementary temporal failure mode: inconsistent Gaussian mapping and noisy facial tracking can produce wobble despite a fixed identity asset [1]. This motivates separate measurements of tracking and deformation error, failed detections, and temporal artifacts, alongside identity and synchronization.

Two recent developments extend the Gaussian recipe along the axes this survey tracks. First, *mobile and wearable deployment*: TaoAvatar delivers full-body talking avatars on augmented-reality glasses by baking a StyleUNet-learned deformation into a lighter MLP and reports 90 FPS on the Apple Vision Pro [25], and SqueezeMe distills multiple Gaussian full-body avatars to run simultaneously on a Quest 3 headset by moving corrective computation out of per-frame networks [84]. These examples support operation on wearable devices under the reported conditions; sustained power and thermal behavior require separate measurement. Second, *one-shot Gaussian construction*: LAM builds an animatable Gaussian head from a single image in one feed-forward pass, explicitly targeting cross-platform real-time animation and rendering [67]; GUAVA extends fast feed-forward construction to upper-body avatars [232]; and VASA-3D lifts VASA-1's motion space to a Gaussian head built from one portrait [210]. One-shot construction attacks the route's historical weakness, hours of C_{enroll}, by amortizing enrollment into a generalizable network. The reduced input coverage leaves unseen appearance and subject-specific detail to the learned prior, creating a fidelity trade-off that depends on the task.

SDTalk extends one-shot Gaussian construction by separating visible-surface reconstruction from prior-based completion of occluded facial regions, then using coarse and fine motion fields [87]. It requires no personalized fine-tuning, but the teeth and other unseen structures still depend on the learned prior. Lower enrollment cost should therefore be assessed alongside the fidelity of newly exposed regions.

The generalizable head-avatar line makes the enrollment-versus-representation trade especially legible through two works by the same authors. GPAvatar reconstructs a controllable head from one or several images in a single forward pass using a tri-plane feature field with NeRF-style volume rendering, driven by a FLAME point-based expression field and a multi-image attention module; it renders a 128×128 feature map super-resolved to 512×512 and reports about 15 FPS on an A100, which the authors explicitly label pre-real-time [35]. GAGAvatar keeps the one-shot, generalizable, FLAME-conditioned formulation but replaces the volumetric field with 3D Gaussians reconstructed by dual-lifting, reporting 67 FPS at 512×512 on an A100 using 2.5 GB of memory (driving-render only, excluding pre-computable driving parameters) [34]. Both use an A100 and report 512×512 output, but they change the reconstruction and rendering pipeline together. Their timing difference is therefore not a controlled ablation of representation alone. GAGAvatar also serves as a reusable backend for controllers such as REA-Listener (Section 6.4).

6.3 Implicit Image-Space Appearance Reuse

Image-space methods store identity primarily in a source image or its encoded features and predict correspondences to each target pose. FOMM estimates ten keypoints and local Jacobians, constructs dense backward flow and an occlusion map, and uses an inpainting decoder for content that cannot be copied directly [168], building on earlier keypoint-based transfer [167] and refined by articulated region modeling [169]. TPS Motion Model replaces the local affine model with ten thin-plate-spline transformations plus an affine background transformation, combined by multi-resolution masks before inpainting [246]. Both were evaluated as source-to-driving-video animation without runtime disclosure. The lineage acquired deployment weight with face-vid2vid, which learns a 3D-aware keypoint representation for one-shot talking-head synthesis explicitly motivated by video-conferencing bandwidth: transmitting keypoints instead of pixels makes the warp both the codec and the renderer [198]. Latent-space navigation offers an alternative implicit interface [199], and audio-driven variants animate a still image directly from speech [196].

LivePortrait provides an efficiency-oriented instance of this family. It learns implicit keypoints and motion, warps a source feature volume, and decodes with a SPADE-based generator, supporting cross-identity animation and paste-back compositing [62]. The paper reports 12.8 ms for model generation/inference on one RTX 4090 in naive PyTorch, without binding the timer to a stated resolution or decomposing preprocessing, paste-back, and I/O. Its one-shot appearance interface is reused by several systems discussed in Section 6.4. Large pose changes, disocclusion, mouth interiors, hair, and crop-boundary compositing remain relevant failure cases when evaluating these composed systems.

Reference reuse can also target oral detail. EfficientSync selects source-video references, warps their features, mixes their textures, and decodes the mouth region with a SPADE generator [70]. The transported reference appearance places it in the render route despite adversarial training. Reference selection belongs to preparation cost, and missing views or mouth states can limit the texture information available at inference.

6.4 Motion Controllers over Reused Appearance

Reusable appearance permits motion control to be studied separately from image synthesis. A compact motion model can drive a one-shot warp or Gaussian asset, changing temporal behavior without changing the visual owner. The controller and renderer have separate costs, and either can determine system latency.

VASA-1 illustrates this separation. A diffusion Transformer generates holistic facial dynamics and head motion in a learned face latent space, conditioned on audio and optional control signals, and drives a one-shot appearance decoder; the paper reports online generation of 512×512 video at up to 40 FPS with 170 ms/95 ms latency on one/two RTX 4090 GPUs under its stated protocol [209]. Sampling compact motion can reduce the size of the iterative state, while the appearance decoder reuses identity information. The cost relative to dense-video generation still depends on the networks, sampling schedule, and decoder. FLOAT replaces the sampler with motion-latent flow matching [95]; Ditto structures the motion space for controllable real-time synthesis [110]; ChatAnyone stacks hierarchical motion diffusion for stylized upper-body video with real-time intent [151]; MirrorMe reports real-time audio-driven half-body animation [133]; and FlowTalk pushes motion-space flow matching toward the same operating point [47]. MoDA stays in this compact-controller family while keeping a

diffusion motion generator: it reuses LivePortrait's disentangled facial representation as the appearance renderer and generates motion in a joint parameter space with a rectified-flow model, reducing denoising from 50 to 10 steps and reporting a real-time factor of 0.846 on in-the-wild inputs (below one indicates faster than playback; inference GPU NR) [112]. It is one-shot from a single reference image, and under the ownership test its LivePortrait renderer, not the motion diffuser, owns appearance. GAIA scales a zero-shot two-stage design (motion generation, then appearance) [66], JoyVASA extends the recipe to animal faces [18], and style-controllable variants [130, 197] enrich the control space. The compact interface also connects to speech-to-3D animation research that predicts vertices or coefficients without owning pixels [54, 248]; such motion-only models are boundary components under the ownership rule.

Compact controllers now separate articulation from other aspects of personal motion. VibeAvatar generates motion in LIA's warp space, adapting phonetic conditions before preference-based optimization of motion aesthetics [195]. Talk Like You predicts habit-conditioned motion in one sampling step and uses frozen LivePortrait rendering modules [187]. Both remain appearance-reuse systems: richer motion objectives change the controller without turning it into a dense visual generator. Evaluation should distinguish synchronization, motion preference, and reproduction of a target speaking habit.

Interactive-agent systems assemble the same parts with session state. Teller processes 200-ms audio blocks with an autoregressive residual-vector-quantized motion model; an implicit-keypoint warper and a single-pass enhancer produce RGB, with reported stage times of 7, 106, and 71 ms on a machine containing four H800 GPUs [251]. ARIG predicts continuous head-motion coefficients with a 15-step DiffusionMLP over a framewise autoregressive controller and renders through LivePortrait-style warping; it reports a 31 FPS generation-process rate (hardware and included components NR) and maintains a 512-slot interaction summary cache, a bounded controller memory whose Stable effect is unmeasured [64]. DyStream couples an AR/flow motion module (60 ms speaker-audio buffer) to a reference-conditioned appearance volume and reports 29.24 FPS with 34.2 ms audio-packet-to-motion delay on one H200; the delay ends at facial-motion generation rather than final rendering [22]. REA-Listener predicts listener motion with a lightweight Transformer and renders with GAGAvatar [34]/3DGS plus inpainting [249]. PortraitDirector adds hierarchical disentanglement for controllable real-time reenactment [85], and VASA-Rig connects audio-driven rigging to virtual-reality use [140].

EmbodiedHead further illustrates the cost of this composition. Its source reports over 900 FPS for motion-coefficient generation but 59 FPS when using GAGAvatar in the full pipeline on an RTX 3090 [240]. The difference is a within-source example of why controller timing cannot stand in for completed video generation.

6.5 Fast Design Guidance

The route's levers map onto Equation (6) as follows:

1. **Amortize appearance at enrollment** ($C_{\text{repr}}/C_{\text{render}} \downarrow$ via $C_{\text{enroll}} \uparrow$). Subject-specific fields and Gaussian sets reuse fitted appearance across frames. One-shot warps and feed-forward Gaussian builders, such as LivePortrait and LAM, reduce preparation cost but rely more on learned priors for unseen content.
2. **Structure the queries** ($C_{\text{render}} \downarrow$). Hash grids and occupancy pruning reduce field-query work, while Gaussian representations permit rasterization. Mobile and wearable examples include SplattingAvatar, TaoAvatar, and SqueezeMe. Their reported frame rates establish particular device configurations, not a measured speed-per-watt ranking.
3. **Keep the controller compact** ($C_{\text{motion}} \downarrow$). Motion-space diffusion or flow reduces the representation processed by an iterative sampler. The renderer and compositor remain on the inference path and require their own component measurements.
4. **Reuse source pixels** (C_{compose} trade). Paste-back and outside-mask retention (GaussianTalker, LivePortrait) avoid re-synthesizing static content, at the price of seams and crop-boundary failures.

Table 11 shows why these levers must be evaluated separately: reported quantities span renderer-only throughput, animation-plus-render time, model-generation time, motion-level packet delay, and block latency, and no cross-paper rank is supported when anatomy, personalization, device, resolution, and compositing differ. The route-specific accounting rule is to include $C_{\text{track/cond}}$ wherever runtime face tracking, FLAME fitting, keypoint estimation, or audio conditioning is required. Precomputed controls and runtime-estimated controls define different boundaries.

6.6 Stable Design Guidance

An enrolled appearance asset provides a persistent identity anchor (L1 in Section 2.8), but does not guarantee stable rendered output. Tracking, controller state, deformation, disocclusion, and compositing can introduce

Table 11 Selected rendering systems and reported operating conditions. Frame or rendering time differs from TTFF; the final column identifies which components are measured.

Method	Appearance / enrollment	Source timing / device	Measured stages and temporal evaluation
AD-NeRF [63]	Head/torso fields; 3–5-min capture	∼12 s/frame, 450^2, RTX 3090	Frame rendering; ∼36 h training
StyleAvatar [193]	Person-specific UV/background; adversarial decode	35 FPS / 28 ms live; RTX 3090	20 ms model-only; hours-scale enrollment; longitudinal metrics NR
FlashAvatar [205]	FLAME-surface Gaussians; 1–3-min capture	304 FPS, 512^2, RTX 3090	Animation/render only; tracking excluded
TaoAvatar [25]	Full-body Gaussians; multi-view capture	90 FPS, Apple Vision Pro	Complete pipeline coverage NR
LivePortrait [62]	Source feature volume; one-shot	12.8 ms model generation; RTX 4090	Warping/decoding; preprocessing and paste-back separate
VASA-1 [209]	Reused face appearance; one-shot	Up to 40 FPS, 512^2; RTX 4090	170/95 ms for one/two devices under source protocol
Teller [251]	Source image features	7/106/71 ms components; four-H800 machine	Total < 200 ms; TTFF and full RGB boundary NR
DyStream [22]	Reference appearance volume; one-shot	29.24 FPS; 34.2 ms packet delay; H200	Timer ends at motion; 180-s pose-drift ablation
AvatarReX [253]	Body/hand/face fields; 22-camera capture	25 FPS, 1024^2, RTX 3090	Animation/render; enrollment and tracking separate
GAGAvatar [34]	Feed-forward Gaussian head; one-shot	67 FPS, 512^2, A100	Driving/render; driving-parameter acquisition excluded
MoDA [112]	Motion flow + LivePortrait; one-shot	RTF 0.846; inference device NR	Wild-data protocol; training hardware is not inference hardware

temporal errors even when the canonical asset is unchanged. Controller memories, such as ARIG's 512-slot cache, provide another form of persistent state. Position handling and training with generated histories are relevant when these controllers are sequential; their longitudinal effects are not established by renderer throughput or asset reuse alone.

A representative duration-based result highlighted in Table 11 is DyStream's 180-s head-pose drift ablation, which shows its appearance anchor reducing pose drift over ten three-minute clips [22]. This is a component ablation of anchor conditioning for pose drift, rather than a validation of every L1 mechanism or a full stability protocol. These studies do not establish joint, duration-binned reliability across identity, appearance, synchronization, geometry, flicker, memory, and latency. Partial evaluations remain informative about the properties they test.

A useful comparison would evaluate a tracked-and-rendered avatar and an anchored dense generator under the same long-session conditions. Identity retention, motion and deformation error, temporal wobble, and compositing failures should be reported separately. GaussianHeadTalk motivates the distinction between identity similarity and temporal wobble [1]; the reviewed studies do not establish which complete pipeline is more stable over an hour.

Table 12 Rendering-route design options and their trade-offs. Potential extensions require evaluation in the intended setting.

Target	**Design options**	**Trade-offs to check**
Per-subject reuse	Gaussian assets and baked deformation	Hours of enrollment; capture requirements
One-shot generality	LivePortrait warps, LAM Gaussians, VASA-1 motion controller	Disocclusion, large poses, mouth interiors; limited input-view coverage
Mobile/wearable deployment	splatting and baked correctives (TaoAvatar, SqueezeMe)	Per-subject assets; thermal/battery limits untimed
Minute-scale stability	enrolled L1 asset; Potential: joint long-session reliability	Temporal errors in deformation and composition; limited joint long-session evaluation
Expressive control	motion-space diffusion/flow controller	Controller expressiveness bounded by the asset's deformation space

6.7 Synthesis and Open Questions

The central trade-off is not simply fidelity versus throughput. Subject-specific neural fields and Gaussians amortize a substantial enrollment cost over many frames; one-shot feature warpers reduce enrollment but depend heavily on source-view coverage and learned disocclusion; parametric controls provide stable, editable interfaces whose base space cannot express motion it does not encode. Explicit scenes must handle topology, primitive count, memory, and view coverage. Image-space reuse must synthesize newly visible content and preserve crop seams.

Three directions follow from these trade-offs. First, joint design of feed-forward appearance construction and compact motion control could reduce both preparation and runtime costs; one-shot construction still has a nonzero setup cost. Second, controller training should be evaluated under generated histories to test whether it reduces drift in recurrent systems such as ARIG and DyStream. Third, matched evaluations should report enrollment data and time, asset size, hardware, precision, all runtime components, and quality over duration. These measurements would show whether a change reduces complete-system cost or shifts work to another stage.

Render-based methods reuse an explicit scene or source-conditioned appearance representation under time-varying control. Efficient queries, compact controllers, and appearance reuse support reported desktop, mobile, and wearable deployments. The reused asset anchors identity, while tracking, deformation, and compositing remain independent sources of temporal error. The comparison in Table 11 therefore records enrollment, runtime, and temporal evaluation separately.

7 Cross-Route Analysis and Design Trade-offs

The four routes reduce computation and maintain temporal consistency in different ways, but they share several design trade-offs. This section examines decoder choice, pipeline parallelism, history management, and appearance reuse through representative studies. Numerical comparisons within each case use the configurations reported in the same paper.

7.1 A Faster Decoder Can Change the Quality Trade-off

PersonaLive isolates a concrete decoder choice in its appendix, Table 4 [113]. Replacing SD-VAE with TinyVAE increases the reported rate from 15.8 to 20.0 FPS, a

Table 13 PersonaLive decoder ablation, reproduced from supplementary Table 4 [113]: one H100, 512^2. ID is identity similarity; tLP is temporal LPIPS.

Decoder	FPS ↑	ID ↑	tLP ↓
SVD-VAE	11.4	0.6920	12.97
SD-VAE	15.8	0.6983	12.83
TinyVAE	20.0	0.6758	14.66

derived 26.6% increase. Under that same ablation, identity similarity decreases from 0.6983 to 0.6758, average expression distance increases from 0.7028 to 0.7593, and temporal LPIPS increases from 12.83 to 14.66. These measurements establish a local speed–quality trade-off, rather than a universally preferable decoder.

The deduction is that reducing denoiser work changes the value of optimizing other stages. A decoder replacement can improve total throughput while altering the very identity and temporal properties the system is meant to preserve. This experiment does not provide a complete component-time breakdown, so it does not by itself prove that decoding dominates runtime. The useful design rule is to profile the remaining path and measure quality at the chosen operating point.

7.2 Throughput and Startup Follow Different System Paths

Live Avatar's efficiency ablation changes several deployment terms in controlled stages [81]. We use arXiv v6, Table 3, which specifies H100 hardware. Distillation changes the one-GPU rate from 0.29 to 3.66 FPS. With four GPUs, timestep-forcing pipeline parallelism reaches 10.16 FPS but increases TTFF from 4.56 to 4.73 s relative to the one-GPU distilled configuration. Adding a fifth GPU for VAE parallelism reaches 20.88 FPS; the final kernel-optimized configuration reaches 45.2 FPS and 1.21 s TTFF (Table 14).

The measured conclusion is that the final operating point depends on both algorithm and serving topology. The 12.3× ratio between 45.2 and 3.66 FPS is a system-configuration ratio, not a same-resource algorithmic speedup. The pipeline stage illustrates how greater throughput can coexist with a longer startup interval. Even the final kernel row bundles several implementation changes and does not isolate the contribution of each kernel.

Enrollment changes the accounting horizon. For additive resource cost, the amortized cost per frame is $\bar{C}(N) = C_{\text{enroll}}/N + C_{\text{runtime}}$. As an illustrative calculation on one fixed device, a reusable system with 3600 GPU-seconds of enrollment and 0.02 GPU-seconds per

Table 14 Live Avatar serving configurations (arXiv v6, Table 3, PDF p.12; 720 × 400). TTFF excludes networking. NFE includes the cache-refresh call. †: Table 10 gives 0.94 s for the stated default three-latent block; this configuration discrepancy remains unresolved.

Configuration	GPUs	NFE	FPS	TTFF (s)
Teacher	1	80	0.29	45.50
Distilled	1	5	3.66	4.56
Sequence parallel	4	5	4.50	3.94
Timestep pipeline	4	4	10.16	4.73
+ VAE parallel	5	4	20.88	2.89
+ kernel optimization	5	4	45.20	1.21†

frame breaks even with a 0.10-GPU-second direct generator after $3600/(0.10 - 0.02) = 45{,}000$ frames. This is 30 minutes of content at 25 FPS. The numbers are hypothetical; the calculation shows why enrollment feasibility, expected reuse, and representation coverage must all enter a deployment decision.

7.3 An Ablation Supports a Mechanism Within Its Tested Horizon

EARTalking provides a component-level test of history training and anchoring [203]. Its Table 3 compares corrupted-history training (L4-C) and reference anchoring (L1). The complete model has FID 18.981, FVD 363.909, and Sync-C 5.707; removing corruption gives 20.643, 401.354, and 4.277; removing the anchor gives 42.298, 489.197, and 4.067. The accompanying qualitative example extends to three training-window lengths. This supports the two components under the source protocol, while numeric inference throughput, TTFF, hardware, and minute-scale time-binned stability remain NR in the v1 paper.

The ablation shows that anchoring and history corruption improve the tested outputs. Measuring how these gains evolve during a longer session requires additional time-dependent evaluation. Similarly, Live Avatar's sampled windows under repeated audio support a specific extrapolation test, not every intervening instant or every changing-input session. Bounded state establishes a resource property, while the outputs establish the tested stability property.

7.4 Reusing Appearance Leaves Other Failure Paths

Rendering stores appearance separately from the motion and tracking stages that can introduce temporal errors. DyStream's anchor ablation evaluates head-pose drift across ten three-minute clips [22]; GaussianHeadTalk measures temporal wobble and illustrates mouth-detail errors under shorter reenactment protocols [1]. These studies examine different temporal failures. Identity, synchronization, and sustained service reliability require additional measurements.

The architectural deduction is that a persistent appearance representation narrows what must be regenerated but leaves the controller, tracker, deformation field, and compositor responsible for temporal behavior. The corresponding research hypothesis is that generated-history training and error-triggered refresh could improve sequential controllers without changing the appearance owner. This proposed transfer requires controlled testing. Figure 10 grounds the failure vocabulary in source-reported examples rather than schematic degradation.

8 Datasets and Evaluation

8.1 Datasets for Generation and Temporal Evaluation

A dataset determines which conditional relationships a model can learn and which claims an evaluation can support. For efficient virtual-human generation, the useful axes are therefore not venue and year, but the generated anatomy, conditioning modalities, identity and interaction structure, media properties, split policy, and continuous observation horizon. These axes expose task mismatch: a high-quality face-video corpus may support appearance learning without providing audio, a multi-view capture may support reconstruction without cross-identity generation, and a dyadic collection may represent turn taking without providing a single continuous session long enough to test drift.

Single-person resources span distinct purposes. HDTF provides in-the-wild high-resolution audiovisual portraits, whereas VFHQ is a video-face super-resolution corpus whose source paper does not establish retained or synchronized audio [206,243]. CelebV-HQ instead targets unconditional face-video generation and attribute editing, with manually annotated appearance, action, and emotion attributes but no source-defined audio-driving protocol [259]. MEAD adds controlled speech, synchronized views, discrete emotions, and intensity levels [191]. TalkVerse scales synchronized single-person audio–video curation and pose annotation, but its source clips remain short [201]. OpenHumanVid supplies large-scale human-centric video with caption, skeleton, and audio conditions for pretraining human video generators [106], and EchoMimicV2's half-body corpus extends conditioning to hand and gesture coverage [136]. These resources are complementary rather than interchangeable benchmarks for lip synchronization, identity, expression, or temporal behavior.

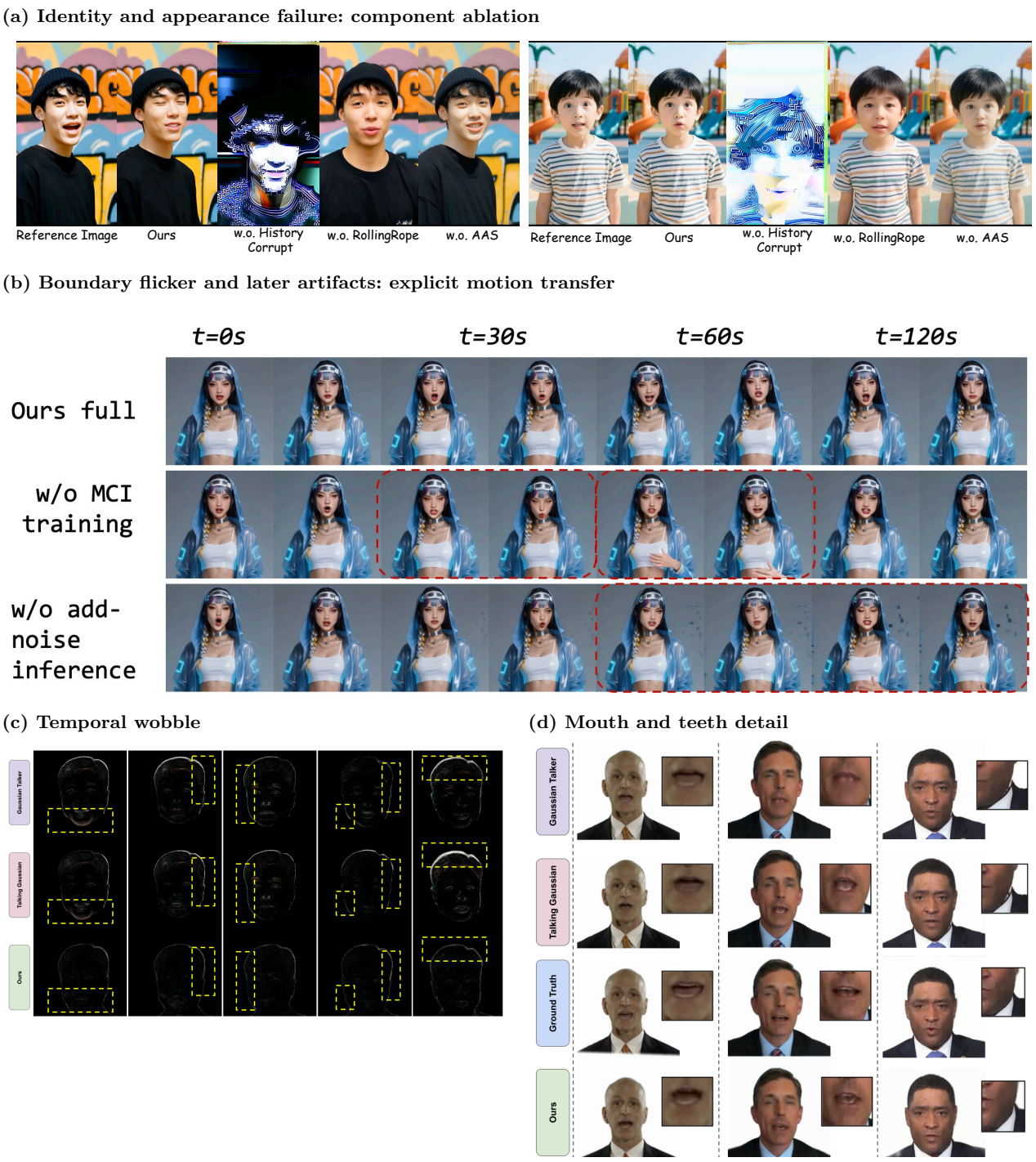


Fig. 10 Observed examples reproduced from primary papers; no examples were generated for this survey. (a) Live Avatar v6, Fig.4, reports identity/appearance degradation when components are removed; the displayed frames have no source timestamp [81]. (b) JoyStreamer-Flash, Fig.7, reports block-transition flicker without Motion Condition Injection and later artifacts without conditioning noise; the source labels 0, 30, 60, and 120 s [105]. Static samples illustrate the reported effects but cannot display flicker directly. (c,d) GaussianHeadTalk, Figs.3–4, shows image-difference wobble at ten-frame intervals and mouth-detail comparisons; absolute timestamps are NR [1]. Source labels and comparison methods are preserved. Suppression of an intended change by an anchor remains a separately proposed test, not an observed panel in this gallery.

Interaction and reconstruction data introduce different supervision. ViCo pairs a visible and audible speaker with a listener's non-verbal facial response and supplies both seen-identity and identity-disjoint test regimes [257]. ResponseNet adds synchronized speaker–listener video, separated audio, and word-level transcripts for online multimodal response generation [126]. SpeakerVid-5M organizes single-speaker clips into dialogue, listening, and multi-turn structures, but its pre-training/supervised fine-tuning (SFT) tiers are quality strata rather than a documented leakage-free train/test partition [238]. Seamless Interaction records long active segments from two participants with separate audio channels and participant-level partitions, while NeRSemble supplies synchronized high-resolution multi-view head capture for sequence-specific novel-view reconstruction [2,97]. For this survey's lenses, the relevant property of dyadic data is its *continuous-horizon* value: long uninterrupted sessions are exactly the material a stability protocol needs, independent of any interaction claim.

New interaction datasets make the preceding context part of the learning target. Face-to-Face pairs a partner's preceding observations with a subsequent audiovisual response, preserving natural reaction delays [32]. InteracVid organizes livestream material into context–query–response triplets; genuine live-chat queries and queries reconstructed from responses must remain distinguishable in evaluation [231]. These collections support context-sensitive behavior, while source-stream overlap, identity separation, and the origin of the query still determine what a test split measures.

The OmniHuman dataset and its OHBench benchmark broaden coverage to scene-level audiovisual quality, person–person and person–object relations, and individual fidelity [261]. This is distinct from the OmniHuman-1 generation method [116]. Its hierarchical evaluation helps localize failures that face-only metrics miss; benchmark clip scores still need to be complemented by continuous-session measurements when studying drift.

Table 15 distinguishes corpus scale from continuous observation length. Capture or normalized frame rates describe the data rather than generation throughput, and aggregate hours can combine many subjects, views, or short clips. The listed resources do not establish a matched hour-long continuous-generation benchmark. Some long-horizon evaluations therefore extend shorter inputs through repetition, as in Live Avatar [81], which limits the diversity of conditions tested. A public benchmark of continuous 10–60-minute single-subject and dyadic sessions would help measure drift under sustained, changing inputs.

Dataset governance is also part of evaluation validity. Web-derived collections report filtering and source-tracking procedures, but public availability of source videos does not by itself establish participant consent or redistribution rights [201, 206, 238, 257]. Controlled captures provide stronger acquisition records: NeRSemble reports General Data Protection Regulation (GDPR)-compliant participant agreements and deletion rights, and Seamless Interaction reports adult informed consent and compensation [2, 97]. Their primary papers nevertheless do not specify a complete dataset license. Reproducible reporting should therefore separate media provenance, consent, access terms, and technical split integrity rather than compressing them into a binary *public* field.

8.2 Evaluation Scope and Protocols

Evaluation must preserve the distinction introduced in Section 2.1: visual quality, identity fidelity, driving fidelity, temporal fidelity, and deployment behavior are different observables. A low image distance does not establish audiovisual synchronization, and high steady-state throughput does not establish low time to first frame (TTFF) or stable long-horizon operation. We therefore organize evaluation by the property being measured and the conditions under which the result is meaningful.

Four classes of evaluation measures are useful. *Full-reference measures* compare an output with a paired and aligned target. *Distributional measures* compare sets of generated and real samples. *Task-specific evaluators* estimate an attribute such as face identity or audiovisual synchronization. *System and longitudinal measures* characterize the visible pipeline boundary and change over an evaluated session. These classes complement rather than replace one another.

8.3 Reference-Based and Distributional Visual Quality

Pixel errors, peak signal-to-noise ratio (PSNR), and the structural similarity index (SSIM) require a paired reference. SSIM compares local luminance, contrast, and structure before spatial aggregation [200], and its relationship to PSNR is itself protocol-dependent [72]. Its original validation concerns aligned still images; a video-level report must therefore state the frame pairing and temporal aggregation, and it should not interpret frame-wise SSIM as a temporal-consistency result. Learned perceptual image patch similarity (LPIPS) instead aggregates channel-normalized deep-feature differences across layers [235]; ensemble variants improve its robustness to transformations [92]. A lower value means that the candidate is closer to the paired reference under the

Table 15 Properties of representative datasets. Media frame rate is a capture or preprocessing rate, not model throughput. Aggregate corpus duration is not a continuous generation horizon. NR denotes not reported in the primary paper.

Dataset	Task and modalities	Scale and media	Data partition	Continuous-horizon boundary
HDTF [243]	In-the-wild audiovisual talking face	362 videos; 15.8 h; 300+ subjects; 10k+ sentences; 720p/1080p	Fixed split and identity separation NR	Longest clip NR; 15.8 h is aggregate corpus time
VFHQ [206]	Video-face super-resolution; audio retention NR	16,827 clips from 7,228 source videos; roughly 700–1000 px faces; 512^2, $\times 4$ benchmark	50 test sequences exclude corresponding training clips; identity separation NR	Tracks retain 100–2000 frames, but frame rate and total hours are NR
CelebV-HQ [259]	Face-video generation and 83-attribute editing; audio NR	35,666 clips; 15,653 identities; at least 512^2; source reports approximately 65 h in text versus 68 h in a table	Fixed identity-disjoint split NR	Each clip is 3–20 s and is truncated at 20 s
MEAD [191]	Controlled emotional audiovisual talking faces; seven synchronized views	60 actors; 281,400 clips; 1920×1080 at a 30 FPS capture rate; 48-kHz audio; neutral plus seven emotions and three non-neutral intensities	Paper-specific 20% holdout; split unit and identity separation NR	38 min 57 s per speaker aggregates many short utterances, not one session
ViCo [257]	Dyadic speaker audio–video to listener facial response	483 pairs; 92 identities; 95 min 22 s; 384^2 face crops; source video 1080p at a 30 FPS capture rate and 44.1-kHz audio	D_{test} identities occur in training; D_{ood} is identity-disjoint	Clips span 1–71 s; no minute-scale drift protocol
Response Net [126]	Dyadic video, separated audio, word-level transcripts, and facial coefficients	696 pairs; 161 identities; 14.2 h; 1024^2 face streams	417/139/140 train/validation/test pairs; participant-pair disjointness stated; individual-identity disjointness NR	Clips span 27.13–863.13 s; data duration does not establish generated-session stability
SpeakerVid-5M [238]	Single, dialogue, listening, and multi-turn audiovisual organizations	Single branch: 5.2M clips/8.7k h; dialogue branch: 770k pairs/1.8k h; branches are not proven disjoint	Pretraining/SFT are quality tiers; VidChatBench has 500 unseen-speaker pairs, but global source/identity leakage control is NR	Basic clips span 3–14 s; stored order does not establish a continuous long-horizon sample
TalkVerse [201]	Single-person synchronized audio–video with 2D poses and structured annotations	2.3M clips; 6.3k h; 720p/1080p; unified to a 25 FPS content rate	Fixed identity-disjoint split NR	Source clips span 5–50 s; aggregate hours do not establish long-session samples
OpenHuman Vid [106]	Human-centric video with caption, skeleton, and audio conditions	13.2M clips reported by the source; large-scale pretraining corpus	Source-defined filtering tiers; identity-disjoint split NR	Clip-scale samples; not a continuous-session resource
NeRSemble [97]	Multi-view human-head capture and sequence-specific novel-view synthesis	222 participants; 4,734 sequences; 31.7M camera images; 7.5 h; 16 views at 3208×2200 and a 73 FPS capture rate	Dataset split NR; experiments use 12 input and four held-out views within each sequence	Roughly 3 min per participant is divided among 25 short sequences; longest sequence NR
Seamless Interaction [2]	Dyadic speech, face, upper-body, hand, transcript, and interaction metadata	4,065.04 h; 64,739 interactions; 5,098 sessions; 4,284 participants; dual 48-kHz audio and 2160×3840 video at a 30 FPS capture rate	Public train/dev/test plus private test; participant partition is functionally clique-level	Released active segments span 2–10 min; one-hour sessions include unreleased meta time; generation evaluation uses about 20 s

Abbreviations: SFT, supervised fine-tuning; D_{test} and D_{ood}, test and out-of-distribution subsets; 2D, two-dimensional; p, progressive-scan resolution; px, pixels; k, thousand; M, million; kHz, kilohertz; h, hours; min, minutes; s, seconds; dev, development; FPS, frames per second.

selected backbone and calibration variant. Comparisons must retain the same network, layer set, preprocessing, crop, dynamic range, and aggregation.

The Fréchet Inception Distance (FID) fits Gaussian statistics to pretrained Inception features from real and generated image sets and computes their squared Fréchet distance [68]. Because this construction discards sample order, FID is an image-distribution measure that does not measure temporal stability. Fréchet Video Distance (FVD) applies the same statistical form to spatiotemporal features; the original study selected Kinetics-400 RGB Inflated 3D ConvNet (I3D) logits and showed that finite-sample values depend on sample count [185]. Consequently, an FID or FVD number is comparable only after the reference set, feature network and checkpoint, activation layer, preprocessing, sample count, and, for FVD, clip construction and content frame rate have been fixed. Neither distance isolates identity, lip synchronization, or deployment speed. The same protocol dependence applies to similarity scores computed with other pretrained encoders, such as CLIP-based text or image similarity [154]: the encoder checkpoint and preprocessing are part of the metric's identity.

8.4 Human-Specific Fidelity

Human-specific evaluators require equally explicit boundaries. ArcFace is a face-recognition representation trained with an additive angular margin [46]; it is not itself a standardized virtual-human quality score. When a paper reports ArcFace-based cosine similarity (CSIM), the report should identify the embedding model and checkpoint, face detector and alignment, reference image or set, handling of failed detections, and frame or track aggregation. Recognition accuracy reported by the

Table 16 Properties and comparison settings of representative visual-quality measures. PSNR denotes peak signal-to-noise ratio; SSIM, structural similarity index; LPIPS, learned perceptual image patch similarity; FID, Fréchet Inception Distance; and FVD, Fréchet Video Distance. The arrows ↑ and ↓ indicate that higher and lower values, respectively, are preferable only within a matched protocol.

Measure	Comparison unit	Reference	Minimum fields to report	Does not by itself establish
PSNR / SSIM ↑	Aligned image or frame pair	Paired target	Crop and alignment, color space, dynamic range, resolution, implementation, frame aggregation	Temporal coherence, identity, lip sync, or system timing
LPIPS ↓	Aligned image or patch pair	Paired target	Backbone, calibrated variant, layer set, normalization, crop, resolution, aggregation	Attribute-specific fidelity or temporal behavior
FID ↓	Image sets	Real image set	Reference set, feature checkpoint/layer, preprocessing, sample count, random sampling	Sample order, temporal stability, identity, or lip sync
FVD ↓	Video-clip sets	Real clip set	Video encoder/checkpoint/layer, clip length and sampling, frame rate, preprocessing, sample count	Failure localization, long-horizon stability, or runtime

ArcFace source cannot be transferred into a threshold for generated-video identity fidelity.

For audiovisual synchronization, the underlying evaluator is SyncNet, which learned a joint audio–video embedding for lip-sync assessment in the wild [37]. Wav2Lip introduced lip-sync error distance (LSE-D; lower is better) and lip-sync error confidence (LSE-C; higher is better) on top of a pretrained SyncNet [150]; the evaluator differs from the synchronization expert used to train its generator. Cross-paper comparison therefore requires the same evaluator checkpoint, audio–video pairing, temporal window, face and mouth crop, preprocessing, and averaging rule. These scores test the selected audiovisual correspondence protocol; they do not measure expression quality, identity preservation, or latency.

Landmark Distance (LMD) was defined for paired lip-motion evaluation using Dlib detections on generated and ground-truth video [26]. The original construction uses 20 lip landmarks, aligns the mean landmark positions, computes corresponding Euclidean distances, and averages over points and frames; lower is better. LMD is thus a paired landmark-accuracy measure, not a no-reference audio–lip synchronization score. Detector version, landmark subset, crop and coordinate scale, alignment, frame correspondence, failure policy, and aggregation must remain fixed for numerical comparison.

Identity, synchronization, and landmark measures cover different failure modes. They should be paired with task-appropriate motion measures only when the underlying detector, normalization, reference, and aggregation are documented. A learned evaluator's name alone is not a reproducible protocol.

Temporal alignment changes the question asked by a metric. Temporally-Aligned Evaluation uses Soft-DTW to compare feature trajectories while allowing local temporal offsets [244]. This is useful when motion has several plausible phase alignments, but an alignment that tolerates delay should not replace unwarped synchronization or measured reaction latency. The feature extractor, allowed alignment, and aggregation must remain explicit.

Fine oral appearance requires a complementary reference. RGOR uses same-person enrollment frames and oral-region supervision to preserve teeth and mouth identity [181]. Its comparison of cross-clip and target-frame references shows why paired reconstruction can credit access to the answer. Separating enrollment from query clips and inspecting oral detail helps distinguish reference fidelity from lip synchronization or whole-face identity similarity.

8.5 Whole-Body, Interaction, and Cross-Scene Evaluation

Recent avatar systems expand the evaluation surface beyond face-only fidelity. Whole-body generation calls for separate face lip-sync, hand-keypoint consistency, gesture–audio alignment, body-motion diversity, face–body temporal coherence, and identity consistency across body regions. Listening-oriented systems additionally require response appropriateness, listener-motion diversity, speaking/listening transition smoothness, turn-taking responsiveness, and user preference; these should be reported separately from the latency of receiving the partner's signal. Cross-scene identity evaluation should distinguish same-scene retention from changes in background, lighting, framing, or viewpoint, because a persistent reference may preserve identity in one scene while failing under a context shift.

Temporal wobble and tracking stability form another independent axis. A render-based system can preserve a fixed identity asset while introducing jitter through tracking, deformation, Gaussian mapping, or compositing. Reports should therefore specify the temporal artifact detector, reference coordinate system, time bins, failed-detection policy, and aggregation, and should not substitute a single-frame identity score or renderer FPS

for a longitudinal wobble measurement. These dimensions are complementary to ArcFace, SyncNet, and LMD and should be stratified by anatomy, interaction mode, scene constancy, and failure type [1, 36, 101].

Human-centered preference models provide another complementary view. HVEval+ collects scores and pairwise preferences for spatial quality, temporal quality, and text–video correspondence; MoE-Rater predicts these dimensions [204]. Their separation avoids treating a visually sharp frame as proof of natural motion. Such clip-level ratings characterize the presented stimuli and do not automatically quantify deterioration over a long session.

A complete protocol should disclose which of these measurements are paired, no-reference, or human-rated, together with the continuous duration and reset policy. At present, the literature reports subsets under incompatible settings rather than a unified whole-body, dyadic, cross-scene benchmark.

8.6 System Timing

System evaluation follows the boundary in Section 2.7. At minimum, a speed claim should state hardware and device count, output resolution and content frame rate, precision, batch size or concurrency, warm-up, measurement window, and every included component. Frames per second (FPS) is a frame-throughput measure only when frames are counted over a disclosed wall-clock generation boundary; otherwise an FPS value may denote content or capture frame rate. Real-time factor (RTF) compares wall-clock time with generated-media duration; TTFF starts at a declared input-arrival event; and inter-chunk latency measures the time between successive committed chunks under a declared buffering policy. Number of function evaluations (NFE) records iterative model calls for a stated generated unit. FPS and RTF can be converted when frame count, media rate, elapsed interval, and boundary match (Section 2.7); neither determines TTFF or NFE. Measurements with different boundaries do not define a controlled cross-route ranking.

8.7 Common-Environment Runtime and Memory Evaluation

We plan to benchmark all open-source methods covered in this survey on the same machine with a controlled software environment. For each method and configuration, the comparison will record the total number of frames produced in one generation run, output resolution, preprocessing time, generation time, postprocessing time, and peak GPU memory. These measurements are pending. Runtime values elsewhere in the survey remain those reported in the original papers.

Generation length and resolution. Frames per run denotes the total completed output of one request, including all chunks for a chunked method. We will record the generated width and height and identify methods that synthesize a face region before compositing it into a larger canvas. Comparisons will group compatible tasks and inputs and use matching lengths and resolutions where supported; native settings will remain explicit when they differ.

Three timing stages. Preprocessing includes the input preparation and conditioning required for the request, such as face detection and alignment, audio-feature extraction, and reference encoding. Generation includes motion or visual-state prediction and the decoding or rendering needed to produce RGB frames. Postprocessing includes operations on those frames, such as paste-back, blending, and output conversion; inclusion of video encoding or file writing will be stated consistently. Model loading and subject-specific training or fitting are separate setup costs. Reused preparation is identified so that its cost is not silently omitted from a first-use comparison.

Runtime and memory settings. The experiment description will state the machine, software versions, precision, batch size, warm-up policy, repeated-run aggregation, and GPU-memory measurement method. Stage timers must account for asynchronous device execution. For sequential stages, their sum gives the processing time for the request; when stages overlap, elapsed request time is measured separately. Peak GPU memory covers the measured request with the model resident. Multiple-GPU configurations retain the device count and per-device memory use. Table entries and numerical comparisons will be added after the measurements are completed.

8.8 Longitudinal and Human Evaluation

Longitudinal evaluation measures how output quality and system behavior change during continuous generation. A reproducible protocol should specify session content and duration, initialization, retained context, and reset policy. Identity and appearance should be measured in predeclared time bins relative to an initial or short-horizon reference, alongside synchronization, motion, temporal artifacts, memory, and latency. Failures, resets, and recovery intervals require separate reporting

Table 17 System and temporal measurements used in virtual-human evaluation. FPS denotes frames per second; RTF, real-time factor; TTFF, time to first frame; NFE, number of function evaluations; and NR, not reported in the primary paper.

Record	Required fields	Interpretation boundary
Measured frame throughput (FPS)	Frames counted, wall-clock elapsed interval, resolution, batch or concurrency, warm-up, hardware, precision, included pipeline	Establishes FPS throughput only for the disclosed generation boundary; does not establish latency or content/capture frame rate
RTF	Wall-clock numerator, generated-duration denominator, clip or stream length, pipeline boundary	Values below one establish faster-than-playback only for that convention
TTFF	Input-arrival event, buffering, first observable frame	First-frame service metric; does not establish steady-state inter-chunk latency or throughput
Inter-chunk latency	Successive committed-output timestamps, commitment interval, buffering, jitter summary	Steady-state service metric; does not establish TTFF or throughput
NFE / serial calls	Generated unit, solver steps, denoiser or vector-field calls, guidance multiplier, decoder calls	Call count does not establish wall-clock speed
Memory	Peak GPU memory, measurement tool, loading/warm-up policy, frame count, resolution, batch size, and device count	Use the same memory definition across methods; distinguish inference from loading and subject-specific fitting
Stability protocol	Evaluated continuous duration, persistent state, context bound, time bins, identity and appearance drift variables, failure counts, reset policy, and resource trajectories	A long demonstration or aggregate dataset duration does not quantify temporal stability

because averages can conceal them. Ablating the active L1–L5 mechanisms under matched conditions would further distinguish their individual contributions from the behavior of the complete system.

We recommend human studies when automatic measures do not isolate visible artifacts or preference. Reports should state the question, stimuli, duration, and presentation order. They should also report rater recruitment, repeated judgments, aggregation, uncertainty, and the statistical test. A preference percentage without these fields is descriptive rather than a transferable quality scale.

8.9 Matching Tasks to Evaluation Data

The evaluation data should reflect the task and input conditions of each method. Audio-driven portraits need synchronized speech and face video; video-driven reenactment needs motion targets; subject-specific renderers require enrollment data; and body-generation tasks need measurements beyond the face. Table 18 summarizes these relationships. Continuous-session evaluation additionally needs a declared duration and input sequence, with repeated and varied inputs analyzed separately.

9 Discussion, Challenges, and Future Directions

The four routes offer different ways to distribute appearance, motion, and temporal state across a generation pipeline. Their practical value depends on the task, available resources, and evaluation conditions. The methods reviewed in Sections 3–8 motivate four common research problems: composing modules with explicit interfaces, measuring complete-system cost, maintaining consistency with bounded state, and evaluating deployment safeguards.

The synthesis in Section 7 separates measured effects from architectural deductions. The following questions build on those findings.

9.1 Composition Requires Explicit Interfaces

Hybrid systems often divide work among a condition or motion model, a visual representation owner, a decoder or renderer, and a compositor. Under the ownership rule in Section 2.3, the route label follows the module that produces or persistently stores dense visible content [22, 64]. Secondary modules remain part of the system boundary. Replacing one module can change motion control without changing the visual owner; it can also reduce visual-model calls while leaving condition encoding, decoding, and transport on the critical path.

Composition requires both a representation interface and a state-update policy. Each module should specify its coordinate system, identity reference, subject-specific assets, retained history, reset behavior, and recovery procedure. These fields determine when output becomes available and which errors can propagate. For example, Teller's audio buffering precedes motion prediction and rendering, while LivePortrait's reused appearance still requires motion estimation and decoding [62, 251].

Compact motion interfaces can reduce the state presented to a dense visual owner, but they also restrict appearance changes to those expressible by the interface. Dense latent interfaces offer more freedom while making identity ownership and failure localization harder to analyze. These are testable architectural trade-offs rather than universal advantages of a route. A falsifiable research question is whether a hybrid can expose typed representation, state, and recovery contracts such that the latency and failure contribution of every module is reproducible in isolation and in composition.

Table 18 Example task–dataset pairings and suitable measurements. Continuous-session tests require sufficiently long inputs, and whole-body tasks extend evaluation beyond face-only measures.

Task	Candidate source material	Measurements and limits
Audio-driven portrait	HDTF; MEAD	Identity, lip synchronization, expression/control fidelity, artifact ratings, and continuous service traces. Paired image metrics require aligned targets.
Video-driven reenactment	VFHQ; CelebV-HQ	Pose/landmark target error, identity, temporal artifacts, and disocclusion. Unpaired cross-identity reenactment lacks a pixel-matched RGB target.
Subject-specific rendering	NeRSemble; declared subject capture	Novel-view/reenactment quality, tracking/deformation wobble, capture and fitting cost. Multi-view capture is an enrollment condition.
Selected upper/full-body animation	OpenHumanVid; task-specific video/pose sequences	Face synchronization, body/hand keypoint fidelity where annotations permit, gesture–audio correspondence, and face–body consistency. Face identity scores do not cover hands or body motion.

9.2 Complete-System Cost and Timing

The source literature reports FPS, wall-clock latency, real-time factor, function evaluations, and relative speedups under different boundaries. The route-specific tables identify the settings and measured components of each reported result. A renderer-only measurement is not an end-to-end system result [62, 81]. The planned common-environment comparison in Section 8.7 will measure frames per run, output resolution, preprocessing, generation and postprocessing time, and peak GPU memory for the open-source methods. Its numerical results will be incorporated after the runs are completed.

A deployment record should account separately for subject enrollment or asset preparation. The runtime breakdown should time condition acquisition and encoding, motion or state generation, dense visual synthesis, decoding or rendering, compositing, transfer, and I/O. User-visible measures include time to first frame, intercommit latency and jitter, steady throughput, memory, and recovery cost. The critical path moves with the operating point. The within-source analyses in Section 7 illustrate two consequences: decoder choice can change both throughput and output quality, and multi-device pipelining can raise throughput while increasing the wait for the first committed output. Component timings are needed to establish which stage is the bottleneck. A falsifiable research question is which component owns the critical path at each resolution, chunk size, concurrency level, and session duration, and whether the same conclusion survives when enrollment, I/O, and recovery are included.

9.3 Bounded-State Long-Horizon Generation

The five patterns in Section 2.8 provide a common vocabulary for mechanisms whose evaluation remains uneven. The reviewed diffusion systems contain examples of all five, including Live Avatar's sampled observations beyond an hour under looped audio [81]. Render-based systems reuse an identity asset but can drift through their controllers or rendering pipeline; DyStream's 180-s pose-drift ablation provides a limited longitudinal example [22]. Native visual AR explicitly retains history, yet the reviewed systems lack a complete joint stability protocol. Framewise GAN generators avoid recursive visual feedback when earlier outputs are not reused, but can still exhibit jitter or inherit controller errors.

This comparison suggests studying which mechanisms transfer across representations. Training with generated histories could be evaluated in both native visual AR and recurrent motion controllers. The anchor-regulated updates of TaoMate and the selection policy of InteractiveAvatar offer concrete alternatives to fixed windows [56, 173]. Comparisons with compression and resets should now test recovery after corrupted history and requested reference changes. DynaForcing and Routed Forcing also motivate testing motion amplitude and diversity alongside appearance consistency [82, 175]. A matched study should isolate each mechanism's effect on identity, synchronization, motion diversity, and legitimate scene changes over predeclared time bins. The reviewed studies cover different tasks and durations, leaving sustained operation under varied conditions an open challenge.

Audio-to-deformation and expanded Stable axes. Recent audio-driven Gaussian systems expose a shift from the conventional audio → pose or SMPL → renderer cascade toward direct audio → Gaussian or particle deformation → renderer paths. This can preserve high-frequency facial, hand, and body dynamics that an intermediate pose interface may quantize, but it moves complexity into learned deformation fields and their tracking boundary. The resulting Stable problem is multidimensional: appearance identity, motion and deformation, audio–visual synchronization, interaction response, and system/resource behavior can fail independently. Generalizable Gaussian priors reduce enrollment cost but

introduce adaptation and out-of-distribution trade-offs; fixed Gaussian scenes anchor appearance but do not prevent wobble, tracking failure, or compositing drift. These systems motivate joint evaluation of enrollment and runtime costs with time-binned identity, motion, synchronization, wobble, failure, and recovery measurements [1, 4, 74, 101, 115].

9.4 Route Selection Under Deployment Constraints

Table 19 summarizes overlapping suitability conditions. The design guidance of Sections 3–6 supports a first set of conditional guidelines. These summarize documented design options under stated constraints; they do not rank individual methods, and each guideline inherits the measurement caveats of its underlying sources.

When per-subject enrollment is acceptable, render-based systems offer reusable appearance and reported high-throughput or wearable configurations [25, 205]. Selection should account for capture and fitting cost, followed by tracking, deformation, and compositing over the intended session duration.

When one-shot operation is required, compact motion generation over reused appearance provides a practical design pattern [62, 67, 209]. The motion module and renderer must both fit the latency budget, while deformation and disocclusion determine which movements the appearance model can reproduce.

When changes extend beyond a reusable appearance model, dense diffusion offers mechanisms for synthesizing broader visual content. The reviewed streaming examples combine distillation, memory, and substantial serving resources [81, 165]. Persistent anchors can also restrict scene changes, so the intended variability must be evaluated directly. Component profiling should determine whether further optimization targets the sampler, decoder, or conditioning path.

When the edit is confined to a small region, feed-forward GAN synthesis can reduce the amount of generated content [150, 239]. Evaluation should include compositing seams, synchronization, and framewise jitter as well as runtime.

When explicit visual-history modeling is central, native visual AR offers a direct formulation. Its deployment evaluation remains limited in the reviewed studies: Table 9 does not contain a system with both full-pipeline timing and a joint time-binned stability evaluation. The representation, local emission head, and state policy should therefore be validated for the intended application.

9.5 Trustworthy Deployment

Fast and Stable systems sharpen deployment risks: low-latency synthesis enables live impersonation, and long-horizon stability with cached identity state extends it. Trustworthiness is not an additional scalar metric. The NIST AI 100-4 report separates provenance-data tracking from synthetic-content detection and discusses watermarks, metadata, filtering, and red-teaming as controls with different lifecycle positions and failure surfaces [20]; it provides no avatar-specific effectiveness or runtime measurement. The C2PA specification defines signed claims, content bindings, and validation states, including segment-level paths for live video [40]. These mechanisms bind a signing credential to covered media bytes and expose specified integrity failures; they do not establish factual truth, verify the depicted identity, or prove authorization to use a likeness, and they carry manifest-stripping, key-management, and privacy risks [38, 39].

Empirical safeguards have equally important boundaries. DeepfakeBench exposes generalization gaps across manipulation types in frame-level evaluation [213], and Video Seal demonstrates faster-than-playback video watermarking under a stated offline boundary without adaptive-attack coverage [55]. A watermark hit, a detector score, and a valid provenance record are different observations; none alone proves authorization or harmlessness. A deployment evaluation should declare the attacker model, benign platform transformations, operating point, authorization and revocation procedures, and the latency cost of each control on the serving path. The last point connects trust directly to this survey's Fast accounting, since provenance signing and watermarking must fit inside the same per-chunk budget the route mechanisms optimize.

9.6 Conclusion

Fast and Stable virtual-human generation depends on how appearance, motion, and temporal state are represented and computed. GAN methods reduce repeated synthesis through feed-forward and regional generation; diffusion methods combine compressed representations, fewer sampling calls, and computation reuse; visual autoregression organizes evolving appearance through explicit history; and render-based methods reuse identity-specific appearance under changing controls. Across these routes, decoder choice, execution schedules, and state management connect efficiency with output quality. Persistent appearance and bounded memory help organize generation, while tracking, control, and temporal

Table 19 Requirements that guide method selection. Suitability depends on the intended task, available resources, and supported controls.

Requirement	Candidate mechanism	Additional condition to check
Regional lip editing	Wav2Lip+GAN, MuseTalk	Source pixels remain usable; inspect seams, synchronization, and jitter.
Reusable identity and controllable pose	Fields, Gaussians, feature warps	Enrollment is feasible *and* the representation supports the intended appearance, views, and motion.
One-shot portrait animation	LivePortrait, VASA-1, LAM	Test disocclusion, pose limits, and complete controller–renderer timing.
Appearance or scene changes	Dense visual generation	Validate the specific change. A static-scene system such as Live Avatar does not establish open-ended scene editing.
Explicit visual-history control	Native visual AR	Profile local sampling and decoding; test long-rollout errors and resets.

feedback introduce their own failure modes. Complete-pipeline runtime comparison and sustained evaluation under varied inputs remain central to understanding these trade-offs.

Data Availability Statement

This survey reviews published methods and datasets. The common-environment runtime and memory measurements described in Section 8.7 are pending. The public research companion provides a searchable literature catalogue.

References

1. Agarwal, M., Zhang, M., Sevilla-Lara, L., McDonagh, S.: GaussianHeadTalk: Wobble-Free 3D Talking Heads with Audio Driven Gaussian Splatting. In: WACV, pp. 8017–8027 (2026). DOI 10.1109/WACV61042.2026.00774. URL https://doi.org/10.1109/WACV61042.2026.00774 32, 35, 37, 38, 42, 45
2. Agrawal, V., Akinyemi, A., Alvero, K., Behrooz, M., Buffalini, J., Carlucci, F.M., Chen, J., Chen, J., Chen, Z., Cheng, S., Chowdary, P., Chuang, J., D'Avirro, A., Daly, J., Dong, N., Duppenthaler, M., Gao, C., Girard, J.M., Gleize, M., Gomez, S., Gong, H., Govindarajan, S., Han, B., He, S., Hernandez, D., Hristov, Y., Huang, R., Inaguma, H., Jain, S., Janardhan, R., Jia, Q., Klaiber, C., Kovachev, D., Kumar, M., Li, H., Li, Y., Litvin, P., Liu, W., Ma, G., Ma, J., Ma, M., Ma, X., Mantovani, L., Miglani, S., Mohan, S., Morency, L.P., Ng, E., Ng, K.W., Nguyen, T.A., Oberai, A., Peloquin, B.N., Pino, J., Popovic, J., Poursaeed, O., Prada, F., Rakotoarison, A., Ranjan, R., Richard, A., Ropers, C., Saleem, S., Sharma, V., Shcherbyna, A., Shen, J., Shen, J., Stathopoulos, A., Sun, A.Y., Tomasello, P., Tran, T., Turkatenko, A., Wan, B., Wang, C., Wang, J., Williamson, M., Wood, C., Xiang, T., Yang, Y., Yao, J., Zhang, C., Zhang, J., Zhang, X., Zheng, J., Zhyzheria, P., Zikes, J., Zollhöfer, M.: Seamless Interaction: Dyadic Audiovisual Motion Modeling and Large-Scale Dataset. CoRR **abs/2506.22554** (2025). DOI 10.48550/ARXIV.2506.22554. URL https://doi.org/10.48550/ARXIV.2506.22554 39, 40
3. An, Y., Wang, R., Wei, W., Bu, R., Wang, Q., Hu, H., Cheng, Z., Xie, R., Song, L., Zhang, W.: Decoupled Self-Forcing Distillation for Streaming Talking Head Generation. CoRR **abs/2609.10317** (2026). DOI 10.48550/ARXIV.2609.10317. URL https://doi.org/10.48550/ARXIV.2609.10317 19
4. Aneja, S., Sevastopolsky, A., Kirschstein, T., Thies, J., Dai, A., Nießner, M.: GaussianSpeech: Audio-Driven Personalized 3D Gaussian Avatars. In: ICCV, pp. 13065–13075 (2025). DOI 10.1109/ICCV51701.2025.01214. URL https://doi.org/10.1109/ICCV51701.2025.01214 32, 45
5. Ao, T.: Body of Her: A Preliminary Study on End-to-End Humanoid Agent. CoRR **abs/2408.02879** (2024). DOI 10.48550/ARXIV.2408.02879. URL https://doi.org/10.48550/ARXIV.2408.02879 2, 4, 9, 25, 26, 27, 28
6. Athar, S., Shu, Z., Samaras, D.: FLAME-in-NeRF: Neural control of Radiance Fields for Free View Face Animation. In: FG, pp. 1–8 (2023). DOI 10.1109/FG57933.2023.10042553. URL https://doi.org/10.1109/FG57933.2023.10042553 30
7. Azari, B., Lim, A.: EmoStyle: One-Shot Facial Expression Editing Using Continuous Emotion Parameters. In: WACV, pp. 6373–6382 (2024). DOI 10.1109/WACV57701.2024.00626. URL https://doi.org/10.1109/WACV57701.2024.00626 14
8. Bai, L., Zhang, T., Shao, S., Tan, D., Zhong, Q., Xie, Z., Li, H., Huang, Q., Shen, D., Ji, T., Wang, W., Wu, P., Zhao, Y., Zhu, X., Luo, W., Yang, S., Xie, Z.: MaineCoon: Pursuing A Real-Time Audio-Visual Social World Model. CoRR **abs/2606.17800** (2026). DOI 10.48550/ARXIV.2606.17800. URL https://doi.org/10.48550/ARXIV.2606.17800 3
9. Blanz, V., Vetter, T.: A Morphable Model for the Synthesis of 3D Faces. In: SIGGRAPH, pp. 187–194 (1999). URL https://dl.acm.org/citation.cfm?id=311556 30
10. Blattmann, A., Dockhorn, T., Kulal, S., Mendelevitch, D., Kilian, M., Lorenz, D., Levi, Y., English, Z., Voleti, V., Letts, A., Jampani, V., Rombach, R.: Stable Video Diffusion: Scaling Latent Video Diffusion Models to Large Datasets. CoRR **abs/2311.15127** (2023). DOI 10.48550/ARXIV.2311.15127. URL https://doi.org/10.48550/ARXIV.2311.15127 19
11. Bounareli, S., Tzelepis, C., Argyriou, V., Patras, I., Tzimiropoulos, G.: HyperReenact: One-Shot Reenactment via Jointly Learning to Refine and Retarget Faces. In: ICCV, pp. 7115–7125 (2023). DOI 10.1109/ICCV51070.2023.00657. URL https://doi.org/10.1109/ICCV51070.2023.00657 14
12. Bounareli, S., Tzelepis, C., Argyriou, V., Patras, I., Tzimiropoulos, G.: StyleMask: Disentangling the Style Space

of StyleGAN2 for Neural Face Reenactment. In: FG, pp. 1–8 (2023). DOI 10.1109/FG57933.2023.10042744. URL https://doi.org/10.1109/FG57933.2023.10042744 14
13. Bounareli, S., Tzelepis, C., Argyriou, V., Patras, I., Tzimiropoulos, G.: One-Shot Neural Face Reenactment via Finding Directions in GAN's Latent Space. Int. J. Comput. Vis. **132**, 3324–3354 (2024). DOI 10.1007/S11263-024-02018-6. URL https://doi.org/10.1007/S11263-024-02018-6 14
14. Brown, T.B., Mann, B., Ryder, N., Subbiah, M., Kaplan, J., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., Agarwal, S., Herbert-Voss, A., Krueger, G., Henighan, T., Child, R., Ramesh, A., Ziegler, D.M., Wu, J., Winter, C., Hesse, C., Chen, M., Sigler, E., Litwin, M., Gray, S., Chess, B., Clark, J., Berner, C., McCandlish, S., Radford, A., Sutskever, I., Amodei, D.: Language Models are Few-Shot Learners. In: NeurIPS (2020). URL https://proceedings.neurips.cc/paper/2020/hash/1457c0d6bfcb4967418bfb8ac142f64a-Abstract.html 9
15. Cai, Y., Chu, X., Gao, X., Gong, S., Huang, Y., Kang, C., Li, K., Liu, H., Liu, R., Liu, Y., Ng, D., Su, Z., Wu, E., Wu, Y., Yan, D., Yan, T., Zeng, C., Zheng, B., Zhou, Y.: Towards Interactive Intelligence for Digital Humans. CoRR **abs/2512.13674** (2025). DOI 10.48550/ARXIV.2512.13674. URL https://doi.org/10.48550/ARXIV.2512.13674 3
16. Cai, Y., Wu, Y., Li, K., Zhou, Y., Zheng, B., Liu, H.: FloodDiffusion: Tailored Diffusion Forcing for Streaming Motion Generation. In: CVPR, pp. 2295–2304 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Cai_FloodDiffusion_Tailored_Diffusion_Forcing_for_Streaming_Motion_Generation_CVPR_2026_paper.html 17, 18
17. Canfes, Z., Atasoy, M.F., Dirik, A., Yanardag, P.: Text and Image Guided 3D Avatar Generation and Manipulation. In: WACV, pp. 4410–4420 (2023). DOI 10.1109/WACV56688.2023.00440. URL https://doi.org/10.1109/WACV56688.2023.00440 14
18. Cao, X., Wang, G., Shi, S., Zhao, J., Yao, Y., Fei, J., Gao, M.: JoyVASA: Portrait and Animal Image Animation with Diffusion-Based Audio-Driven Facial Dynamics and Head Motion Generation. CoRR **abs/2411.09209** (2024). DOI 10.48550/ARXIV.2411.09209. URL https://doi.org/10.48550/ARXIV.2411.09209 34
19. Cao, Z., Hidalgo, G., Simon, T., Wei, S.E., Sheikh, Y.: OpenPose: Realtime Multi-Person 2D Pose Estimation Using Part Affinity Fields. IEEE Trans. Pattern Anal. Mach. Intell. **43**, 172–186 (2021). DOI 10.1109/TPAMI.2019.2929257. URL https://doi.org/10.1109/TPAMI.2019.2929257 10
20. Chandra, B., Dunietz, J., Roberts, K., Lee, Y., Fontana, P., Awad, G.: Reducing risks posed by synthetic content: An overview of technical approaches to digital content transparency. NIST Trustworthy and Responsible AI NIST AI 100-4, National Institute of Standards and Technology (2024). DOI 10.6028/NIST.AI.100-4 45
21. Chang, Q., Ding, Y.X., Zhou, K.: Enhancing Identity-Deformation Disentanglement in StyleGAN for One-Shot Face Video Re-Enactment. In: AAAI, pp. 1247–1255 (2025). DOI 10.1609/AAAI.V39I2.32113. URL https://doi.org/10.1609/AAAI.V39I2.32113 14
22. Chen, B., Liu, H.: DyStream: Streaming Dyadic Talking Heads Generation via Flow Matching-based Autoregressive Model. CoRR **abs/2512.24408** (2025). DOI 10.48550/ARXIV.2512.24408. URL https://doi.org/10.48550/ARXIV.2512.24408 10, 27, 28, 29, 34, 35, 37, 43, 44
23. Chen, B., Monso, D.M., Du, Y., Simchowitz, M., Tedrake, R., Sitzmann, V.: Diffusion Forcing: Next-token Prediction Meets Full-Sequence Diffusion. In: NeurIPS (2024). URL http://papers.nips.cc/paper_files/paper/2024/hash/2aee1c4159e48407d68fe16ae8e6e49e-Abstract-Conference.html 22
24. Chen, J., Cai, H., Chen, J., Xie, E., Yang, S., Tang, H., Li, M., Han, S.: Deep Compression Autoencoder for Efficient High-Resolution Diffusion Models. In: ICLR (2025). URL https://openreview.net/forum?id=wH8XXUOUZU 9, 26, 27
25. Chen, J., Hu, J., Wang, G., Jiang, Z., Zhou, T., Chen, Z., Lv, C.: TaoAvatar: Real-Time Lifelike Full-Body Talking Avatars for Augmented Reality via 3D Gaussian Splatting. In: CVPR, pp. 10723–10734 (2025). DOI 10.1109/CVPR52734.2025.01002. URL https://openaccess.thecvf.com/content/CVPR2025/html/Chen_TaoAvatar_Real-Time_Lifelike_Full-Body_Talking_Avatars_for_Augmented_Reality_via_CVPR_2025_paper.html 32, 35, 45
26. Chen, L., Li, Z., Maddox, R.K., Duan, Z., Xu, C.: Lip Movements Generation at a Glance. In: ECCV (7), pp. 538–553 (2018). DOI 10.1007/978-3-030-01234-2_32. URL https://doi.org/10.1007/978-3-030-01234-2_32 41
27. Chen, M., Cui, L., Zhang, W., Zhang, H., Zhou, Y., Li, X., Tang, S., Liu, J., Liao, B., Chen, H., Liu, X., Wan, P.: MIDAS: Multimodal Interactive Digital-humAn Synthesis via Real-time Autoregressive Video Generation. CoRR **abs/2508.19320** (2025). DOI 10.48550/ARXIV.2508.19320. URL https://doi.org/10.48550/ARXIV.2508.19320 1, 2, 4, 9, 10, 12, 25, 26, 27, 28
28. Cheng, K., Cun, X., Zhang, Y., Xia, M., Yin, F., Zhu, M., Wang, X., Wang, J., Wang, N.: VideoReTalking: Audio-based Lip Synchronization for Talking Head Video Editing In the Wild. In: SIGGRAPH Asia, pp. 30:1–30:9 (2022). DOI 10.1145/3550469.3555399. URL https://doi.org/10.1145/3550469.3555399 14
29. Chern, E., Hu, Z., Tang, B., Su, J., Chern, S., Deng, Z., Liu, P.: LiveTalk: Real-Time Multimodal Interactive Video Diffusion via Improved On-Policy Distillation. CoRR **abs/2512.23576** (2025). DOI 10.48550/ARXIV.2512.23576. URL https://doi.org/10.48550/ARXIV.2512.23576 21, 22, 25
30. Cho, K., Lee, J., Yoon, H., Hong, Y., Ko, J., Ahn, S., Kim, S.: GaussianTalker: Real-Time High-Fidelity Talking Head Synthesis with Audio-Driven 3D Gaussian Splatting. CoRR **abs/2404.16012** (2024). DOI 10.48550/ARXIV.2404.16012. URL https://doi.org/10.48550/ARXIV.2404.16012 32
31. Cho, P.H., Jang, J., Lee, S., Lee, J., Jin, S., Shin, H., Yi, J., Park, Y., Park, C., Kim, S.: Lip Forcing: Few-Step Autoregressive Diffusion for Real-time Lip Synchronization. CoRR **abs/2606.11180** (2026). DOI 10.48550/ARXIV.2606.11180. URL https://doi.org/10.48550/ARXIV.2606.11180 21, 22, 23, 25
32. Chu, E., Patel, V.M.: Face-to-Face: A Video Dataset for Multi-Person Interaction Modeling. CoRR **abs/2603.14794** (2026). DOI 10.48550/ARXIV.2603.14794. URL https://doi.org/10.48550/ARXIV.2603.14794 39
33. Chu, H., Ma, S., la Torre, F.D., Fidler, S., Sheikh, Y.: Expressive Telepresence via Modular Codec Avatars. In: ECCV (12), pp. 330–345 (2020). DOI 10.1007/978-3-030-58610-2_20. URL https://doi.org/10.1007/978-3-030-58610-2_20 30
34. Chu, X., Harada, T.: Generalizable and Animatable Gaussian Head Avatar. In: NeurIPS (2024). URL http://papers.nips.cc/paper_files/paper/2024/hash/6a14c7f9f

b3f42645cfa6bd5aa446819-Abstract-Conference.html 33, 34, 35
35. Chu, X., Li, Y., Zeng, A., Yang, T., Lin, L., Liu, Y., Harada, T.: GPAvatar: Generalizable and Precise Head Avatar from Image(s). In: ICLR (2024). URL https://openreview.net/forum?id=hgehGq2bDv 33
36. Chu, X., Liu, R., Huang, Y., Liu, Y., Peng, Y., Zheng, B.: UniLS: End-to-End Audio-Driven Avatars for Unified Listening and Speaking. In: CVPR, pp. 25142–25152 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Chu_UniLS_End-to-End_Audio-Driven_Avatars_for_Unified_Listening_and_Speaking_CVPR_2026_paper.html 27, 42
37. Chung, J.S., Zisserman, A.: Out of Time: Automated Lip Sync in the Wild. In: ACCV Workshops (2), pp. 251–263 (2016). DOI 10.1007/978-3-319-54427-4_19. URL https://doi.org/10.1007/978-3-319-54427-4_19 41
38. Coalition for Content Provenance and Authenticity: C2PA harms modelling. Technical Report Version 2.0, Coalition for Content Provenance and Authenticity (2026). Release 2026-03-30; https://spec.c2pa.org/specifications/specifications/2.4/security/Harms_Modelling.html 45
39. Coalition for Content Provenance and Authenticity: C2PA security considerations. Technical Report Version 2.4, Coalition for Content Provenance and Authenticity (2026). Release 2026-03-30; https://spec.c2pa.org/specifications/specifications/2.4/security/Security_Considerations.html 45
40. Coalition for Content Provenance and Authenticity: C2PA technical specification. Technical Specification Version 2.4, Coalition for Content Provenance and Authenticity (2026). Release 2026-04-01; https://spec.c2pa.org/specifications/specifications/2.4/specs/C2PA_Specification.html 45
41. Cui, J., Li, H., Yao, Y., Zhu, H., Shang, H., Cheng, K., Zhou, H., Zhu, S., Wang, J.: Hallo2: Long-Duration and High-Resolution Audio-Driven Portrait Image Animation. In: ICLR (2025). URL https://openreview.net/forum?id=rkzabmWl5k 12, 25
42. Cui, J., Wu, J., Li, M., Yang, T., Li, X., Wang, R., Bai, A., Ban, Y., Hsieh, C.J.: Self-Forcing++: Towards Minute-Scale High-Quality Video Generation. In: ICLR (2026). URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/8a30aba6514b56d02976f49797f6338a-Abstract-Conference.html 23
43. Cui, L., Hu, W., Zhang, W., Yang, Z., Shi, F., Liu, X.: AvatarForcing: One-Step Streaming Talking Avatars via Local-Future Sliding-Window Denoising. CoRR **abs/2603.14331** (2026). DOI 10.48550/ARXIV.2603.14331. URL https://doi.org/10.48550/ARXIV.2603.14331 20
44. Dao, T.: FlashAttention-2: Faster Attention with Better Parallelism and Work Partitioning. In: ICLR (2024). URL https://openreview.net/forum?id=mZn2Xyh9Ec 11
45. Dao, T., Fu, D.Y., Ermon, S., Rudra, A., Ré, C.: FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness. In: NeurIPS (2022). URL http://papers.nips.cc/paper_files/paper/2022/hash/67d57c32e20fd0a7a302cb81d36e40d5-Abstract-Conference.html 11
46. Deng, J., Guo, J., Xue, N., Zafeiriou, S.: ArcFace: Additive Angular Margin Loss for Deep Face Recognition. In: CVPR, pp. 4690–4699 (2019). DOI 10.1109/CVPR.2019.00482. URL http://openaccess.thecvf.com/content_CVPR_2019/html/Deng_ArcFace_Additive_Angular_Margin_Loss_for_Deep_Face_Recognition_CVPR_2019_paper.html 40
47. Deng, K., Guo, Y., Shen, L.: FlowTalk: Real-Time Audio-Driven Talking Head Synthesis via Motion-Space Flow Matching. In: MMAsia Workshops, pp. 23:1–23:6 (2025). DOI 10.1145/3769748.3773363. URL https://doi.org/10.1145/3769748.3773363 33
48. Deng, X., Zheng, Z., Zhang, Y., Sun, J., Xu, C., Yang, X., Wang, L., Liu, Y.: RAM-Avatar: Real-time Photo-Realistic Avatar from Monocular Videos with Full-body Control. In: CVPR, pp. 1996–2007 (2024). DOI 10.1109/CVPR52733.2024.00195. URL https://doi.org/10.1109/CVPR52733.2024.00195 15
49. Dey, A., Alcoverro, M., Zabaleta, I., Fernández, C.: Interactive real-time visual avatar generation: A survey. Unpublished manuscript, Unith Research Lab (2026) 3, 4
50. Dhamo, H., Nie, Y., Moreau, A., Song, J., Shaw, R., Zhou, Y., Pérez-Pellitero, E.: HeadGaS: Real-Time Animatable Head Avatars via 3D Gaussian Splatting. In: ECCV (2), pp. 459–476 (2024). DOI 10.1007/978-3-031-72627-9_26. URL https://doi.org/10.1007/978-3-031-72627-9_26 32
51. Du, F., Li, T., Zhang, Z., Qiao, Q., Yu, T., Zhen, D., Jia, X., Yang, Y., Yin, S., Liu, S.: RAP: Real-time Audio-driven Portrait Animation with Video Diffusion Transformer. CoRR **abs/2508.05115** (2025). DOI 10.48550/ARXIV.2508.05115. URL https://doi.org/10.48550/ARXIV.2508.05115 17, 19, 23
52. Ekman, P., Friesen, W.V.: Facial action coding system. Environmental Psychology & Nonverbal Behavior (1978) 10
53. Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., Levi, Y., Lorenz, D., Sauer, A., Boesel, F., Podell, D., Dockhorn, T., English, Z., Rombach, R.: Scaling Rectified Flow Transformers for High-Resolution Image Synthesis. In: ICML, pp. 12606–12633 (2024). URL https://proceedings.mlr.press/v235/esser24a.html 9, 17
54. Fan, Y., Lin, Z., Saito, J., Wang, W., Komura, T.: FaceFormer: Speech-Driven 3D Facial Animation with Transformers. In: CVPR, pp. 18749–18758 (2022). DOI 10.1109/CVPR52688.2022.01821. URL https://doi.org/10.1109/CVPR52688.2022.01821 34
55. Fernandez, P., Elsahar, H., Yalniz, I.Z., Mourachko, A.: Video Seal: Open and Efficient Video Watermarking. CoRR **abs/2412.09492** (2024). DOI 10.48550/ARXIV.2412.09492. URL https://doi.org/10.48550/ARXIV.2412.09492 45
56. Gan, Q., Zhang, C., Jin, M., Ma, J., Shen, Q.: TaoMate: Anchor-Guided Memory Bridging Evolving and Reference States for Real-Time Audio-Video Digital Human Generation. CoRR **abs/2607.24359** (2026). DOI 10.48550/ARXIV.2607.24359. URL https://doi.org/10.48550/ARXIV.2607.24359 24, 25, 44
57. Gan, Y., Yang, Z., Yue, X., Sun, L., Yang, Y.: Efficient Emotional Adaptation for Audio-Driven Talking-Head Generation. In: ICCV, pp. 22577–22588 (2023). DOI 10.1109/ICCV51070.2023.02069. URL https://doi.org/10.1109/ICCV51070.2023.02069 14
58. Goodfellow, I.J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A.C., Bengio, Y.: Generative adversarial networks. Commun. ACM **63**, 139–144 (2020). DOI 10.1145/3422622. URL https://doi.org/10.1145/3422622 4, 8, 13
59. Gu, A., Dao, T.: Mamba: Linear-Time Sequence Modeling with Selective State Spaces. CoRR **abs/2312.00752** (2023). DOI 10.48550/ARXIV.2312.00752. URL https://doi.org/10.48550/ARXIV.2312.00752 29
60. Guan, J., Xu, Z., Zhou, H., Wang, K., He, S., Zhang, Z., Liang, B., Feng, H., Ding, E., Liu, J., Wang, J., Zhao, Y., Liu, Z.: ReSyncer: Rewiring Style-Based Generator

for Unified Audio-Visually Synced Facial Performer. In: ECCV (41), pp. 348–367 (2024). DOI 10.1007/978-3-031-72940-9_20. URL https://doi.org/10.1007/978-3-031-72940-9_20 14

61. Guan, J., Zhang, Z., Zhou, H., Hu, T., Wang, K., He, D., Feng, H., Liu, J., Ding, E., Liu, Z., Wang, J.: StyleSync: High-Fidelity Generalized and Personalized Lip Sync in Style-Based Generator. In: CVPR, pp. 1505–1515 (2023). DOI 10.1109/CVPR52729.2023.00151. URL https://doi.org/10.1109/CVPR52729.2023.00151 14
62. Guo, J., Zhang, D., Liu, X., Zhong, Z., Zhang, Y., Wan, P., Zhang, D.: LivePortrait: Efficient Portrait Animation with Stitching and Retargeting Control. CoRR **abs/2407.03168** (2024). DOI 10.48550/ARXIV.2407.03168. URL https://doi.org/10.48550/ARXIV.2407.03168 1, 2, 6, 9, 10, 29, 31, 33, 35, 43, 44, 45
63. Guo, Y., Chen, K., Liang, S., Liu, Y.J., Bao, H., Zhang, J.: AD-NeRF: Audio Driven Neural Radiance Fields for Talking Head Synthesis. In: ICCV, pp. 5764–5774 (2021). DOI 10.1109/ICCV48922.2021.00573. URL https://doi.org/10.1109/ICCV48922.2021.00573 1, 2, 29, 30, 31, 35
64. Guo, Y., Liu, X., Zhen, C., Yan, P., Wei, X.: ARIG: Autoregressive Interactive Head Generation for Real-Time Conversations. In: ICCV, pp. 12956–12965 (2025). DOI 10.1109/ICCV51701.2025.01204. URL https://doi.org/10.1109/ICCV51701.2025.01204 24, 27, 34, 43
65. HaCohen, Y., Chiprut, N., Brazowski, B., Shalem, D., Moshe, D., Richardson, E., Levin, E., Shiran, G., Zabari, N., Gordon, O., Panet, P., Weissbuch, S., Kulikov, V., Bitterman, Y., Melumian, Z., Bibi, O.: LTX-Video: Realtime Video Latent Diffusion. CoRR **abs/2501.00103** (2025). DOI 10.48550/ARXIV.2501.00103. URL https://doi.org/10.48550/ARXIV.2501.00103 19
66. He, T., Guo, J., Yu, R., Wang, Y., Zhu, J., An, K., Li, L., Tan, X., Wang, C., Hu, H., Wu, H., Zhao, S., Bian, J.: GAIA: Zero-shot Talking Avatar Generation. In: ICLR (2024). URL https://openreview.net/forum?id=ATEawsFUj4 34
67. He, Y., Gu, X., Ye, X., Xu, C., Zhao, Z., Dong, Y., Yuan, W., Dong, Z., Bo, L.: LAM: Large Avatar Model for One-shot Animatable Gaussian Head. In: SIGGRAPH (Conference Paper Track), pp. 27:1–27:13 (2025). DOI 10.1145/3721238.3730706. URL https://doi.org/10.1145/3721238.3730706 32, 45
68. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium. In: NIPS, pp. 6626–6637 (2017). URL https://proceedings.neurips.cc/paper/2017/hash/8a1d694707eb0fefe65871369074926d-Abstract.html 40
69. Ho, J., Jain, A., Abbeel, P.: Denoising Diffusion Probabilistic Models. In: NeurIPS (2020). URL https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-Abstract.html 4, 9, 16, 17
70. Hong, F.T., Liu, R., Song, L., Cai, H., Xian, C.: EfficientSync: Real-Time Lip Synchronization via Deformation-Based Reference Texture Mixing. CoRR **abs/2608.18832** (2026). DOI 10.48550/ARXIV.2608.18832. URL https://doi.org/10.48550/ARXIV.2608.18832 33
71. Hong, Y., Peng, B., Xiao, H., Liu, L., Zhang, J.: HeadNeRF: A Realtime NeRF-based Parametric Head Model. In: CVPR, pp. 20342–20352 (2022). DOI 10.1109/CVPR52688.2022.01973. URL https://doi.org/10.1109/CVPR52688.2022.01973 30
72. Horé, A., Ziou, D.: Image Quality Metrics: PSNR vs. SSIM. In: ICPR, pp. 2366–2369 (2010). DOI 10.1109/ICPR.2010.579. URL https://doi.org/10.1109/ICPR.2010.579 39
73. Hu, L.: Animate Anyone: Consistent and Controllable Image-to-Video Synthesis for Character Animation. In: CVPR, pp. 8153–8163 (2024). DOI 10.1109/CVPR52733.2024.00779. URL https://doi.org/10.1109/CVPR52733.2024.00779 4, 16
74. Hu, W., Li, S., Peng, Z., Zhang, H., Shi, F., Liu, X., Wan, P., Zhang, D., Tian, H.: GGTalker: Talking Head Systhesis with Generalizable Gaussian Priors and Identity-Specific Adaptation. In: ICCV, pp. 10108–10117 (2025). DOI 10.1109/ICCV51701.2025.00942. URL https://doi.org/10.1109/ICCV51701.2025.00942 32, 45
75. Hu, Y., Liu, R., Ren, Y., Yin, X., Li, H.: UniTalker: Conversational Speech-Visual Synthesis. In: ACM Multimedia, pp. 10248–10257 (2025). DOI 10.1145/3746027.3755502. URL https://doi.org/10.1145/3746027.3755502 27
76. Hu, Y., Zhang, J.: Speculative Decoding for Autoregressive Video Generation. CoRR **abs/2604.17397** (2026). DOI 10.48550/ARXIV.2604.17397. URL https://doi.org/10.48550/ARXIV.2604.17397 21, 26, 28
77. Huang, L., Wu, Z.F., Shi, Y., Wang, W., Feng, M., He, J., Xie, C.W., Liu, Y., Zhou, J., Wang, A., Zhang, B., Ai, B., Liang, C., Yu, C., Zhong, C., Qi, J., Zhu, K., Li, P., Zhang, P., Zhang, W., Cheng, X., Huang, Y., Zheng, Y., Bao, Y., Wang, Y., Bi, Z.: Wan-Streamer v0.2: Higher Resolution, Same Latency. CoRR **abs/2607.04443** (2026). DOI 10.48550/ARXIV.2607.04443. URL https://doi.org/10.48550/ARXIV.2607.04443 22, 28, 29
78. Huang, L., Wu, Z.F., Shi, Y., Wang, W., Feng, M., Yu, C., Liang, C., He, J., Xie, C.W., Liu, Y., Zhou, J., Wang, A., Zhang, B., Ai, B., Zhong, C., Qi, J., Zhu, K., Li, P., Zhang, P., Zhang, W., Cheng, X., Huang, Y., Zheng, Y., Bao, Y., Wang, Y., Lin, Z., Bi, Z.: Video = World + Event Stream. CoRR **abs/2607.15038** (2026). DOI 10.48550/ARXIV.2607.15038. URL https://doi.org/10.48550/ARXIV.2607.15038 28
79. Huang, L., Wu, Z.F., Wang, W., Shi, Y., Feng, M., He, J., Xie, C.W., Liu, Y., Zhou, J., Wang, A., Zhang, B., Ai, B., Liang, C., Yu, C., Zhong, C., Qi, J., Zhu, K., Li, P., Zhang, P., Zhang, W., Cheng, X., Huang, Y., Zheng, Y., Wang, Y., Bi, Z.: Wan-Streamer v0.1: End-to-end Real-time Interactive Foundation Models. CoRR **abs/2606.25041** (2026). DOI 10.48550/ARXIV.2606.25041. URL https://doi.org/10.48550/ARXIV.2606.25041 22
80. Huang, X., Li, Z., He, G., Zhou, M., Shechtman, E.: Self Forcing: Bridging the Train-Test Gap in Autoregressive Video Diffusion. In: NeurIPS (2025). URL http://papers.nips.cc/paper_files/paper/2025/hash/f4823f831af67a3ef15e41a85434422a-Abstract-Conference.html 12, 22
81. Huang, Y., Guo, H., Wu, F., Zhang, S., Huang, S., Gan, Q., Liu, L., Zhao, S., Chen, E., Liu, J., Hoi, S.: Live Avatar: Streaming Real-time Audio-Driven Avatar Generation with Infinite Length. CoRR **abs/2512.04677** (2025). DOI 10.48550/ARXIV.2512.04677. URL https://doi.org/10.48550/ARXIV.2512.04677 1, 2, 3, 10, 12, 17, 20, 21, 22, 23, 24, 25, 36, 38, 39, 44, 45
82. Huang, Y., Zhao, S., Yao, X., Zhang, Z., Huang, J., Cui, F., Wu, S., Chen, E.: DynaForcing: Overcoming Dynamic Collapse in Self-Forcing Distillation for Streaming Avatar Generation. CoRR **abs/2608.17707** (2026). DOI 10.48550/ARXIV.2608.17707. URL https://doi.org/10.48550/ARXIV.2608.17707 21, 44
83. Huang, Z., Rong, Z., Qin, S., Liang, S., Hu, W., Luo, Y., Zhang, Y., Gao, M.: FlowAct-R2: Beyond Talking Avatar

via Streaming Multimodal References and Proactive Agent Planning. arXiv preprint arXiv:2609.35728 (2026). URL https://arxiv.org/abs/2609.35728v1 23
84. Iandola, F., Pidhorskyi, S., Santesteban, I., Gupta, D., Pahuja, A., Bartolovic, N., Yu, F., Garbin, E., Simon, T., Saito, S.: SqueezeMe: Mobile-Ready Distillation of Gaussian Full-Body Avatars. In: SIGGRAPH (Conference Paper Track), pp. 91:1–91:11 (2025). DOI 10.1145/3721238.3730599. URL https://doi.org/10.1145/3721238.3730599 32
85. Ji, C., Qi, J., Xu, S., Zhang, P., Zhang, B.: PortraitDirector: A Hierarchical Disentanglement Framework for Controllable and Real-time Facial Reenactment. In: CVPR, pp. 32377–32388 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Ji_PortraitDirector_A_Hierarchical_Disentanglement_Framework_for_Controllable_and_Real-time_Facial_CVPR_2026_paper.html 34
86. Ji, Y., Zhong, Z., Zhang, J., Yang, Q., Jin, X., Qin, Y., Luo, W., Mao, S., Liu, W., Li, H.: Forcing-KV: Hybrid KV Cache Compression for Efficient Autoregressive Video Diffusion Models. CoRR **abs/2605.09681** (2026). DOI 10.48550/ARXIV.2605.09681. URL https://doi.org/10.48550/ARXIV.2605.09681 21
87. Jia, P., Xiao, Z., Li, J., Liu, X., Hu, Z., Yu, L.: SDTalk: Structured Facial Priors and Dual-Branch Motion Fields for Generalizable Gaussian Talking Head Synthesis. CoRR **abs/2605.09956** (2026). DOI 10.48550/ARXIV.2605.09956. URL https://doi.org/10.48550/ARXIV.2605.09956 33
88. Kang, T., Oh, J., Lee, J., Park, S., Choo, J.: Expression Domain Translation Network for Cross-Domain Head Reenactment. In: ICASSP, pp. 7356–7359 (2024). DOI 10.1109/ICASSP48485.2024.10447878. URL https://doi.org/10.1109/ICASSP48485.2024.10447878 14
89. Karras, T., Laine, S., Aila, T.: A Style-Based Generator Architecture for Generative Adversarial Networks. In: CVPR, pp. 4401–4410 (2019). DOI 10.1109/CVPR.2019.00453. URL http://openaccess.thecvf.com/content_CVPR_2019/html/Karras_A_Style-Based_Generator_Architecture_for_Generative_Adversarial_Networks_CVPR_2019_paper.html 4, 8, 13
90. Karras, T., Laine, S., Aittala, M., Hellsten, J., Lehtinen, J., Aila, T.: Analyzing and Improving the Image Quality of StyleGAN. In: CVPR, pp. 8107–8116 (2020). DOI 10.1109/CVPR42600.2020.00813. URL https://openaccess.thecvf.com/content_CVPR_2020/html/Karras_Analyzing_and_Improving_the_Image_Quality_of_StyleGAN_CVPR_2020_paper.html 8, 13
91. Kerbl, B., Kopanas, G., Leimkühler, T., Drettakis, G.: 3D Gaussian Splatting for Real-Time Radiance Field Rendering. ACM Trans. Graph. **42**, 139:1–139:14 (2023). DOI 10.1145/3592433. URL https://doi.org/10.1145/3592433 3, 6, 9, 10, 29, 30
92. Kettunen, M., Härkönen, E., Lehtinen, J.: E-LPIPS: Robust Perceptual Image Similarity via Random Transformation Ensembles. CoRR **abs/1906.03973** (2019). DOI 10.48550/ARXIV.1906.03973. URL https://doi.org/10.48550/ARXIV.1906.03973 39
93. Ki, T., Jang, S., Jo, J., Yoon, J., Hwang, S.J.: Avatar Forcing: Real-Time Interactive Head Avatar Generation for Natural Conversation. In: CVPR, pp. 18074–18084 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Ki_Avatar_Forcing_Real-Time_Interactive_Head_Avatar_Generation_for_Natural_Conversation_CVPR_2026_paper.html 17, 18, 20
94. Ki, T., Min, D.: StyleLipSync: Style-based Personalized Lip-sync Video Generation. In: ICCV, pp. 22784–22793 (2023). DOI 10.1109/ICCV51070.2023.02088. URL https://doi.org/10.1109/ICCV51070.2023.02088 14
95. Ki, T., Min, D., Chae, G.: FLOAT: Generative Motion Latent Flow Matching for Audio-Driven Talking Portrait. In: ICCV, pp. 14699–14710 (2025). DOI 10.1109/ICCV51701.2025.01364. URL https://doi.org/10.1109/ICCV51701.2025.01364 33
96. Kingma, D.P., Welling, M.: Auto-Encoding Variational Bayes. In: ICLR (2014). DOI 10.48550/ARXIV.1312.6114. URL https://doi.org/10.48550/ARXIV.1312.6114 9
97. Kirschstein, T., Qian, S., Giebenhain, S., Walter, T., Nießner, M.: NeRSemble: Multi-view Radiance Field Reconstruction of Human Heads. ACM Trans. Graph. **42**, 161:1–161:14 (2023). DOI 10.1145/3592455. URL https://doi.org/10.1145/3592455 39, 40
98. Kong, W., Tian, Q., Zhang, Z., Min, R., Dai, Z., Zhou, J., Xiong, J., Li, X., Wu, B., Zhang, J., Wu, K., Lin, Q., Yuan, J., Long, Y., Wang, A., Wang, A., Li, C., Huang, D., Yang, F., Tan, H., Wang, H., Song, J., Bai, J., Wu, J., Xue, J., Wang, J., Wang, K., Liu, M., Li, P., Li, S., Wang, W., Yu, W., Deng, X., Li, Y., Chen, Y., Cui, Y., Peng, Y., Yu, Z., He, Z., Xu, Z., Zhou, Z., Xu, Z., Tao, Y., Lu, Q., Liu, S., Zhou, D., Wang, H., Yang, Y., Wang, D., Liu, Y., Jiang, J., Zhong, C.: HunyuanVideo: A Systematic Framework For Large Video Generative Models. CoRR **abs/2412.03603** (2024). DOI 10.48550/ARXIV.2412.03603. URL https://doi.org/10.48550/ARXIV.2412.03603 19
99. Kravtsov, A., Ziganshin, D., Poletaev, V., Balitskiy, G., Tikhonova, A., Burkov, E., Lebedev, V.: AVTR-1: Open Stack for Real-Time Interactive Avatars. CoRR **abs/2609.22913** (2026). DOI 10.48550/ARXIV.2609.22913. URL https://doi.org/10.48550/ARXIV.2609.22913 32
100. Lee, J., Li, C., Tran, L., Wei, S.E., Saragih, J.M., Richard, A., Joo, H., Bai, S.: Audio Driven Real-Time Facial Animation for Social Telepresence. In: SIGGRAPH Asia, pp. 51:1–51:12 (2025). DOI 10.1145/3757377.3763854. URL https://doi.org/10.1145/3757377.3763854 17, 18
101. Lee, S., Moon, S., Lew, H.M., Kang, J.S., Park, G.M.: AudioAvatar: Personalized Audio-driven Whole-body Talking Avatars. In: CVPR, pp. 3998–4010 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Lee_AudioAvatar_Personalized_Audio-driven_Whole-body_Talking_Avatars_CVPR_2026_paper.html 32, 42, 45
102. Leviathan, Y., Kalman, M., Matias, Y.: Fast Inference from Transformers via Speculative Decoding. In: ICML, pp. 19274–19286 (2023). URL https://proceedings.mlr.press/v202/leviathan23a.html 1, 9, 10, 21, 26, 28
103. Li, C., Huang, D., Lu, Z., Xiao, Y., Pei, Q., Bai, L.: A Survey on Long Video Generation: Challenges, Methods, and Prospects. CoRR **abs/2403.16407** (2024). DOI 10.48550/ARXIV.2403.16407. URL https://doi.org/10.48550/ARXIV.2403.16407 4
104. Li, C., Li, J., Mei, R., Xia, H., Zhu, H., Wang, J., Zhu, S.: Hallo-Live: Real-Time Streaming Joint Audio-Video Avatar Generation with Asynchronous Dual-Stream and Human-Centric Preference Distillation. CoRR **abs/2604.23632** (2026). DOI 10.48550/ARXIV.2604.23632. URL https://doi.org/10.48550/ARXIV.2604.23632 21, 23
105. Li, C., Wang, R., Zhou, L., Feng, J., Luo, H., Zhang, H., Wu, Y., He, X.: JoyAvatar: Real-time and Infinite Audio-Driven Avatar Generation with Autoregressive Diffusion. CoRR **abs/2512.11423** (2025). DOI 10.48550/ARXIV.2

512.11423. URL https://doi.org/10.48550/ARXIV.2512.11423 12, 25, 38

106. Li, H., Xu, M., Zhan, Y., Mu, S., Li, J., Cheng, K., Chen, Y., Chen, T., Ye, M., Wang, J., Zhu, S.: OpenHumanVid: A Large-Scale High-Quality Dataset for Enhancing Human-Centric Video Generation. In: CVPR, pp. 7752–7762 (2025). DOI 10.1109/CVPR52734.2025.00726. URL https://openaccess.thecvf.com/content/CVPR2025/html/Li_OpenHumanVid_A_Large-Scale_High-Quality_Dataset_for_Enhancing_Human-Centric_Video_Generation_CVPR_2025_paper.html 37, 40
107. Li, J., Zhang, J., Bai, X., Zhou, J., Gu, L.: Efficient Region-Aware Neural Radiance Fields for High-Fidelity Talking Portrait Synthesis. In: ICCV, pp. 7534–7544 (2023). DOI 10.1109/ICCV51070.2023.00696. URL https://doi.org/10.1109/ICCV51070.2023.00696 30, 31
108. Li, R., Yang, T., Ma, Z., Ai, F., Wen, S., Zhang, L.: AvatarForever: Decoupled Parallel Training for High-Quality Real-Time Infinite Avatars. CoRR **abs/2608.12107** (2026). DOI 10.48550/ARXIV.2608.12107. URL https://doi.org/10.48550/ARXIV.2608.12107 22, 25
109. Li, T., Bolkart, T., Black, M.J., Li, H., Romero, J.: Learning a model of facial shape and expression from 4D scans. ACM Trans. Graph. **36**, 194:1–194:17 (2017). DOI 10.1145/3130800.3130813. URL https://doi.org/10.1145/3130800.3130813 6, 10, 30, 31
110. Li, T., Zheng, R., Yang, M., Chen, J., Yang, M.: Ditto: Motion-Space Diffusion for Controllable Realtime Talking Head Synthesis. In: ACM Multimedia, pp. 9704–9713 (2025). DOI 10.1145/3746027.3755075. URL https://doi.org/10.1145/3746027.3755075 33
111. Li, W., Gao, Y., Hassan, M., Feng, L., Pan, W., Luan, P.C., Alahi, A.: EverAnimate: Minute-Scale Human Animation via Latent Flow Restoration. CoRR **abs/2605.15042** (2026). DOI 10.48550/ARXIV.2605.15042. URL https://doi.org/10.48550/ARXIV.2605.15042 24, 25
112. Li, X., Li, G., Lin, Z., Qian, Y., Yao, G., Jia, W., Wang, A., Chen, W., Wang, F.: MoDA: Multi-modal Diffusion Architecture for Talking Head Generation. CoRR **abs/2507.03256** (2025). DOI 10.48550/ARXIV.2507.03256. URL https://doi.org/10.48550/ARXIV.2507.03256 34, 35
113. Li, Z., Pun, C.M., Fang, C., Wang, J., Cun, X.: PersonaLive! Expressive Portrait Image Animation for Live Streaming. In: CVPR, pp. 18118–18128 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Li_PersonaLive_Expressive_Portrait_Image_Animation_for_Live_Streaming_CVPR_2026_paper.html 1, 3, 10, 12, 19, 22, 23, 25, 36
114. Liang, B., Chen, C., Lin, D., Somov, I., Zhao, J., Yuan, J., Zhang, J., Huang, J., Nolte, N., Haqiqi, P., Wang, P., Yan, R., Zhang, R., Prokopchuk, S., Wang, S., Goriachko, V., Ren, Y., Li, Y., Chen, Y., Ye, Z., Hong, Z., Nie, Z., Guo, Z.: Avatar V: Scaling Video-Reference Avatar Video Generation. CoRR **abs/2606.13872** (2026). DOI 10.48550/ARXIV.2606.13872. URL https://doi.org/10.48550/ARXIV.2606.13872 12, 17, 19, 21, 23, 25
115. Liang, X., Fan, Y., Yang, Q., Wang, X., Gao, W., Li, G.: DGTalker: Disentangled Generative Latent Space Learning for Audio-Driven Gaussian Talking Heads. In: ICCV, pp. 11079–11088 (2025). DOI 10.1109/ICCV51701.2025.01031. URL https://doi.org/10.1109/ICCV51701.2025.01031 32, 45
116. Lin, G., Jiang, J., Yang, J., Zheng, Z., Liang, C., Zhang, Y., Liu, J.: OmniHuman-1: Rethinking the Scaling-Up of One-Stage Conditioned Human Animation Models. In: ICCV, pp. 13848–13858 (2025). DOI 10.1109/ICCV51701.2025.01285. URL https://doi.org/10.1109/ICCV51701.2025.01285 16, 39
117. Lipman, Y., Chen, R.T.Q., Ben-Hamu, H., Nickel, M., Le, M.: Flow Matching for Generative Modeling. In: ICLR (2023). URL https://openreview.net/forum?id=PqvMRDCJT9t 4, 9, 16, 17
118. Liu, F., Zhang, S., Wang, X., Wei, Y., Qiu, H., Zhao, Y., Zhang, Y., Ye, Q., Wan, F.: Timestep Embedding Tells: It's Time to Cache for Video Diffusion Model. In: CVPR, pp. 7353–7363 (2025). DOI 10.1109/CVPR52734.2025.00689. URL https://openaccess.thecvf.com/content/CVPR2025/html/Liu_Timestep_Embedding_Tells_Its_Time_to_Cache_for_Video_Diffusion_CVPR_2025_paper.html 1, 11
119. Liu, K., Hu, W., Xu, J., Shan, Y., Lu, S.: Rolling Forcing: Autoregressive Long Video Diffusion in Real Time. In: ICLR (2026). URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/935151cc6cb5d8b6816133b75233775a-Abstract-Conference.html 22
120. Liu, X., Gao, F., Zhang, X., Zhang, Y., Wei, X., Lei, Z., Zhu, X.: TurboTalk: Progressive Distillation for One-Step Audio-Driven Talking Avatar Generation. CoRR **abs/2604.14580** (2026). DOI 10.48550/ARXIV.2604.14580. URL https://doi.org/10.48550/ARXIV.2604.14580 20
121. Lombardi, S., Saragih, J.M., Simon, T., Sheikh, Y.: Deep appearance models for face rendering. ACM Trans. Graph. **37**, 68 (2018). DOI 10.1145/3197517.3201401. URL https://doi.org/10.1145/3197517.3201401 30
122. Long, J., Sun, W., Tu, R.C., Tao, D.: Lightning Fast Caching-based Parallel Denoising Prediction for Accelerating Talking Head Generation. In: AAAI, pp. 7503–7511 (2026). DOI 10.1609/AAAI.V40I9.37690. URL https://doi.org/10.1609/AAAI.V40I9.37690 21, 23
123. Loper, M., Mahmood, N., Romero, J., Pons-Moll, G., Black, M.J.: SMPL: a skinned multi-person linear model. ACM Trans. Graph. **34**, 248:1–248:16 (2015). DOI 10.1145/2816795.2818013. URL https://doi.org/10.1145/2816795.2818013 10, 30
124. Lu, Y., Chai, J., Cao, X.: Live speech portraits: real-time photorealistic talking-head animation. ACM Trans. Graph. **40**, 220:1–220:17 (2021). DOI 10.1145/3478513.3480484. URL https://doi.org/10.1145/3478513.3480484 2, 15, 16
125. Lu, Y., Sun, J., Zhu, G., Cao, M.: AsymTalker: Identity-Consistent Long-Term Talking Head Generation via Asymmetric Distillation. CoRR **abs/2605.02948** (2026). DOI 10.48550/ARXIV.2605.02948. URL https://doi.org/10.48550/ARXIV.2605.02948 21, 25
126. Luo, C., Wang, J., Li, B., Song, S., Ghanem, B.: OmniResponse: Online Multimodal Conversational Response Generation in Dyadic Interactions. In: NeurIPS (2025). URL http://papers.nips.cc/paper_files/paper/2025/hash/b734c30b9c955c535e333f0301f5e45c-Abstract-Conference.html 10, 27, 39, 40
127. Luo, S., Tan, Y., Patil, S., Gu, D., von Platen, P., Passos, A., Huang, L., Li, J., Zhao, H.: LCM-LoRA: A Universal Stable-Diffusion Acceleration Module. CoRR **abs/2311.05556** (2023). DOI 10.48550/ARXIV.2311.05556. URL https://doi.org/10.48550/ARXIV.2311.05556 1, 9, 10, 20
128. Ma, J., Du, Y., Sun, Y., Xing, Z., Li, C., Tang, Z., Li, B., Jiang, P.T., Yuan, L., Zhou, D., Tian, Y.: SyncCache: Exploiting Asymmetric Dynamics for Fast Audio-Driven Portrait Animation. In: ECCV (55), pp. 601–618 (2026). DOI 10.1007/978-3-032-37261-1_34. URL https://doi.org/10.1007/978-3-032-37261-1_34 17, 21

129. Ma, X., Fang, G., Wang, X.: DeepCache: Accelerating Diffusion Models for Free. In: CVPR, pp. 15762–15772 (2024). DOI 10.1109/CVPR52733.2024.01492. URL https://doi.org/10.1109/CVPR52733.2024.01492 1, 11
130. Ma, Y., Wang, S., Hu, Z., Fan, C., Lv, T., Ding, Y., Deng, Z., Yu, X.: StyleTalk: One-Shot Talking Head Generation with Controllable Speaking Styles. In: AAAI, pp. 1896–1904 (2023). DOI 10.1609/AAAI.V37I2.25280. URL https://doi.org/10.1609/AAAI.V37I2.25280 34
131. Ma, Y., Yan, Z., Liu, H., Wang, H., Pan, H., He, Y., Yuan, J., Zeng, A., Cai, C., Shum, H.Y., Li, Z., Liu, W., Zhang, L., Chen, Q.: Follow-Your-Emoji-Faster: Towards Efficient, Fine-Controllable, and Expressive Freestyle Portrait Animation. Int. J. Comput. Vis. **134**, 130 (2026). DOI 10.1007/S11263-025-02685-Z. URL https://doi.org/10.1007/S11263-025-02685-Z 21, 23
132. Ma, Y., Zheng, X., Xu, J., Xu, X., Ling, F., Zheng, X., Kuang, H., Li, H., Wang, X., Xiao, X., Chao, F., Ji, R.: Flow Caching for Autoregressive Video Generation. In: ICLR (2026). URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/85dc8f85ff978b9c606d3b2f5b0da69a-Abstract-Conference.html 1, 11, 21
133. Meng, D., Xiao, S., Zhang, X., Wang, G., Zhang, P., Wang, Q., Zhang, B., Bo, L.: MirrorMe: Towards Realtime and High Fidelity Audio-Driven Halfbody Animation. CoRR **abs/2506.22065** (2025). DOI 10.48550/ARXIV.2506.22065. URL https://doi.org/10.48550/ARXIV.2506.22065 33
134. Meng, M., Zhao, Y., Zhang, B., Zhu, Y., Shi, W., Wen, M., Fan, Z.: A Survey of Talking Head Synthesis Techniques: Portrait Generation, Driving Mechanisms, and Editing. ACM Comput. Surv. **58**, 188:1–188:43 (2026). DOI 10.1145/3785656. URL https://doi.org/10.1145/3785656 1, 4
135. Meng, R., Yin, Y., Li, Y., Ma, C.: EchoTorrent: Towards Swift, Sustained, and Streaming Multi-Modal Video Generation. CoRR **abs/2602.13669** (2026). DOI 10.48550/ARXIV.2602.13669. URL https://doi.org/10.48550/ARXIV.2602.13669 22, 25
136. Meng, R., Zhang, X., Li, Y., Ma, C.: EchoMimicV2: Towards Striking, Simplified, and Semi-Body Human Animation. In: CVPR, pp. 5489–5498 (2025). DOI 10.1109/CVPR52734.2025.00516. URL https://openaccess.thecvf.com/content/CVPR2025/html/Meng_EchoMimicV2_Towards_Striking_Simplified_and_Semi-Body_Human_Animation_CVPR_2025_paper.html 37
137. Mildenhall, B., Srinivasan, P.P., Tancik, M., Barron, J.T., Ramamoorthi, R., Ng, R.: NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis. In: ECCV (1), pp. 405–421 (2020). DOI 10.1007/978-3-030-58452-8_24. URL https://doi.org/10.1007/978-3-030-58452-8_24 3, 6, 9, 10, 30
138. Müller, T., Evans, A., Schied, C., Keller, A.: Instant neural graphics primitives with a multiresolution hash encoding. ACM Trans. Graph. **41**, 102:1–102:15 (2022). DOI 10.1145/3528223.3530127. URL https://doi.org/10.1145/3528223.3530127 9, 30
139. van den Oord, A., Vinyals, O., Kavukcuoglu, K.: Neural Discrete Representation Learning. In: NIPS, pp. 6306–6315 (2017). URL https://proceedings.neurips.cc/paper/2017/hash/7a98af17e63a0ac09ce2e96d03992fbc-Abstract.html 9, 26
140. Pan, Y., Liu, C., Xu, S., Tan, S., Yang, J.: VASA-Rig: Audio-Driven 3D Facial Animation with 'Live' Mood Dynamics in Virtual Reality. IEEE Trans. Vis. Comput. Graph. **31**, 2416–2425 (2025). DOI 10.1109/TVCG.2025.3549168. URL https://doi.org/10.1109/TVCG.2025.3549168 34
141. Pang, Y., Zhang, Y., Shao, R., Deng, X., Gao, F., Xiaoming, X., Wei, X., Liu, Y.: UniMo: Unifying 2D Video and 3D Human Motion with an Autoregressive Framework. CoRR **abs/2512.03918** (2025). DOI 10.48550/ARXIV.2512.03918. URL https://doi.org/10.48550/ARXIV.2512.03918 27, 28
142. Papamakarios, G., Nalisnick, E.T., Rezende, D.J., Mohamed, S., Lakshminarayanan, B.: Normalizing Flows for Probabilistic Modeling and Inference. J. Mach. Learn. Res. **22**, 57:1–57:64 (2021). URL https://jmlr.org/papers/v22/19-1028.html 17
143. Papantoniou, F.P., Filntisis, P.P., Maragos, P., Roussos, A.: Neural Emotion Director: Speech-preserving semantic control of facial expressions in "in-the-wild" videos. In: CVPR, pp. 18759–18768 (2022). DOI 10.1109/CVPR52688.2022.01822. URL https://doi.org/10.1109/CVPR52688.2022.01822 14
144. Park, S.J., Kim, M., Hong, J., Choi, J., Ro, Y.M.: SyncTalkFace: Talking Face Generation with Precise Lip-Syncing via Audio-Lip Memory. In: AAAI, pp. 2062–2070 (2022). DOI 10.1609/AAAI.V36I2.20102. URL https://doi.org/10.1609/AAAI.V36I2.20102 14
145. Pavlakos, G., Choutas, V., Ghorbani, N., Bolkart, T., Osman, A.A.A., Tzionas, D., Black, M.J.: Expressive Body Capture: 3D Hands, Face, and Body From a Single Image. In: CVPR, pp. 10975–10985 (2019). DOI 10.1109/CVPR.2019.01123. URL http://openaccess.thecvf.com/content_CVPR_2019/html/Pavlakos_Expressive_Body_Capture_3D_Hands_Face_and_Body_From_a_CVPR_2019_paper.html 10, 30
146. Peebles, W., Xie, S.: Scalable Diffusion Models with Transformers. In: ICCV, pp. 4172–4182 (2023). DOI 10.1109/ICCV51070.2023.00387. URL https://doi.org/10.1109/ICCV51070.2023.00387 17, 19
147. Peng, Z., Hu, W., Ma, J., Zhu, X., Zhang, X., Zhao, H., Tian, H., He, J., Liu, H., Fan, Z.: SyncTalk++: High-Fidelity and Efficient Synchronized Talking Heads Synthesis Using Gaussian Splatting. CoRR **abs/2506.14742** (2025). DOI 10.48550/ARXIV.2506.14742. URL https://doi.org/10.48550/ARXIV.2506.14742 32
148. Peng, Z., Hu, W., Shi, Y., Zhu, X., Zhang, X., Zhao, H., He, J., Liu, H., Fan, Z.: SyncTalk: The Devil is in the Synchronization for Talking Head Synthesis. In: CVPR, pp. 666–676 (2024). DOI 10.1109/CVPR52733.2024.00070. URL https://doi.org/10.1109/CVPR52733.2024.00070 31
149. Podell, D., English, Z., Lacey, K., Blattmann, A., Dockhorn, T., Müller, J., Penna, J., Rombach, R.: SDXL: Improving Latent Diffusion Models for High-Resolution Image Synthesis. In: ICLR (2024). URL https://openreview.net/forum?id=di52zR8xgf 9
150. Prajwal, K.R., Mukhopadhyay, R., Namboodiri, V.P., Jawahar, C.V.: A Lip Sync Expert Is All You Need for Speech to Lip Generation In the Wild. In: ACM Multimedia, pp. 484–492 (2020). DOI 10.1145/3394171.3413532. URL https://doi.org/10.1145/3394171.3413532 2, 4, 10, 13, 14, 16, 41, 45
151. Qi, J., Ji, C., Xu, S., Zhang, P., Zhang, B., Bo, L.: ChatAnyone: Stylized Real-time Portrait Video Generation with Hierarchical Motion Diffusion Model. CoRR **abs/2503.21144** (2025). DOI 10.48550/ARXIV.2503.21144. URL https://doi.org/10.48550/ARXIV.2503.21144 33
152. Qian, S., Kirschstein, T., Schoneveld, L., Davoli, D., Giebenhain, S., Nießner, M.: GaussianAvatars: Photorealis-

tic Head Avatars with Rigged 3D Gaussians. In: CVPR, pp. 20299–20309 (2024). DOI 10.1109/CVPR52733.2024.01919. URL https://doi.org/10.1109/CVPR52733.2024.01919 31

153. Qian, Z., Wang, S., Mihajlovic, M., Geiger, A., Tang, S.: 3DGS-Avatar: Animatable Avatars via Deformable 3D Gaussian Splatting. In: CVPR, pp. 5020–5030 (2024). DOI 10.1109/CVPR52733.2024.00480. URL https://doi.org/10.1109/CVPR52733.2024.00480 32
154. Radford, A., Kim, J.W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., Sutskever, I.: Learning Transferable Visual Models From Natural Language Supervision. In: ICML, pp. 8748–8763 (2021). URL http://proceedings.mlr.press/v139/radford21a.html 40
155. Radford, A., Narasimhan, K., Salimans, T., Sutskever, I., et al.: Improving language understanding by generative pre-training (2018) 9
156. Rakesh, V.K., Mazumdar, S., Maity, R.P., Pal, S., Das, A., Samanta, T.: Advancements in talking head generation: a comprehensive review of techniques, metrics, and challenges. Vis. Comput. **42**, 9 (2026). DOI 10.1007/S00371-025-04232-W. URL https://doi.org/10.1007/S00371-025-04232-W 4
157. Richard, A., Lea, C., Ma, S., Gall, J., la Torre, F.D., Sheikh, Y.: Audio- and Gaze-driven Facial Animation of Codec Avatars. In: WACV, pp. 41–50 (2021). DOI 10.1109/WACV48630.2021.00009. URL https://doi.org/10.1109/WACV48630.2021.00009 30
158. Rombach, R., Blattmann, A., Lorenz, D., Esser, P., Ommer, B.: High-Resolution Image Synthesis with Latent Diffusion Models. In: CVPR, pp. 10674–10685 (2022). DOI 10.1109/CVPR52688.2022.01042. URL https://doi.org/10.1109/CVPR52688.2022.01042 9
159. Saggar, A., Darling, J.C., Sarikaya, D., Hogg, D.C.: EmbedTalk: Triplane-Free Talking Head Synthesis using Embedding-Driven Gaussian Deformation. CoRR **abs/2603.07604** (2026). DOI 10.48550/ARXIV.2603.07604. URL https://doi.org/10.48550/ARXIV.2603.07604 32
160. Salimans, T., Ho, J.: Progressive Distillation for Fast Sampling of Diffusion Models. In: ICLR (2022). URL https://openreview.net/forum?id=TIdIXIpzhoI 9
161. Sha, T., Zhang, W., Shen, T., Li, Z., Mei, T.: Deep Person Generation: A Survey from the Perspective of Face, Pose, and Cloth Synthesis. ACM Comput. Surv. **55**, 257:1–257:37 (2023). DOI 10.1145/3575656. URL https://doi.org/10.1145/3575656 4
162. Shah, J., Bikshandi, G., Zhang, Y., Thakkar, V., Ramani, P., Dao, T.: FlashAttention-3: Fast and Accurate Attention with Asynchrony and Low-precision. In: NeurIPS (2024). URL http://papers.nips.cc/paper_files/paper/2024/hash/7ede97c3e082c6df10a8d6103a2eebd2-Abstract-Conference.html 11
163. Shao, Z., Wang, Z., Li, Z., Wang, D., Lin, X., Zhang, Y., Fan, M., Wang, Z.: SplattingAvatar: Realistic Real-Time Human Avatars With Mesh-Embedded Gaussian Splatting. In: CVPR, pp. 1606–1616 (2024). DOI 10.1109/CVPR52733.2024.00159. URL https://doi.org/10.1109/CVPR52733.2024.00159 31
164. Shen, H., Zhang, J., Xiong, B., Hu, R., Chen, S., Wan, Z., Wang, X., Zhang, Y., Gong, Z., Bao, G., Tao, C., Huang, Y., Yuan, Y., Zhang, M.: Efficient Diffusion Models: A Survey. Trans. Mach. Learn. Res. **2025** (2025). URL https://openreview.net/forum?id=wHECkBOwyt 4, 5, 10
165. Shen, L., Qian, Q., Yu, T., Zhou, K., Yu, T., Zhan, Y., Wang, Z., Tao, M., Yin, S., Liu, S.: SoulX-FlashTalk: Real-Time Infinite Streaming of Audio-Driven Avatars via Self-Correcting Bidirectional Distillation. CoRR **abs/2512.23379** (2025). DOI 10.48550/ARXIV.2512.23379. URL https://doi.org/10.48550/ARXIV.2512.23379 1, 17, 21, 22, 23, 25, 45
166. Shin, J., Li, Z., Zhang, R., Zhu, J.Y., Park, J., Shechtman, E., Huang, X.: MotionStream: Real-Time Video Generation with Interactive Motion Controls. In: ICLR (2026). URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/0cece806cd3d1dfad4a893f016ad3d7d-Abstract-Conference.html 19, 21, 23, 25
167. Siarohin, A., Lathuilière, S., Tulyakov, S., Ricci, E., Sebe, N.: Animating Arbitrary Objects via Deep Motion Transfer. In: CVPR, pp. 2377–2386 (2019). DOI 10.1109/CVPR.2019.00248. URL http://openaccess.thecvf.com/content_CVPR_2019/html/Siarohin_Animating_Arbitrary_Objects_via_Deep_Motion_Transfer_CVPR_2019_paper.html 33
168. Siarohin, A., Lathuilière, S., Tulyakov, S., Ricci, E., Sebe, N.: First Order Motion Model for Image Animation. In: NeurIPS, pp. 7135–7145 (2019). URL https://proceedings.neurips.cc/paper/2019/hash/31c0b36aef265d9221af80872ceb62f9-Abstract.html 31, 33
169. Siarohin, A., Woodford, O.J., Ren, J., Chai, M., Tulyakov, S.: Motion Representations for Articulated Animation. In: CVPR, pp. 13653–13662 (2021). DOI 10.1109/CVPR46437.2021.01344. URL https://openaccess.thecvf.com/content/CVPR2021/html/Siarohin_Motion_Representations_for_Articulated_Animation_CVPR_2021_paper.html 33
170. Song, H., Yoon, B., Cho, W., Woo, W.: RC-SMPL: Real-time Cumulative SMPL-based Avatar Body Generation. In: ISMAR, pp. 89–98 (2023). DOI 10.1109/ISMAR59233.2023.00023. URL https://doi.org/10.1109/ISMAR59233.2023.00023 30
171. Song, K., Cui, J., Zhang, J.: ProgressiveAvatars: Progressive Animatable 3D Gaussian Avatars. In: CVPR, pp. 32518–32527 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Song_ProgressiveAvatars_Progressive_Animatable_3D_Gaussian_Avatars_CVPR_2026_paper.html 32
172. Song, L., Chen, L., Liu, C., Liu, P., Xu, C.: TextToon: Real-Time Text Toonify Head Avatar from Single Video. In: SIGGRAPH Asia, pp. 107:1–107:11 (2024). DOI 10.1145/3680528.3687632. URL https://doi.org/10.1145/3680528.3687632 32
173. Song, Q., He, Y., Zhang, Y., Cheng, S., He, Z., Guo, Z., Zhang, C., Li, X., Jiang, C.: InteractiveAvatar: Real-Time Streaming Video Generation for Consistent and Intent-Aware Avatars. In: ECCV (75), pp. 680–697 (2026). DOI 10.1007/978-3-032-37032-7_37. URL https://doi.org/10.1007/978-3-032-37032-7_37 12, 24, 44
174. Song, Y., Dhariwal, P., Chen, M., Sutskever, I.: Consistency Models. In: ICML, pp. 32211–32252 (2023). URL https://proceedings.mlr.press/v202/song23a.html 1, 9, 10, 20
175. Su, Z., Lu, S., Zhuang, J., Xue, Z., Huang, H., Li, G., Tan, X., Yuan, C., Duan, N.: Where and When to Force: Routed Forcing for Streaming Avatars. arXiv preprint arXiv:2609.30963 (2026). URL https://arxiv.org/abs/2609.30963v1 21, 44
176. Sun, Z., Peng, Z., Ma, Y., Chen, Y., Zhou, Z., Zhou, Z., Zhang, G., Zhang, Y., Zhou, Y., Lu, Q., Liu, Y.J.: StreamAvatar: Streaming Diffusion Models for Real-Time Interactive Human Avatars. In: CVPR, pp. 10887–10897 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Sun_StreamAvatar_Streaming_Diffusion_Mod

els_for_Real-Time_Interactive_Human_Avatars_CVPR_2026_paper.html 17, 21, 22, 23, 25
177. Tan, S., Gong, B., Ji, B., Pan, Y.: FixTalk: Taming Identity Leakage for High-Quality Talking Head Generation in Extreme Cases. In: ICCV, pp. 24–36 (2025). DOI 10.1109/ICCV51701.2025.00010. URL https://doi.org/10.1109/ICCV51701.2025.00010 14, 16
178. Tan, S., Ji, B., Bi, M., Pan, Y.: EDTalk: Efficient Disentanglement for Emotional Talking Head Synthesis. In: ECCV (6), pp. 398–416 (2024). DOI 10.1007/978-3-031-72658-3_23. URL https://doi.org/10.1007/978-3-031-72658-3_23 4, 14, 16
179. Tan, S., Ji, B., Pan, Y.: FlowVQTalker: High-Quality Emotional Talking Face Generation through Normalizing Flow and Quantization. In: CVPR, pp. 26307–26317 (2024). DOI 10.1109/CVPR52733.2024.02486. URL https://doi.org/10.1109/CVPR52733.2024.02486 14, 16
180. Tang, A., He, T., Tan, X., Ling, J., Li, R., Zhao, S., Bian, J., Song, L.: Memories are One-to-Many Mapping Alleviators in Talking Face Generation. IEEE Trans. Pattern Anal. Mach. Intell. **46**, 8758–8770 (2024). DOI 10.1109/TPAMI.2024.3409380. URL https://doi.org/10.1109/TPAMI.2024.3409380 14
181. Tang, B.: Beyond Lip Sync: Reference-Grounded Oral Refinement for Audio-Driven Portrait Animation. arXiv preprint arXiv:2609.38019 (2026). URL https://arxiv.org/abs/2609.38019v1 41
182. Tang, J., Wang, K., Zhou, H., Chen, X., He, D., Hu, T., Liu, J., Liu, Z., Zeng, G., Wang, J.: Real-Time Neural Radiance Talking Portrait Synthesis via Audio-Spatial Decomposition. Int. J. Comput. Vis. **133**, 6362–6373 (2025). DOI 10.1007/S11263-025-02481-9. URL https://doi.org/10.1007/S11263-025-02481-9 30
183. Tian, L., Wang, Q., Zhang, B., Bo, L.: EMO: Emote Portrait Alive Generating Expressive Portrait Videos with Audio2Video Diffusion Model Under Weak Conditions. In: ECCV (83), pp. 244–260 (2024). DOI 10.1007/978-3-031-73010-8_15. URL https://doi.org/10.1007/978-3-031-73010-8_15 1, 2, 4, 16
184. Tu, S., Pan, Y., Huang, Y., Han, X., Xing, Z., Dai, Q., Luo, C., Wu, Z., Jiang, Y.G.: StableAvatar: Infinite-Length Audio-Driven Avatar Video Generation. CoRR **abs/2508.08248** (2025). DOI 10.48550/ARXIV.2508.08248. URL https://doi.org/10.48550/ARXIV.2508.08248 23, 25
185. Unterthiner, T., van Steenkiste, S., Kurach, K., Marinier, R., Michalski, M., Gelly, S.: Towards Accurate Generative Models of Video: A New Metric & Challenges. CoRR **abs/1812.01717** (2018). DOI 10.48550/ARXIV.1812.01717. URL https://doi.org/10.48550/ARXIV.1812.01717 40
186. Wang, A., Ai, B., Wen, B., Mao, C., Xie, C.W., Chen, D., Yu, F., Zhao, H., Yang, J., Zeng, J., Wang, J., Zhang, J., Zhou, J., Wang, J., Chen, J., Zhu, K., Zhao, K., Yan, K., Huang, L., Meng, X., Zhang, N., Li, P., Wu, P., Chu, R., Feng, R., Zhang, S., Sun, S., Fang, T., Wang, T., Gui, T., Weng, T., Shen, T., Lin, W., Wang, W., Wang, W., Zhou, W., Wang, W., Shen, W., Yu, W., Shi, X., Huang, X., Xu, X., Kou, Y., Lv, Y., Li, Y., Liu, Y., Wang, Y., Zhang, Y., Huang, Y., Li, Y., Wu, Y., Liu, Y., Pan, Y., Zheng, Y., Hong, Y., Shi, Y., Feng, Y., Jiang, Z., Han, Z., Wu, Z.F., Liu, Z.: Wan: Open and Advanced Large-Scale Video Generative Models. CoRR **abs/2503.20314** (2025). DOI 10.48550/ARXIV.2503.20314. URL https://doi.org/10.48550/ARXIV.2503.20314 19
187. Wang, B., Ding, Z., Li, J., Zhao, J., Lei, Z., Zhu, X.: Talk Like You: Imitating How You Speak in Real-Time Talking Head Generation. arXiv preprint arXiv:2610.06658 (2026). URL https://arxiv.org/abs/2610.06658v1 34
188. Wang, H., Weng, Y., Du, J., Xu, H., Wu, X., He, S., Yin, B., Liu, C., Gao, J., Liu, Q.: READ: Real-time and Efficient Asynchronous Diffusion for Audio-driven Talking Head Generation. In: AAAI, pp. 9766–9774 (2026). DOI 10.1609/AAAI.V40I12.37940. URL https://doi.org/10.1609/AAAI.V40I12.37940 17, 19
189. Wang, H., Weng, Y., Li, Y., Guo, Z., Du, J., Niu, S., Ma, J., He, S., Wu, X., Hu, Q., Yin, B., Liu, C., Liu, Q.: EmotiveTalk: Expressive Talking Head Generation through Audio Information Decoupling and Emotional Video Diffusion. In: CVPR, pp. 26212–26221 (2025). DOI 10.1109/CVPR52734.2025.02441. URL https://openaccess.thecvf.com/content/CVPR2025/html/Wang_EmotiveTalk_Expressive_Talking_Head_Generation_through_Audio_Information_Decoupling_and_CVPR_2025_paper.html 19
190. Wang, J., Qian, X., Zhang, M., Tan, R.T., Li, H.: Seeing What You Said: Talking Face Generation Guided by a Lip Reading Expert. In: CVPR, pp. 14653–14662 (2023). DOI 10.1109/CVPR52729.2023.01408. URL https://doi.org/10.1109/CVPR52729.2023.01408 14, 16
191. Wang, K., Wu, Q., Song, L., Yang, Z., Wu, W., Qian, C., He, R., Qiao, Y., Loy, C.C.: MEAD: A Large-Scale Audio-Visual Dataset for Emotional Talking-Face Generation. In: ECCV (21), pp. 700–717 (2020). DOI 10.1007/978-3-030-58589-1_42. URL https://doi.org/10.1007/978-3-030-58589-1_42 37, 40
192. Wang, K., Zhou, H., Wu, Q., Tang, J., Xu, Z., Liang, B., Hu, T., Ding, E., Liu, J., Liu, Z., Wang, J.: Efficient Video Portrait Reenactment via Grid-based Codebook. In: SIGGRAPH (Conference Paper Track), pp. 66:1–66:9 (2023). DOI 10.1145/3588432.3591509. URL https://doi.org/10.1145/3588432.3591509 14
193. Wang, L., Zhao, X., Sun, J., Zhang, Y., Zhang, H., Yu, T., Liu, Y.: StyleAvatar: Real-time Photo-realistic Portrait Avatar from a Single Video. In: SIGGRAPH (Conference Paper Track), pp. 67:1–67:10 (2023). DOI 10.1145/3588432.3591517. URL https://doi.org/10.1145/3588432.3591517 2, 9, 10, 15, 35
194. Wang, L., Zhu, Y., Ge, Z., Zheng, Y., Zhang, L., Hu, T., Qin, S., Luo, M., Zhang, J., Chen, X., Wang, Y., Zheng, Z., Jiang, J., Liang, C., Chen, W., Wang, X., Zhang, Y., Gao, M.: FlowAct-R1: Towards Interactive Humanoid Video Generation. CoRR **abs/2601.10103** (2026). DOI 10.48550/ARXIV.2601.10103. URL https://doi.org/10.48550/ARXIV.2601.10103 22, 23, 24, 25
195. Wang, Q., Li, M., Tang, H.: VibeAvatar: Aligning Phonetic Kinematics and Human Aesthetics for High-Fidelity Talking Avatar Synthesis. CoRR **abs/2609.18632** (2026). DOI 10.48550/ARXIV.2609.18632. URL https://doi.org/10.48550/ARXIV.2609.18632 34
196. Wang, S., Li, L., Ding, Y., Fan, C., Yu, X.: Audio2Head: Audio-driven One-shot Talking-head Generation with Natural Head Motion. In: IJCAI, pp. 1098–1105 (2021). DOI 10.24963/IJCAI.2021/152. URL https://doi.org/10.24963/IJCAI.2021/152 33
197. Wang, S., Ma, Y., Ding, Y., Hu, Z., Fan, C., Lv, T., Deng, Z., Yu, X.: StyleTalk++: A Unified Framework for Controlling the Speaking Styles of Talking Heads. IEEE Trans. Pattern Anal. Mach. Intell. **46**, 4331–4347 (2024). DOI 10.1109/TPAMI.2024.3357808. URL https://doi.org/10.1109/TPAMI.2024.3357808 34

198. Wang, T.C., Mallya, A., Liu, M.Y.: One-Shot Free-View Neural Talking-Head Synthesis for Video Conferencing. In: CVPR, pp. 10039–10049 (2021). DOI 10.1109/CVPR46437.2021.00991. URL https://openaccess.thecvf.com/content/CVPR2021/html/Wang_One-Shot_Free-View_Neural_Talking-Head_Synthesis_for_Video_Conferencing_CVPR_2021_paper.html 33
199. Wang, Y., Yang, D., Brémond, F., Dantcheva, A.: Latent Image Animator: Learning to Animate Images via Latent Space Navigation. In: ICLR (2022). URL https://openreview.net/forum?id=7r6kDq0mK_ 33
200. Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P.: Image quality assessment: from error visibility to structural similarity. IEEE Trans. Image Process. **13**, 600–612 (2004). DOI 10.1109/TIP.2003.819861. URL https://doi.org/10.1109/TIP.2003.819861 39
201. Wang, Z., Wang, J., Ma, K., Lin, D., Zhou, B.: TalkVerse: Democratizing Minute-Long Audio-Driven Video Generation. CoRR **abs/2512.14938** (2025). DOI 10.48550/ARXIV.2512.14938. URL https://doi.org/10.48550/ARXIV.2512.14938 37, 39, 40
202. Wang, Z., Zhang, P., Qi, J., Wang, Y., Xu, S., Zhang, B.: OmniTalker: One-shot Real-time Text-Driven Talking Audio-Video Generation With Multimodal Style Mimicking. In: NeurIPS (2025). URL http://papers.nips.cc/paper_files/paper/2025/hash/1f7034b0d7183fa95f1acc4c2bf23c28-Abstract-Conference.html 17
203. Weng, Y., Wang, H., Yu, Y., Du, J., He, S., Wu, X., Xu, H.: EARTalking: End-to-end GPT-style Autoregressive Talking Head Synthesis with Frame-wise Control. CoRR **abs/2603.20307** (2026). DOI 10.48550/ARXIV.2603.20307. URL https://doi.org/10.48550/ARXIV.2603.20307 2, 4, 9, 10, 12, 27, 28, 37
204. Wu, S., Li, Y., Duan, H., Zhu, Y., Min, X., Callet, P.L., Zhai, G.: Multi-Dimensional Quality Assessment for AI-Generated Human-Centric Videos: Dataset and Model. CoRR **abs/2607.16742** (2026). DOI 10.48550/ARXIV.2607.16742. URL https://doi.org/10.48550/ARXIV.2607.16742 42
205. Xiang, J., Gao, X., Guo, Y., Zhang, J.: FlashAvatar: High-Fidelity Head Avatar with Efficient Gaussian Embedding. In: CVPR, pp. 1802–1812 (2024). DOI 10.1109/CVPR52733.2024.00177. URL https://doi.org/10.1109/CVPR52733.2024.00177 1, 31, 32, 35, 45
206. Xie, L., Wang, X., Zhang, H., Dong, C., Shan, Y.: VFHQ: A High-Quality Dataset and Benchmark for Video Face Super-Resolution. In: CVPR Workshops, pp. 656–665 (2022). DOI 10.1109/CVPRW56347.2022.00081. URL https://doi.org/10.1109/CVPRW56347.2022.00081 37, 39, 40
207. Xie, Y., Gu, T., Li, Z., Zhang, C., Song, G., Zhao, X., Liang, C., Jiang, J., Xu, H., Luo, L.: X-Streamer: Unified Human World Modeling with Audiovisual Interaction. CoRR **abs/2509.21574** (2025). DOI 10.48550/ARXIV.2509.21574. URL https://doi.org/10.48550/ARXIV.2509.21574 9, 28, 29
208. Xiong, J., Liu, G., Huang, L., Wu, C., Wu, T., Mu, Y., Yao, Y., Shen, H., Wan, Z., Huang, J., Tao, C., Yan, S., Yao, H., Kong, L., Yang, H., Zhang, M., Sapiro, G., Luo, J., Luo, P., Wong, N.: Autoregressive Models in Vision: A Survey. Trans. Mach. Learn. Res. **2025** (2025). URL https://openreview.net/forum?id=1BqXkjNEGP 1, 4, 9, 26
209. Xu, S., Chen, G., Guo, Y.X., Yang, J., Li, C., Zang, Z., Zhang, Y., Tong, X., Guo, B.: VASA-1: Lifelike Audio-Driven Talking Faces Generated in Real Time. In: NeurIPS (2024). URL http://papers.nips.cc/paper_files/paper/2024/hash/014fe398da515cd552fa6e1f33e0565e-Abstract-Conference.html 1, 6, 33, 35, 45
210. Xu, S., Chen, G., Yang, J., Zhang, Y., Deng, Y., Lin, S., Guo, B.: VASA-3D: Lifelike Audio-Driven Gaussian Head Avatars from a Single Image. In: NeurIPS (2025). URL http://papers.nips.cc/paper_files/paper/2025/hash/dd7dde414b41dd1f0c2128a1bc856943-Abstract-Conference.html 32
211. Xu, S., Deng, Y., Hu, S., Wang, Y., Zhang, Y., Chen, Z., Yang, J., Guo, B.: Real-Time Generation of Streamable Talking Portrait Video with Reference-Guided Deep Compression VAEs. In: CVPR, pp. 9284–9295 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Xu_Real-Time_Generation_of_Streamable_Talking_Portrait_Video_with_Reference-Guided_Deep_CVPR_2026_paper.html 19
212. Xue, H., Luo, X., Hu, Z., Zhang, X., Xiang, X., Dai, Y., Liu, J., Zhang, Z., Li, M., Yang, J., Ma, F., Wu, Z., Yang, C., Dai, Z., Yu, F.R.: Human Motion Video Generation: A Survey. IEEE Trans. Pattern Anal. Mach. Intell. **47**, 10709–10730 (2025). DOI 10.1109/TPAMI.2025.3594034. URL https://doi.org/10.1109/TPAMI.2025.3594034 1, 4, 5
213. Yan, Z., Zhang, Y., Yuan, X., Lyu, S., Wu, B.: DeepfakeBench: A Comprehensive Benchmark of Deepfake Detection. In: NeurIPS (2023). URL http://papers.nips.cc/paper_files/paper/2023/hash/0e735e4b4f07de483cbe250130992726-Abstract-Datasets_and_Benchmarks.html 45
214. Yang, H., Zhang, Z., Dong, Y., Qian, J., Yang, J.: GaussianEmoTalker: Real-Time Emotional Talking Head Synthesis with Audio-Driven and Blendshape-Based 3D Gaussian Splatting. CoRR **abs/2607.00959** (2026). DOI 10.48550/ARXIV.2607.00959. URL https://doi.org/10.48550/ARXIV.2607.00959 32
215. Yang, S., Huang, W., Chu, R., Xiao, Y., Zhao, Y., Wang, X., Li, M., Xie, E., Chen, Y., Lu, Y., Han, S., Chen, Y.: LongLive: Real-time Interactive Long Video Generation. In: ICLR (2026). URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/91a1610c6ed9e02d33f826b46f472b92-Abstract-Conference.html 3, 23
216. Yang, S., Kong, Z., Gao, F., Cheng, M., Liu, X., Zhang, Y., Kang, Z., Luo, W., Cai, X., He, R., Wei, X.: InfiniteTalk: Audio-driven Video Generation for Sparse-Frame Video Dubbing. CoRR **abs/2508.14033** (2025). DOI 10.48550/ARXIV.2508.14033. URL https://doi.org/10.48550/ARXIV.2508.14033 23
217. Yang, S., Oh, S.W., Zhou, Y., Kim, S.J.: IF-MDM: Implicit Face Motion Diffusion Model for High-Fidelity Realtime Talking Head Generation. CoRR **abs/2412.04000** (2024). DOI 10.48550/ARXIV.2412.04000. URL https://doi.org/10.48550/ARXIV.2412.04000 17, 18
218. Yang, S., Wang, B., Zhang, Y., Shen, Y., Kim, Y.: Parallelizing Linear Transformers with the Delta Rule over Sequence Length. In: NeurIPS (2024). URL http://papers.nips.cc/paper_files/paper/2024/hash/d13a3eae72366e61dfdc7eea82eeb685-Abstract-Conference.html 29
219. Yang, Y., Zhang, F., Pi, H., Zeng, A., Guo, S., Xu, G., Zhai, W., Cao, Y., Zha, Z.J.: Gloria: Consistent Character Video Generation via Content Anchors. In: CVPR, pp. 36724–36735 (2026). URL https://openaccess.thecvf.com/content/CVPR2026/html/Yang_Gloria_Consistent_Character_Video_Generation_via_Content_Anchors_CVPR_2026_paper.html 24, 25
220. Yang, Z., Teng, J., Zheng, W., Ding, M., Huang, S., Xu, J., Yang, Y., Hong, W., Zhang, X., Feng, G., Yin, D.,

Zhang, Y., Wang, W., Cheng, Y., Xu, B., Gu, X., Dong, Y., Tang, J.: CogVideoX: Text-to-Video Diffusion Models with An Expert Transformer. In: ICLR (2025). URL https://openreview.net/forum?id=LQzN6TRFg9 17, 19
221. Yang, Z., Zeng, A., Yuan, C., Li, Y.: Effective Whole-body Pose Estimation with Two-stages Distillation. In: ICCV (Workshops), pp. 4212–4222 (2023). DOI 10.1109/ICCVW60793.2023.00455. URL https://doi.org/10.1109/ICCVW60793.2023.00455 10
222. Ye, Z., Tan, X., Yin, A., Lin, H., Zhang, G., Sun, P., Li, Y., Chan, C.M., Ye, W., Zhang, S., Xue, W.: Talker-T2AV: Joint Talking Audio-Video Generation with Autoregressive Diffusion Modeling. CoRR **abs/2604.23586** (2026). DOI 10.48550/ARXIV.2604.23586. URL https://doi.org/10.48550/ARXIV.2604.23586 27
223. Yin, T., Gharbi, M., Park, T., Zhang, R., Shechtman, E., Durand, F., Freeman, B.: Improved Distribution Matching Distillation for Fast Image Synthesis. In: NeurIPS (2024). URL http://papers.nips.cc/paper_files/paper/2024/hash/54dcf25318f9de5a7a01f0a4125c541e-Abstract-Conference.html 20
224. Yin, T., Gharbi, M., Zhang, R., Shechtman, E., Durand, F., Freeman, W.T., Park, T.: One-Step Diffusion with Distribution Matching Distillation. In: CVPR, pp. 6613–6623 (2024). DOI 10.1109/CVPR52733.2024.00632. URL https://doi.org/10.1109/CVPR52733.2024.00632 1, 10, 20
225. Yin, T., Zhang, Q., Zhang, R., Freeman, W.T., Durand, F., Shechtman, E., Huang, X.: From Slow Bidirectional to Fast Autoregressive Video Diffusion Models. In: CVPR, pp. 22963–22974 (2025). DOI 10.1109/CVPR52734.2025.02138. URL https://openaccess.thecvf.com/content/CVPR2025/html/Yin_From_Slow_Bidirectional_to_Fast_Autoregressive_Video_Diffusion_Models_CVPR_2025_paper.html 1, 22
226. Yu, H., Qu, Z., Yu, Q., Chen, J., Jiang, Z., Chen, Z., Zhang, S., Xu, J., Wu, F., Lv, C., Yu, G.: GaussianTalker: Speaker-specific Talking Head Synthesis via 3D Gaussian Splatting. In: ACM Multimedia, pp. 3548–3557 (2024). DOI 10.1145/3664647.3681675. URL https://doi.org/10.1145/3664647.3681675 31, 32
227. Yu, H., Wang, Z., Pan, Y., Cheng, M., Yang, H., Wang, C., Xie, T., Xu, X., Wei, X., Cai, X.: LLIA - Enabling Low-Latency Interactive Avatars: Real-Time Audio-Driven Portrait Video Generation with Diffusion Models. CoRR **abs/2506.05806** (2025). DOI 10.48550/ARXIV.2506.05806. URL https://doi.org/10.48550/ARXIV.2506.05806 20, 23
228. Yu, J., Qin, Y., Che, H., Liu, Q., Wang, X., Wan, P., Zhang, D., Gai, K., Chen, H., Liu, X.: A Survey of Interactive Generative Video. CoRR **abs/2504.21853** (2025). DOI 10.48550/ARXIV.2504.21853. URL https://doi.org/10.48550/ARXIV.2504.21853 3, 4
229. Yu, T., Qiao, Q., Shen, L., Zhou, K., Hu, J., Sheng, D., Hu, B., Qin, H., Gao, J., Zhou, C., Yin, S., Liu, S.: SoulX-FlashHead: Oracle-guided Generation of Infinite Real-time Streaming Talking Heads. CoRR **abs/2602.07449** (2026). DOI 10.48550/ARXIV.2602.07449. URL https://doi.org/10.48550/ARXIV.2602.07449 23, 25
230. Zeng, A., Yang, C., Ge, C., Zhang, E., Xu, G., Lin, G., Gu, G., Pi, J., Li, L., Shi, M., Wang, S., Bi, S., Tang, S., Hang, T., Guo, T., Li, V., Tong, X., Li, Y., Sun, Y., Zhao, Y., Lu, Y., Li, Y., Zhang, Z., Yang, Z., Ye, Z.: LPM 1.0: Video-based Character Performance Model. CoRR **abs/2604.07823** (2026). DOI 10.48550/ARXIV.2604.07823. URL https://doi.org/10.48550/ARXIV.2604.07823 3, 24, 25
231. Zhang, C., Shi, H., Liu, Y., Yan, Z., Yin, Y., Wu, Y., Liu, M.: InteracVid: Building a Real Interactive Audio-Visual Response Dataset from Live-Chat Videos. CoRR **abs/2608.01157** (2026). DOI 10.48550/ARXIV.2608.01157. URL https://doi.org/10.48550/ARXIV.2608.01157 39
232. Zhang, D., Liu, Y., Lin, L., Zhu, Y., Li, Y., Qin, M., Li, Y., Wang, H.: GUAVA: Generalizable Upper Body 3D Gaussian Avatar. In: ICCV, pp. 14205–14217 (2025). DOI 10.1109/ICCV51701.2025.01318. URL https://doi.org/10.1109/ICCV51701.2025.01318 32
233. Zhang, J., Jiang, K., Chen, J., Wang, X., Liu, D., Li, J., Chen, D., Lin, M., Zhou, J., Jin, H., Jia, Q., Wang, X., Wang, Y., Zhang, Z., Li, R., Huang, Z., Xiong, S., Wang, Y., Dai, Z., He, H., Luo, Y., Ning, M., Feng, W., Ye, C., Lin, X., Zhao, M., Zhu, H., Tan, H., Wang, Z., Xiang, C., Zheng, K., Deng, Z., Bao, F., Chen, J., Zhu, J.: Vidu S2: Real-Time Interactive, Editable, and Spatial Video Generation. CoRR **abs/2609.11638** (2026). DOI 10.48550/ARXIV.2609.11638. URL https://doi.org/10.48550/ARXIV.2609.11638 23
234. Zhang, J., Jiang, K., Chen, J., Wang, X., Luo, Y., Wang, Y., Chen, D., Li, J., Ye, C., Chen, M., Zhu, H., Zhao, M., Jiang, Y., Huang, Z., Xiang, C., Zheng, K., Wang, H., Wang, X., Jia, Q., Chen, X., Chen, Y., Jiang, Y., Fu, F., Deng, Z., Bao, F., Chen, J., Zhu, J.: Vidu S1: A Real-Time Interactive Video Generation Model. CoRR **abs/2607.03118** (2026). DOI 10.48550/ARXIV.2607.03118. URL https://doi.org/10.48550/ARXIV.2607.03118 3, 22
235. Zhang, R., Isola, P., Efros, A.A., Shechtman, E., Wang, O.: The Unreasonable Effectiveness of Deep Features as a Perceptual Metric. In: CVPR, pp. 586–595 (2018). DOI 10.1109/CVPR.2018.00068. URL http://openaccess.thecvf.com/content_cvpr_2018/html/Zhang_The_Unreasonable_Effectiveness_CVPR_2018_paper.html 39
236. Zhang, R., Liu, S.: LeapTalk: Breaking the Latency-Quality Trade-off in Talking Head Generation. CoRR **abs/2608.00079** (2026). DOI 10.48550/ARXIV.2608.00079. URL https://doi.org/10.48550/ARXIV.2608.00079 20
237. Zhang, W., Cun, X., Wang, X., Zhang, Y., Shen, X., Guo, Y., Shan, Y., Wang, F.: SadTalker: Learning Realistic 3D Motion Coefficients for Stylized Audio-Driven Single Image Talking Face Animation. In: CVPR, pp. 8652–8661 (2023). DOI 10.1109/CVPR52729.2023.00836. URL https://doi.org/10.1109/CVPR52729.2023.00836 1
238. Zhang, Y., Li, Z., Wang, D., Zhang, J., Zhou, D., Yin, Z., Dai, X., Yu, G., Li, X.: SpeakerVid-5M: A Large-Scale High-Quality Dataset for Audio-Visual Dyadic Interactive Human Generation. In: ICLR (2026). URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/bf7dbac50ed7f6e12ad529c5b9396bc4-Abstract-Conference.html 39, 40
239. Zhang, Y., Liu, M., Chen, Z., Wu, B., Zeng, Y., Zhan, C., He, Y., Huang, J., Zhou, W.: MuseTalk: Real-Time High Quality Lip Synchronization with Latent Space Inpainting. CoRR **abs/2410.10122** (2024). DOI 10.48550/ARXIV.2410.10122. URL https://doi.org/10.48550/ARXIV.2410.10122 2, 4, 10, 13, 14, 16, 45
240. Zhang, Y., Shen, K., Li, Y.: EmbodiedHead: Real-Time Listening and Speaking Avatar for Conversational Agents. CoRR **abs/2604.17211** (2026). DOI 10.48550/ARXIV.2604.17211. URL https://doi.org/10.48550/ARXIV.2604.17211 17, 18, 34
241. Zhang, Y., Xiong, H., Huang, Y., Song, J., Yu, J., Wang, H., Liu, J., Huang, R., Wang, L.: LiveAnimate: Stable

Long-Form Streaming Human Animation in Real-Time. CoRR **abs/2608.11745** (2026). DOI 10.48550/ARXIV.2608.11745. URL https://doi.org/10.48550/ARXIV.2608.11745 24, 25

242. Zhang, Z., Hu, Z., Deng, W., Fan, C., Lv, T., Ding, Y.: DINet: Deformation Inpainting Network for Realistic Face Visually Dubbing on High Resolution Video. In: AAAI, pp. 3543–3551 (2023). DOI 10.1609/AAAI.V37I3.25464. URL https://doi.org/10.1609/AAAI.V37I3.25464 14
243. Zhang, Z., Li, L., Ding, Y., Fan, C.: Flow-Guided One-Shot Talking Face Generation With a High-Resolution Audio-Visual Dataset. In: CVPR, pp. 3661–3670 (2021). DOI 10.1109/CVPR46437.2021.00366. URL https://openaccess.thecvf.com/content/CVPR2021/html/Zhang_Flow-Guided_One-Shot_Talking_Face_Generation_With_a_High-Resolution_Audio-Visual_Dataset_CVPR_2021_paper.html 37, 40
244. Zhang, Z., Wang, L., Zhang, Y., Gao, Y.: Temporally-Aligned Evaluation for Audio-Driven Talking Head Generation. CoRR **abs/2606.01031** (2026). DOI 10.48550/ARXIV.2606.01031. URL https://doi.org/10.48550/ARXIV.2606.01031 41
245. Zhang, Z., Wang, L., Zhang, Y., Gao, Y.: Test-Time Self-Adaptive Conditioning for Stable Audio-Driven Talking-Head Generation. CoRR **abs/2605.25488** (2026). DOI 10.48550/ARXIV.2605.25488. URL https://doi.org/10.48550/ARXIV.2605.25488 24
246. Zhao, J., Zhang, H.: Thin-Plate Spline Motion Model for Image Animation. In: CVPR, pp. 3647–3656 (2022). DOI 10.1109/CVPR52688.2022.00364. URL https://doi.org/10.1109/CVPR52688.2022.00364 31, 33
247. Zhao, L., Lu, X., Fan, R., Im, S.K., Wang, L.: GaussianHand: Real-Time 3D Gaussian Rendering for Hand Avatar Animation. IEEE Trans. Vis. Comput. Graph. **31**, 6484–6496 (2025). DOI 10.1109/TVCG.2024.3516778. URL https://doi.org/10.1109/TVCG.2024.3516778 32
248. Zhao, Q., Long, P., Zhang, Q., Qin, D., Liang, H., Zhang, L., Zhang, Y., Yu, J., Xu, L.: Media2Face: Co-speech Facial Animation Generation With Multi-Modality Guidance. In: SIGGRAPH (Conference Paper Track) (2024). DOI 10.1145/3641519.3657413. URL https://doi.org/10.1145/3641519.3657413 34
249. Zhao, S., Wang, C., Zhao, W., Li, Z., Li, M., Zhang, S.: REA-Listener: Real-Time Listening Head Generation with Dynamic Emotion Modeling and Flexible Modality Adaptation. In: ACM Multimedia, pp. 9733–9742 (2025). DOI 10.1145/3746027.3755093. URL https://doi.org/10.1145/3746027.3755093 34
250. Zhao, Z., Bao, Z., Li, Q., Qiu, G., Liu, K.: PSAvatar: A Point-Based Shape Model for Real-Time Head Avatar Animation With 3D Gaussian Splatting. IEEE Trans. Vis. Comput. Graph. **32**, 6031–6045 (2026). DOI 10.1109/TVCG.2026.3676544. URL https://doi.org/10.1109/TVCG.2026.3676544 32
251. Zhen, D., Yin, S., Qin, S., Yi, H., Zhang, Z., Liu, S., Qi, G., Tao, M.: Teller: Real-Time Streaming Audio-Driven Portrait Animation with Autoregressive Motion Generation. In: CVPR, pp. 21075–21085 (2025). DOI 10.1109/CVPR52734.2025.01963. URL https://openaccess.thecvf.com/content/CVPR2025/html/Zhen_Teller_Real-Time_Streaming_Audio-Driven_Portrait_Animation_with_Autoregressive_Motion_Generation_CVPR_2025_paper.html 23, 27, 34, 35, 43
252. Zhen, D., Zheng, X., Zhang, R., Jiang, Z., Yan, Y., Tao, M., Yin, S.: SoulX-LiveAct: Towards Hour-Scale Real-Time Human Animation with Neighbor Forcing and ConvKV Memory. CoRR **abs/2603.11746** (2026). DOI 10.48550/ARXIV.2603.11746. URL https://doi.org/10.48550/ARXIV.2603.11746 12, 21, 22, 23, 25
253. Zheng, Z., Zhao, X., Zhang, H., Liu, B., Liu, Y.: AvatarReX: Real-time Expressive Full-body Avatars. ACM Trans. Graph. **42**, 158:1–158:19 (2023). DOI 10.1145/3592101. URL https://doi.org/10.1145/3592101 31, 35
254. Zhong, T., Liang, C., Jiang, J., Lin, G., Yang, J., Zhao, Z.: FADA: Fast Diffusion Avatar Synthesis with Mixed-Supervised Multi-CFG Distillation. In: CVPR, pp. 3101–3110 (2025). DOI 10.1109/CVPR52734.2025.00295. URL https://openaccess.thecvf.com/content/CVPR2025/html/Zhong_FADA_Fast_Diffusion_Avatar_Synthesis_with_Mixed-Supervised_Multi-CFG_Distillation_CVPR_2025_paper.html 2, 17, 20, 23
255. Zhong, W., Fang, C., Cai, Y., Wei, P., Zhao, G., Lin, L., Li, G.: Identity-Preserving Talking Face Generation with Landmark and Appearance Priors. In: CVPR, pp. 9729–9738 (2023). DOI 10.1109/CVPR52729.2023.00938. URL https://doi.org/10.1109/CVPR52729.2023.00938 14
256. Zhou, H., Sun, Y., Wu, W., Loy, C.C., Wang, X., Liu, Z.: Pose-Controllable Talking Face Generation by Implicitly Modularized Audio-Visual Representation. In: CVPR, pp. 4176–4186 (2021). DOI 10.1109/CVPR46437.2021.00416. URL https://openaccess.thecvf.com/content/CVPR2021/html/Zhou_Pose-Controllable_Talking_Face_Generation_by_Implicitly_Modularized_Audio-Visual_Representation_CVPR_2021_paper.html 14
257. Zhou, M., Bai, Y., Zhang, W., Yao, T., Zhao, T., Mei, T.: Responsive Listening Head Generation: A Benchmark Dataset and Baseline. In: ECCV (38), pp. 124–142 (2022). DOI 10.1007/978-3-031-19839-7_8. URL https://doi.org/10.1007/978-3-031-19839-7_8 39, 40
258. Zhou, Y., Han, X., Shechtman, E., Echevarria, J., Kalogerakis, E., Li, D.: MakeItTalk: speaker-aware talking-head animation. ACM Trans. Graph. **39**, 221:1–221:15 (2020). DOI 10.1145/3414685.3417774. URL https://doi.org/10.1145/3414685.3417774 14
259. Zhu, H., Wu, W., Zhu, W., Jiang, L., Tang, S., Zhang, L., Liu, Z., Loy, C.C.: CelebV-HQ: A Large-Scale Video Facial Attributes Dataset. In: ECCV (7), pp. 650–667 (2022). DOI 10.1007/978-3-031-20071-7_38. URL https://doi.org/10.1007/978-3-031-20071-7_38 37, 40
260. Zhu, H., Zhao, M., He, G., Su, H., Li, C., Zhu, J.: Causal Forcing: Autoregressive Diffusion Distillation Done Right for High-Quality Real-Time Interactive Video Generation. CoRR **abs/2602.02214** (2026). DOI 10.48550/ARXIV.2602.02214. URL https://doi.org/10.48550/ARXIV.2602.02214 22
261. Zhu, L., Cai, X., Chen, Y., Li, Y., Yang, B., Liu, H., Chen, J., Li, C., Lyu, J.: OmniHuman: A Large-Scale Dataset and Benchmark for Human-Centric Video Generation. In: ECCV (61), pp. 92–110 (2026). DOI 10.1007/978-3-032-37095-2_6. URL https://doi.org/10.1007/978-3-032-37095-2_6 39
262. Zhu, Y., Zhang, L., Rong, Z., Hu, T., Liang, S., Ge, Z.: INFP: Audio-Driven Interactive Head Generation in Dyadic Conversations. In: CVPR, pp. 10667–10677 (2025). DOI 10.1109/CVPR52734.2025.00997. URL https://openaccess.thecvf.com/content/CVPR2025/html/Zhu_INFP_Audio-Driven_Interactive_Head_Generation_in_Dyadic_Conversations_CVPR_2025_paper.html 17, 18
263. Zielonka, W., Kirschstein, T., Bolkart, T., Giebenhain, S., Sklyarova, V., Deng, X., Xiang, D., Saito, S., Liu, Y., Nießner, M., Thies, J.: How to Build Digital Humans? From Priors to Photorealistic Avatars. CoRR **abs/2607.04341**

(2026). DOI 10.48550/ARXIV.2607.04341. URL https://doi.org/10.48550/ARXIV.2607.04341 4